

Enhancing Investment Decisions with Sentiment Analysis: A Probabilistic Ranking Framework*

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Abstract

We develop a probabilistic framework to extract sentiment information from text by training a model to predict and rank sentiments in newly encountered documents. Our approach imposes a joint semi-parametric model on text and ordinal response variables, addressing the challenges of sparse sentiment signals and complex response distributions. Through a word screening procedure and the use of normalized ranks, our approach achieves consistent sentiment ranking without estimating the full model. Applying our method to the Dow Jones Newswires, we demonstrate its effectiveness in extracting return-predictive signals.

Keywords: Text Mining, Machine Learning, Return Predictability, Sentiment Analysis, Screening, Topic Modeling, Penalized Likelihood

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1 Introduction

Sentiment analysis, a key task within Natural Language Processing (NLP), focuses on extracting emotional tones and subjective information from written text. It encompasses a range of techniques, including lexicon-based approaches, machine learning, and deep learning, and is widely applied across sectors such as customer feedback evaluation, survey data interpretation, social media monitoring, healthcare management, and financial market prediction [13].

One of the most prevalent techniques within the lexicon-based approach is the dictionary method. This strategy relies on a pre-established sentiment dictionary containing lists of both positive and negative words. A sentiment score is then calculated using a weighted combination of these sentiment-charged words, where the weights potentially reflect the strength of their emotional tone and are adjusted based on word frequency.

Sentiment analysis can also be approached from a machine learning perspective: treating it as either a regression task or a classification problem. These methodologies necessitate a collection of training documents already labeled with sentiment scores. Once a set of document-wise features has been established—such as the counts of dictionary words or embeddings generated by large language models—any prevalent machine learning algorithm can be utilized, like naive Bayes, random forest, or neural networks.

In this paper, we offer a unique statistical perspective on sentiment analysis by framing it as a *ranking* problem. The sentiment score assigned to a training document typically assumes the form of an *ordinal variable*, with scales that can vary arbitrarily, such as “0 to 1” or “-10 to 10.” The precise values of these scores are not inherently meaningful, as they depend on how the training labels are mapped to sentiment categories during data preprocessing. Instead of estimating exact sentiment scores for each document—which may have limited practical value—our primary focus is on ranking the sentiments of a set of test documents in accordance with their ordinal relationships. This approach only requires estimating sentiment scores up to a monotonic transformation. In many downstream tasks, sentiment ranks are more relevant

than the exact values. In Section 5, we demonstrate an application in empirical finance where sentiment ranks derived from financial news articles are utilized for portfolio construction.

Our method is founded on a semi-parametric model that encompasses two interconnected components. Firstly, we employ a fully parametric sparse topic model [7, 2], customized specifically for sentiment analysis, to model the dependence of text documents on their sentiment scores. Secondly, we utilize a nonparametric model with minimal assumptions to capture the relationship between ordinal responses and sentiment scores. Therefore, the link between these two components is established through the sentiment score of the document. Our model offers extensive flexibility, allowing for the incorporation of various types of ordinal responses, ranging from discrete (e.g., star ratings) to continuous (e.g., stock returns).

However, estimating this model poses a significant challenge: learning the exact distribution of ordinal responses is often difficult, and at times even impossible. Fortunately, our primary objective is to rank sentiments rather than predict the specific ordinal responses. Recognizing that the distribution of ordinal responses is essentially a “nuisance,” we propose an algorithm that circumvents the need for estimating this distribution while still achieving consistent sentiment ranking. The core idea of our algorithm is a strategic integration of rank statistics, correlation screening, and topic modeling techniques.

A key distinction of our structural approach, in contrast to reduced-form lexicon-based or machine learning methods, is the adoption of a joint data-generating model for text documents and ordinal responses. This allows for a rigorous examination of the estimator’s asymptotic behavior, providing valuable theoretical insights into how sentiment analysis performs under varying data dimensions and signal-to-noise ratios—insights that cannot be gained from reduced-form approaches. We consider scenarios where the number of documents approaches infinity, and we accommodate settings in which the dictionary size and average document length may either remain bounded or grow with the number of documents. This flexibility ensures that our theory is applicable in high-dimensional settings.

When the dictionary size significantly exceeds the document lengths, we impose a sparsity assumption that only a small fraction of words carry sentiment information. This assumption holds true in practical scenarios and significantly enhances the signal-to-noise ratio. Consequently, we demonstrate that our algorithm produces a consistent estimator of the sentiment score, up to a linear transformation uniquely determined by the training data. This suffices to guarantee the consistent ranking of sentiments for any given set of testing documents. Notably, our theory unveils a somewhat unexpected finding: consistent sentiment ranking can be achieved without the need for consistent parameter estimation. The underlying reason behind this lies in the inherent ordinal nature shared by both our objective of sentiment ranking, which aims to order the true sentiment scores, and the training labels themselves, which also serve as ordinal variables.

We present an application of our methodology in which we address one of the most extensively researched text-based questions in finance: the extent to which business news can explain and predict observed variations in asset prices. To conduct our analysis, we utilize the machine text feed and archive database of the Dow Jones Newswires, a highly regarded information source widely subscribed to and closely monitored by sophisticated investors, serving as a central resource for market participants. This dataset spans over 30 years and contains time-stamped articles tagged with identifiers corresponding to the relevant firms. By leveraging these identifiers, we match the articles with stock return data from CRSP, which act as the ordinal responses in our study.

To demonstrate the economic significance of our model’s statistical performance, we highlight its predictive power through a simple trading strategy. This strategy involves purchasing assets with positive sentiment from recent news and selling those with negative sentiment. The portfolio constructed using our approach delivers exceptional risk-adjusted returns in out-of-sample testing, outperforming not only off-the-shelf sentiment scores from RavenPack, the industry-leading provider of financial news sentiment, but also black-box, reduced-form pre-

dictionaries from transformer-based large language models. Furthermore, we utilize our framework to investigate several economic questions. For example, we investigate the impact of “fresh” versus “stale” news on return behavior and examine how disparities in news assimilation are associated with a range of stock attributes.

Related literature

Traditional approaches to sentiment analysis, such as the dictionary methods, rely on pre-defined sentiment dictionaries. However, these dictionaries have limitations. For instance, general sentiment dictionaries like the Harvard IV dictionary primarily capture the psychological states of words and may not be suitable for applications where sentiment information is conveyed through psychologically neutral words. On the other hand, domain-specific dictionaries like the Loughran-McDonald dictionary [10] tailored for finance are not always readily accessible in practical situations. In contrast, our method leverages ordinal labels in the training data to guide the process of sentiment analysis and automatically learns a domain-specific sentiment dictionary. This data-driven approach ensures adaptability to the specific domain of interest without relying on pre-existing dictionaries.

Ordinal classification [6] has emerged as a valuable approach in sentiment analysis to handle ordinal responses. Unlike standard multi-class classification, which assumes unordered class categories, ordinal classification embraces the ordered nature of the classes and constructs classification boundaries accordingly. An interesting recent study on this problem is presented by [15]. The authors proposed an ordinal classifier specifically designed for unstructured predictors, where imprecise information regarding the strength of associations between predictors is available. One practical application of their method is predicting star ratings based on product reviews. However, it is important to note that classification methods are typically limited to discrete ordinal responses. In our financial application discussed in Section 5, the ordinal responses correspond to continuous stock returns. While it is possible to discretize these returns, doing so would result in a large number of classes, and traditional or-

dinal classification methods tend to perform well with a smaller number of classes. In contrast, our method offers flexibility in accommodating various types of ordinal responses, whether they are discrete or continuous in nature.

The multinomial inverse regression (MNIR) was introduced by [14] to study text-sentiment relationships. This fully parametric model employs logistic functions to map the ordinal response to the population word frequencies. Similarly, we also adopt multinomial distributions for word counts in our approach. However, our method diverges from MNIR in the way we handle the sentiment information. We create two separate models for the text and the ordinary response instead of linking them directly, and the two models are connected by the latent sentiment score. This allows us to create a flexible model for the conditional distribution of ordinal response given its sentiment score, making our approach semi-parametric and more versatile than MNIR. Additionally, our algorithm circumvents the need to estimate the exact distribution of the ordinal response, allowing for a higher degree of flexibility in the modeling process. Because of this, our method can adapt to a broader range of scenarios and better capture the complexities inherent in sentiment analysis.

Deep neural networks (DNNs) have recently garnered significant attention for their impressive achievements across various tasks. The application of DNNs in sentiment analysis has also been widely explored; see [13] for a comprehensive survey. More recently, large-language models (LLMs) have emerged as leading benchmarks in DNN-based methods, see, e.g., [4], [11]. These LLM-based approaches primarily focus on training a general-purpose model that can be applied to downstream tasks, such as sentiment analysis. While our method can be integrated with LLMs by replacing word counts with DNN-learned features, such as embeddings, it is important to note that our current approach already offers several advantages over DNN approaches. Our method is a “white box,” making it easy to interpret and diagnose. It is also computationally more efficient and requires fewer training documents. This not only establishes it as a valuable benchmark for sentiment analysis but also as a theoretical refer-

ence that provides critical statistical insights. Moreover, it avoids issues such as look-ahead bias, which LLMs may encounter due to their opaque use of web-based datasets. Despite its simplicity, our approach delivers performance comparable to that of LLM-based methods.

The rest of the paper is organized as follows. In Section 2, we introduce the semi-parametric model and propose a sentiment ranking algorithm called SESTM. In Section 3, we present the theoretical properties of our method. Section 4 contains numerical experiments, and Section 5 reports an empirical analysis of the stock-level news and returns via SESTM and alternative methods. We conclude the paper with further discussions in Section 6. The mathematical proofs and additional details of empirical analysis are relegated to the supplementary material.

2 A Probabilistic Framework for Sentiment Analysis

Consider a collection of n documents, each composed using a dictionary of m words, where both n and m are large. We represent these documents in their bag-of-words form. Specifically, for any $1 \leq i \leq n$ and $1 \leq j \leq m$, let $d_{j,i}$ represent the frequency of word j in document i . The data matrix, denoted as D , is an $m \times n$ matrix defined as $D = [d_1, d_2, \dots, d_n]$, where each $d_i \in \mathbb{R}_+^m$ is a word count vector of document i . In addition to these documents, we have access to a response variable y_i for each document. This variable captures ordinal values and can either be discrete or continuous. We assume that it contains sentiment-related information. In the case of customer reviews, y_i represents the associated star rating and is discrete. When analyzing financial news articles (refer to Section 5), we identify the mentioned company and utilize its stock return (before and after the news release) as y_i , which is a continuous value.

We assume that each document has a sentiment score $p_i \in [0, 1]$. A score of 1 indicates a *maximally positive* sentiment at the document level, while a score of 0 represents a *maximally negative* sentiment. In Section 2.1, we introduce a joint model for d_i and y_i , where these two quantities are interconnected through p_i . Furthermore, in Section 2.2, we propose a sentiment extraction algorithm, and in Section 2.3, we delve into the theoretical insights of our method.

2.1 Data generating processes

We assume the training data $\{(d_i, y_i)\}_{1 \leq i \leq n}$ are independently generated. Fixing i , we describe the model for d_i and y_i , respectively, starting with the model for d_i . Let $\{1, 2, \dots, m\} = S \cup N$ represent a partition of the dictionary into two sets: S , containing sentiment-charged words, and N , containing sentiment-neutral words. Define $d_{[S],i}$ as the sub-vector of d_i restricted to the indices in S , and similarly define $d_{[N],i}$ for sentiment-neutral words.

We denote s_i and n_i as the total counts of sentiment-charged and sentiment-neutral words in document i , respectively. A vector is referred to as a probability mass function (PMF) if its entries are nonnegative and sum up to 1. For two PMFs $Q_i \in \mathbb{R}_+^{|S|}$ and $\Omega_i \in \mathbb{R}_+^{m-|S|}$, we assume the following multinomial distributions:

$$d_{[S],i} \sim \text{Multinomial}(s_i, Q_i), \quad d_{[N],i} \sim \text{Multinomial}(n_i, \Omega_i). \quad (1)$$

According to this model, we have $Q_i = s_i^{-1} \mathbb{E} d_{[S],i}$ and $\Omega_i = n_i^{-1} \mathbb{E} d_{[N],i}$, which characterize the distributions of sentiment-charged and sentiment-neutral words, respectively.

To allow for greater flexibility in modeling the sentiment-neutral words, we make no structural assumptions on Ω_i .¹ However, for the sentiment-charged words, we impose a two-topic structure on Q_i . Let O_+ and O_- be two PMFs in $\mathbb{R}_+^{|S|}$, referred to as the *positive sentiment topic* and the *negative sentiment topic*, respectively. Recall that $p_i \in [0, 1]$ represents the true sentiment score of document i . We make the assumption:

$$Q_i = p_i O_+ + (1 - p_i) O_-. \quad (2)$$

Here, O_+ and O_- describe the distribution of sentiment-charged words in a maximally positive document and maximally negative document, respectively. For any document i , the vector Q_i represents the population frequencies of sentiment-charged words and is a convex combination of O_+ and O_- . By combining this assumption with the model in (1), we effectively model $d_{[S],i}$ using Hofmann’s topic model [7], with two topics representing positive and negative sentiment.

¹Using the financial application in Section 5 as an example, Ω_i may include industry-specific information. Our framework permits the sentiment score p_i to depend on Ω_i in a flexible manner, allowing the frequencies of sentiment words to vary across different industries. As long as industry topics are diversified in the news corpus, our screening step in Section 2.2 will be able to separate words in S from those in N .

In this model, the value of p_i representing the “neutral” sentiment is unspecified.² For convenience, we define the average p_i of training documents, $\tau = n^{-1} \sum_{i=1}^n p_i$, as the neutral sentiment. By (2), for a neutral document, its word distribution is given by $\tau O_+ + (1 - \tau) O_-$. Motivated by this observation, we define a *vector of frequency*, F , and a *vector of tone*, T :

$$F = \tau O_+ + (1 - \tau) O_-, \quad T = (1 - \tau)(O_+ - O_-). \quad (3)$$

If a word has a larger value in F , it indicates a higher overall frequency of occurrence. The sentiment strength conveyed by a word is captured by its corresponding entry in T . A positive T_j indicates that word j has a greater weight in the positive sentiment topic than in the negative one, meaning that the word has a positive tone. Conversely, a negative T_j indicates a negative tone for word j . The magnitude of T_j reflects the strength of the tone.

Next, we proceed to model the ordinal response variable y_i . Without loss of generality, we assume that y_i takes real values, where $y_i = 0$ signifies the absence of a sentiment signal. To illustrate this, let’s consider the example of a 5-star customer rating. If we interpret a review with 3 stars as being “neutral,” we can assign values to y_i from the set $\{-2, -1, 0, 1, 2\}$, representing 1 star to 5 stars, respectively. In this particular example, y_i assumes discrete values. In Section 5 where we analyze financial news articles, we identify the company mentioned in the news and utilize the stock return of this company (or its idiosyncratic component) as the corresponding y_i . A zero return, denoted by $y_i = 0$, indicates the absence of a sentiment signal. In this case, y_i is continuous. To encompass both examples, we make the assumption:

$$\mathbb{P}(y_i > 0) = g(p_i), \text{ for a monotone increasing function } g(\cdot). \quad (4)$$

This model has flexibility as it does not specify the complete distribution of y_i . The monotone increasing property of $g(\cdot)$ ensures that the sign of y_i is positively associated with p_i .

Our model can be classified as semi-parametric. The parametric component includes Model (1)-(3), which deals with word counts and is a natural adaptation of the topic model for sentiment analysis. Meanwhile, Model (4) for ordinal responses is nonparametric.

²This is related to model identification. Readers are referred to Appendix A for a detailed discussion.

It is important to note that our primary objective is to rank sentiments rather than predict the exact ordinal responses. Therefore, we consider the distribution of y_i as a nuisance and refrain from estimating it within our sentiment analysis framework. It is also worth mentioning that Model (4) differs from a classification model that treats the sign of y_i as a binary class label. In our framework, not only the sign of y_i but also its relative rank holds significance. Specifically, the rank of y_i among $y_1, y_2, \dots, y_n$ contains information regarding the overall signal strength in the i^{th} document. This rank information is employed to “re-weight” the contribution of each document in our estimation procedure.

2.2 SESTM: A sentiment extraction and ranking algorithm

We now present our Sentiment Extraction via Screening and Topic Modeling (SESTM) procedure, comprising three steps. Firstly, we conduct screening to identify the set of sentiment-charged words. Secondly, we estimate the positive and negative sentiment topics O_+ and O_- . Lastly, we employ penalized maximum likelihood to estimate the sentiment score for each test document, and use these scores to rank all test documents.

Although the dictionary size m can be large, the proportion of sentiment-charged words is often small (e.g., see [16]). In the first step of SESTM, we perform screening on the dictionary to identify these sentiment-charged words. Our screening statistic measures the frequency at which word j co-occurs with a positive value of y :

$$f_j = \frac{\text{count of word } j \text{ in documents with } \text{sgn}(y) = +1}{\text{count of word } j \text{ in all documents}}, \quad 1 \leq j \leq m, \quad (5)$$

where $\text{sgn}(x)$ denotes the sign function that returns 1 if $x > 0$ and -1 otherwise. We solely rely on the sign of y rather than its actual value. One reason for this choice is that an ordinal variable can be arbitrarily re-coded (e.g., transforming $y \in \{-1, 0, 1\}$ to $y \in \{-2, 0, 3\}$ while preserving the order of y_i 's). Hence, it is advisable to avoid utilizing the actual values of y .

According to Model (4), the occurrence of a sentiment-charged word is correlated with the sign of y . We denote $\hat{\pi} \in [0, 1]$ as the fraction of documents with a positive value of y , and let

k_j represent the count of word j in all documents (i.e., the denominator in (5)). Given tuning parameters $(\alpha_+, \alpha_-, \kappa)$, we estimate S by

$$\widehat{S} = \{j : f_j \geq \widehat{\pi} + \alpha_+, \text{ or } f_j \leq \widehat{\pi} - \alpha_-\} \cap \{j : k_j \geq \kappa\}. \quad (6)$$

In equation (6), we utilize asymmetric thresholds (i.e., $\alpha_{\pm}$ can have different values) to handle scenarios where the individual signal strengths in positive sentiment terms and negative sentiment terms differ. Furthermore, given that the variance of f_j decreases as k_j increases, we introduce an additional threshold κ for k_j to avoid excessively noisy f_j values.

In Step 2 of SESTM, our attention is directed towards the words in the set $\widehat{S}$, as we aim to estimate the two sentiment topic vectors, denoted as O_+ and O_- . We denote the combined matrix as $O = [O_+, O_-] \in \mathbb{R}_+^{|S| \times 2}$. For document i , we define $d_{[\widehat{S}],i}$ as the sub-vector of d_i that is restricted to the indices found in $\widehat{S}$. Additionally, we define

$$H = [\widehat{h}_1, \widehat{h}_2, \dots, \widehat{h}_n], \quad \text{with} \quad \widehat{h}_i = \widehat{s}_i^{-1} d_{[\widehat{S}],i}, \quad \text{and} \quad \widehat{s}_i = \sum_{j \in \widehat{S}} d_{j,i}. \quad (7)$$

If $\widehat{S} = S$, then by Model (2), $\widehat{h}_i \approx Q_i = p_i O_+ + (1 - p_i) O_-$, or in the matrix form,

$$H \approx OW, \quad \text{with} \quad W = \begin{bmatrix} p_1 & p_2 & \cdots & p_n \\ 1 - p_1 & 1 - p_2 & \cdots & 1 - p_n \end{bmatrix}. \quad (8)$$

It motivates an ideal estimator $\widehat{O}^* = HW'(WW')^{-1}$; however, the values of p_i are unknown.

To address this, we utilize the ordinal responses to construct a proxy for p_i as follows:

$$\widehat{p}_i = \frac{\text{rank of } y_i \text{ in } \{y_l\}_{l=1}^n}{n}, \quad 1 \leq i \leq n. \quad (9)$$

Here, $\widehat{p}_i$ is a normalized rank, ranging from 0 to 1, representing the relative position of y_i among all ordinal responses in the training documents. When y_i 's have ties, we can pick an arbitrary order among the equal-valued y_i 's.³ Our proposed estimator of O is

$$\widehat{O} = H\widehat{W}'(\widehat{W}\widehat{W}')^{-1}, \quad \text{where} \quad \widehat{W} = \begin{bmatrix} \widehat{p}_1 & \widehat{p}_2 & \cdots & \widehat{p}_n \\ 1 - \widehat{p}_1 & 1 - \widehat{p}_2 & \cdots & 1 - \widehat{p}_n \end{bmatrix}. \quad (10)$$

³For example, let's consider y_i as the star rating associated with a customer review. We denote ϵ_k as the fraction of k -star reviews (where we set $\epsilon_0 = 0$ by default). For each $1 \leq k \leq 5$, we require that the normalized ranks of all k -star reviews are within the interval $[\sum_{j=0}^{k-1} \epsilon_j, \sum_{j=0}^k \epsilon_j]$ (e.g., we can either set all these $\widehat{p}_i$ to be equal to the midpoint of this interval or let them uniformly distributed in this interval; see Section B of the supplementary material for a detailed discussion). These normalized ranks are invariant under any re-coding of y (e.g., if $y \in \{1, 2, 3, 4, 5\}$ is re-coded to $y \in \{-1, -2, 0, 1, 2\}$, the $\widehat{p}_i$ values remain the same).

While $\hat{p}_i$'s are not estimates of p_i , interestingly, the estimator in (10) remains meaningful (see Section 2.3 for a detailed explanation). We view $\hat{O}$ as a *linear estimator*, because each of $\hat{O}_\pm$ is a weighted linear combination of the empirical word frequency vectors $\hat{h}_1, \hat{h}_2, \dots, \hat{h}_n$, and the normalized ranks in (9) are used to determine the weights in (10).

In Step 3 of SESTM, we use $\hat{S}$ and $\hat{O}$ to estimate the sentiment p for any new test document. Let $d \in \mathbb{R}_+^m$ be word count vector of the new document and s be the total count of sentiment-charged words. According to Model (1)-(2), $d_{[S]} \sim \text{Multinomial}(s, pO_+ + (1-p)O_-)$. We thus estimate p by penalized maximum likelihood estimation (MLE):

$$\hat{p} = \arg \max_{p \in [0,1]} \left\{ \hat{s}^{-1} \sum_{j \in \hat{S}} d_j \log \left(p\hat{O}_{+,j} + (1-p)\hat{O}_{-,j} \right) + 2\lambda \left[\hat{\tau} \log(p) + (1-\hat{\tau}) \log(1-p) \right] \right\}, \quad (11)$$

where $\hat{\tau} = n^{-1} \sum_{i=1}^n \hat{p}_i$ is the average of normalized ranks of training documents, $\hat{s} = \sum_{j \in \hat{S}} d_j$ is the total count of words from $\hat{S}$, and $(d_j, \hat{O}_{+,j}, \hat{O}_{-,j})$ are the j th entries of the corresponding vectors. The last term, $2\lambda [\hat{\tau} \log(p) + (1-\hat{\tau}) \log(1-p)]$, serves as a penalty that encourages the estimate to move toward $\hat{\tau}$. This penalty can be interpreted as imposing a prior distribution of $\text{Beta}(1 + 2\lambda\hat{\tau}, 1 + 2\lambda(1-\hat{\tau}))$ on p and maximizing the posterior likelihood. In Section 2.3, we will show that $\hat{p}$ is an estimate of the rescaled sentiment score $p^* = \hat{\tau} + \hat{\rho}^{-1}(p - \tau)$, where $\tau = n^{-1} \sum_{i=1}^n p_i$ is the neutral sentiment; as a result, moving $\hat{p}$ toward $\hat{\tau}$ is motivated by the observation that the majority of documents exhibit a tone that is close to neutral.

In the previous step, an estimated sentiment score is generated for each test document. In Step 4 of SESTM, these estimated sentiment scores are utilized to rank the sentiments of all test documents. A larger value of $\hat{p}$ corresponds to a more positive sentiment.

A summary of SESTM is given in Algorithm 1. It has four tuning parameters $(\alpha_+, \alpha_-, \kappa, \lambda)$. The first three parameters, $(\alpha_+, \alpha_-, \kappa)$, are utilized during the training phase and selected via cross-validation. The last parameter, λ , does not significantly impact the sentiment ranking, as will be explained in Section 2.3. Consequently, we set it to a fixed value, say, 10, which suffices to ensure that all estimates remain within the interior of the interval $[0, 1]$.

Algorithm 1 Sentiment Extraction via Screening and Topic Modeling (SESTM).

Training. Input is the word count matrix $D = [d_1, d_2, \dots, d_n]$, ordinal responses $y_1, y_2, \dots, y_n$, and tuning parameters $(\alpha_+, \alpha_-, \kappa, \lambda)$.

1. *Word screening:* Compute f_j for each word j as in (5). Estimate S by $\hat{S}$ in (6).
2. *Sentiment topic estimation:* Construct $\hat{p}_1, \dots, \hat{p}_n$ as in (9) and obtain $\hat{O} = [\hat{O}_+, \hat{O}_-]$ as in (10). Due to noise perturbation, $\hat{O}$ may have negative entries. We set negative entries of this matrix to zero and re-normalize each column to have a unit ℓ^1 -norm. To simplify notations, we still denote the resulting matrix by $\hat{O}$.

Testing: Input is $\hat{O}_\pm$ (from training) and the word count vectors of a set of new documents.

3. *Sentiment scoring:* Estimate the sentiment score of each new document using (11).
 4. *Ranking:* Rank new documents in the descending order of estimated sentiment scores.
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2.3 Some theoretical insights for SESTM

We offer further insights into SESTM, with a specific focus on elucidating the rationale behind the formula (10). While the $\hat{p}_i$ values serve as normalized rank statistics rather than estimates of the p_i values themselves, we explain why this approach remains meaningful.

Same as in (3), let F and T be the *vector of frequency* and the *vector of tone*, respectively. Given $\hat{O}_\pm$, we estimate these two vectors by $\hat{F} = \hat{\tau}\hat{O}_+ + (1 - \hat{\tau})\hat{O}_-$ and $\hat{T} = (1 - \hat{\tau})(\hat{O}_+ - \hat{O}_-)$. In Section 2.2, we have seen $H \approx OW$. Plugging it into (10) gives $\hat{O} \approx OW\widehat{W}'(\widehat{W}\widehat{W}')^{-1}$. In addition, it is implied by (3) that $O_+ = F + T$ and $O_- = F + \frac{\tau}{\tau-1}T$. Combining them gives

$$[\hat{F}, \hat{T}] = \hat{O} \begin{bmatrix} \hat{\tau} & 1 - \hat{\tau} \\ 1 - \hat{\tau} & \hat{\tau} - 1 \end{bmatrix} \approx [F, T] \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & \frac{\tau}{\tau-1} \end{bmatrix} W\widehat{W}'(\widehat{W}\widehat{W}')^{-1}}_{\equiv M} \begin{bmatrix} \hat{\tau} & 1 - \hat{\tau} \\ 1 - \hat{\tau} & \hat{\tau} - 1 \end{bmatrix}. \quad (12)$$

The next lemma is one of our key results, which indicates that M is a diagonal matrix:

Lemma 1. $M = \text{diag}(1, \frac{1-\hat{\tau}}{1-\tau}\hat{\rho})$, where $\hat{\rho} = [\sum_{i=1}^n (p_i - \tau)(\hat{p}_i - \hat{\tau})] / [\sum_{i=1}^n (\hat{p}_i - \hat{\tau})^2]$.

Combining Lemma 1 with (12) gives $\hat{F} \approx F$ and $\hat{T} \approx \frac{1-\hat{\tau}}{1-\tau}\hat{\rho}T$. It shows that the linear estimator in (10) can consistently estimate F ; but for T there is an unknown bias. This bias quantity $\frac{1-\hat{\tau}}{1-\tau}\hat{\rho}$ measures the association between true sentiment scores and normalized rank statistics.⁴

⁴For example, when $\tau = \frac{1}{2}$ and $\hat{p}_i$'s are uniformly distributed in $[0, 1]$, $\hat{\rho} = \frac{12}{n} \sum_{i=1}^n (p_i - \frac{1}{2})(\hat{p}_i - \frac{1}{2}) + O(n^{-1})$.

We then proceed to examine the testing stage. Let d represent the word count vector of a test document, and write $\hat{h} = \hat{s}^{-1}d_{[\hat{s}]}$. The optimization in (11) essentially targets to find $\hat{p}$ such that $\hat{h} \approx \hat{p}\hat{O}_+ + (1 - \hat{p})\hat{O}_-$.⁵ To further analyze this, we introduce a re-parametrization of $\hat{p}$ by $\hat{\eta} = \hat{p} - \hat{\tau}$. Consequently, we can express $\hat{p}\hat{O}_+ + (1 - \hat{p})\hat{O}_- = \hat{F} + \frac{\hat{\eta}}{1 - \hat{\tau}}\hat{T}$. In addition, from the preceding analysis of the training stage, we observe that $\hat{F} \approx F$ and $\hat{T} \approx \frac{1 - \hat{\tau}}{1 - \tau}\hat{\rho}T$. By combining these findings, we can interpret (11) as finding $\hat{\eta}$ such that

$$\hat{h} \approx F + \frac{\hat{\eta}\hat{\rho}}{1 - \tau}T. \quad (13)$$

Meanwhile, if we re-parametrize p by $\eta = p - \tau$, it follows from Model (2) that

$$\hat{h} \approx pO_+ + (1 - p)O_- = F + \frac{\eta}{1 - \tau}T. \quad (14)$$

Comparing (14) with (13), we can immediately deduce that $\hat{\eta} \approx \hat{\rho}^{-1}\eta$, or equivalently speaking, $\hat{p} \approx \hat{\tau} + \hat{\rho}^{-1}(p - \tau)$. We shall refer to $p^* \equiv \hat{\tau} + \hat{\rho}^{-1}(p - \tau)$ as the *re-scaled sentiment score* of the test document. Based on the aforementioned analysis, it can be inferred that SESTM learns the re-scaled sentiment score during the testing stage.

In the ideal case where $n^{-1} \sum_{i=1}^n (\hat{p}_i - p_i)^2$ converges to 0 in probability, it can be shown that $\hat{\tau} = \tau + o_{\mathbb{P}}(1)$ and $\hat{\rho} = 1 + o_{\mathbb{P}}(1)$, so that no bias is introduced (refer to Corollary 1). However, in most practical situations, the ordinal responses only carry weak sentiment signals, leading to $\hat{\rho}$ being a positive number distinct from 1. In such cases, p^* retains the positivity/negativity of the tone but exhibits more (or less) “extremeness” compared to the original p when $\hat{\rho} < 1$ ($\hat{\rho} > 1$). If there exists an additional set of “validation” documents with known true p_i values, we can estimate $\hat{\rho}$ by performing regression of estimated sentiment scores on true sentiment scores in this validation dataset. Consequently, it becomes possible to eliminate the bias in $\hat{p}$.

On the other hand, learning the re-scaled sentiment score is sufficient for many downstream tasks. For example, Section 5 presents an application. The test documents consist of financial news articles from a trading day. The estimated sentiment scores are utilized to identify stocks mentioned in the 50 most positive articles and 50 most negative articles, forming a portfolio based on these stocks. In this application, since the rescaled sentiments and true sentiments yield the same ranks, learning the re-scaled sentiments is sufficient.

Our final observation pertains to the penalty term in (11) and the tuning parameter λ . The optimization process in (11) essentially seeks to determine $\hat{p} = \hat{\tau} + \hat{\eta}$ in such a way that

⁵To see this, consider maximizing $\sum_j \hat{h}_j \log(x_j)$, subject to $x_j \geq 0$ and $\sum_j x_j = 1$. The solution is $x_j = \hat{h}_j$.

$\hat{h} \approx \hat{F} + \frac{\hat{\eta}}{1-\hat{\tau}}\hat{T}$ (see (13) and the paragraph above it). The penalty term aims to shrink the value of $\hat{p}$ toward $\hat{\tau}$, which is equivalent to shrinking $\hat{\eta}$ toward 0. The tuning parameter λ governs the degree of shrinkage applied. Different levels of shrinkage will impact the estimated sentiment scores, but they will have minimal influence on the relative sentiment ranks, because all estimates are similarly shrunk toward $\hat{\tau}$. Hence, the choice of λ is not critical.

3 Statistical Theory

Recall that n is the number of training documents, and that m is the size of full vocabulary. Additionally, $|S|$ denotes the size of the sentiment dictionary, and $s_1, s_2, \dots, s_n$ represent the counts of sentiment-charged words in individual documents. We use n as the driving asymptotic parameter and allow $(m, |S|, s_1, \dots, s_n)$ to depend on n . The training data, denoted as $(d_i, y_i)_{i=1}^n$, is generated from the model specified in equations (1) through (4).

We begin by stating a few assumptions. Firstly, we introduce a measure of the association between the word count and the sign of y . For each $1 \leq j \leq m$, we define δ_j as follows:

$$\delta_j = \mathbb{E} \left[n^{-1} \sum_{i=1}^n (\text{sgn}(y_i) - \overline{\text{sgn}}_y) d_{j,i} \right], \quad \text{where} \quad \overline{\text{sgn}}_y = n^{-1} \sum_{i=1}^n \text{sgn}(y_i).$$

In order for the notions of “sentiment-charged” and “sentiment-neutral” to be well-defined, we expect $\delta_j = 0$ for $j \in N$ and $\delta_j \neq 0$ for $j \in S$. For $j \in S$, it can be shown (e.g., see the proof of Theorem 2) that $\delta_j \neq 0$ if and only if $O_{+,j} - O_{-,j} \neq 0$. For $j \in N$, the quantities $O_{\pm,j}$ are not defined. Therefore, we directly impose the assumption $\delta_j = 0$ as a means to distinguish words in N from words in S .

Assumption 1 (Word Classification). *For any $j \in S$, $O_{+,j} - O_{-,j} \neq 0$; for any $j \in N$, $\delta_j = 0$.*

Finally, we need some regularity conditions. Let s_i and n_i be the respective total count of sentiment-charged words and sentiment-neutral words in document i , for $1 \leq i \leq n$. Write $\bar{s} = n^{-1} \sum_{i=1}^n s_i$ and $\bar{n} = n^{-1} \sum_{i=1}^n n_i$. Let $\Omega_{j,i}$ denote the j^{th} entry of the vector Ω_i , for $j \in N$. Define $q_j := (n\bar{n})^{-1} \sum_{i=1}^n n_i \Omega_{j,i}$. Let (τ, F, T) be the same as in equation (3).

Assumption 2 (Regularity Conditions). *There exist constants $a_0 \in (0, 1/2)$ and $c_0 > 0$ such that the following statements hold:*

- (a) $a_0 \leq \tau \leq 1 - a_0$, and $a_0 \leq \hat{\tau} \leq 1 - a_0$.
- (b) $\omega_n / \log(m) \rightarrow \infty$, where $\omega_n := \min \{ n, n\bar{s} \min_{j \in S} F_j, n\bar{n} \min_{j \in N} q_j \}$

- (c) $s_1, s_2, \dots, s_n$ satisfy that $\sum_{i=1}^n s_i^2 \leq Cn\bar{s}^2$, $\sum_{i=1}^n s_i^{-1} \leq Cn\bar{s}^{-1}$, and $s_{\min}^{-1}\bar{s} \leq C\omega_n$. For each $j \in N$, $\sum_{i=1}^n n_i^2 \Omega_{j,i}^2 \leq Cn\bar{n}^2 q_j^2$.
- (d) $\sum_{i=1}^n s_i [p_i O_{+,j} + (1 - p_i) O_{-,j}] \geq c_0 \sum_{i=1}^n s_i F_j$, for any $j \in S$.
- (e) $|\sum_{i=1}^n (g(p_i) - \bar{g})(s_i - \bar{s})| \leq C\bar{s}\sqrt{n \log(m)}$, where $g(\cdot)$ is as in (4) and $\bar{g} = n^{-1} \sum_{i=1}^n g(p_i)$.

Condition (a) says that the average of p_i 's cannot be too extreme, which prevents extreme imbalance of positive/negative sentiments in training and is necessary for efficient estimation of $O_{\pm}$. It also says that the average of $\hat{p}_i$'s cannot be too extreme. This is not hard to satisfy: When y_i 's have no ties, $\hat{\tau} = 1/2$ always holds; when y_i 's have ties, the user has the freedom to choose $\hat{p}_i$ (see Section B of the supplementary material). In Condition (b), noting that the expected total count of a word in all training documents is of the order $n\bar{s}F_j$ for $j \in S$ and $n\bar{n}q_j$ for $j \in N$, we require these expected counts to dominate $\log(m)$. This assumption is reasonable because, in real data, n is usually very large. Condition (c) imposes mild balance conditions on $\{s_i\}_{i=1}^n$ and on $\{n_i \Omega_{j,i}\}_{i=1}^n$ for each $j \in N$. For example, since $\omega_n \rightarrow \infty$ by Condition (b), the requirement $s_{\min}^{-1}\bar{s} \leq C\omega_n$ still allows $s_{\min} \ll \bar{s}$ to happen. Condition (d) and Condition (e) are high-level technical conditions. The former is satisfied when all p_i 's are bounded away from 0 and 1, whereas the latter holds with probability approaching one if we assume that s_i 's are independently and identically distributed.

3.1 Accuracy of estimating sentiment topics

The screening step estimates the sentiment dictionary by $\hat{S}$. We first establish the consistency of screening by showing that $\hat{S} = S$ with an overwhelming probability. Let $g(\cdot)$ be as in Model (4) and $\bar{g} = n^{-1} \sum_{i=1}^n g(p_i)$. We define a quantity to capture the association between ordinal responses and document sentiments:

$$\theta := \frac{\sum_{i=1}^n s_i (p_i - 1/2) [g(p_i) - \bar{g}]}{\sum_{i=1}^n s_i}. \quad (15)$$

This quantity is increasing with steepness of g and extremeness of training documents' tones.

Lemma 2 (Screening Statistics). *Consider Models (1)-(2) and (4), where Assumptions 1-2 are satisfied. Let ϵ_n be a sequence such that $\epsilon_n = o(1)$ and $\log(\epsilon_n^{-1}) = o(n \min\{1, \bar{s} \min_{j \in S} (O_{+,j} + O_{-,j}), \bar{n} \min_{j \in N} q_j\})$, where $(\bar{s}, \bar{n}, q_j)$ are the same as in Assumption 2. With probability $1 - \epsilon_n$,*

simultaneously for all $1 \leq j \leq m$, the following statements are true: For $j \in S$, $|f_j - \hat{\pi}| \geq \theta \cdot \frac{2\tau|O_{+,j} - O_{-,j}|}{(1+\tau)(O_{+,j} + O_{-,j})} - \frac{C\sqrt{\log(m/\epsilon_n)}}{\sqrt{n \min\{1, \bar{s}(O_{+,j} + O_{-,j})\}}}$. For $j \in N$, $|f_j - \hat{\pi}| \leq \frac{C\sqrt{\log(m/\epsilon_n)}}{\sqrt{n \min\{1, \bar{n}q_j\}}}$.

We recall that $\hat{S}$ is obtained by thresholding $|f_j - \hat{\pi}|$. Theorem 2 justifies that $|f_j - \hat{\pi}|$ is relatively large for sentiment-charged words and relatively small for sentiment-neutral words. For any choice of $\epsilon_n = o(1)$, the statement of this lemma holds with probability approaching 1. In what follows, we assume $m \rightarrow \infty$ as $n \rightarrow \infty$ and fix $\epsilon_n = m^{-1}$; item (c) in Assumption 2 ensures that $\epsilon_n = m^{-1}$ satisfies the requirement of Lemma 2. This choice of ϵ_n greatly simplifies the expressions. The next theorem establishes the consistency of screening:

Theorem 1 (Consistency of Screening). *In Models (1)-(2) and (4), suppose Assumptions 1-2 hold, and additionally,*

$$n\theta^2 \min_{j \in S} \frac{(O_{+,j} - O_{-,j})^2}{(O_{+,j} + O_{-,j})^2} \geq \frac{\log^2(m)}{\min\{1, \bar{s} \min_{j \in S} (O_{+,j} + O_{-,j}), \bar{n} \min_{j \in N} q_j\}}. \quad (16)$$

In the SESTM algorithm, set $\kappa = \log(m)$ and $\alpha_{\pm} = \frac{\sqrt{\log(m) \log(\log(m))}}{\sqrt{n \min\{1, \bar{s} \min_{j \in S} (O_{+,j} + O_{-,j}), \bar{n} \min_{j \in N} q_j\}}}$. As $n, m \rightarrow \infty$, $\mathbb{P}(\hat{S} = S) = 1 - o(1)$.

Remark 1. To ensure consistency for screening, the required number of training documents, n , relies on three factors: 1) the association between ordinal responses and document sentiments, denoted as θ and defined in (15); 2) the minimum among frequency-adjusted sentiments of all words in S , expressed as $\min_{j \in S} \frac{|O_{+,j} - O_{-,j}|}{O_{+,j} + O_{-,j}}$; 3) $\min\{1, \bar{s} \min_{j \in S} (O_{+,j} + O_{-,j}), \bar{n} \min_{j \in N} q_j\}$, where the last two terms inside the minimum account for the per-document count of individual words. In essence, a larger sample size n is required if the association between ordinal responses and document sentiments is insufficient, if the sentiment-charged words do not significantly differ from the neutral ones, or if there is an inadequate number of words per document (i.e., the documents are on average too short).

The sentiment topic estimation step produces $\hat{O}_{\pm}$. Let $\hat{F} = \hat{\tau}\hat{O}_{+} + (1 - \hat{\tau})\hat{O}_{-}$ and $\hat{T} = (1 - \hat{\tau})(\hat{O}_{+} - \hat{O}_{-})$, where $\hat{\tau} = n^{-1} \sum_{i=1}^n \hat{p}_i$. They are the respective estimate of the vector of frequency, F , and the vector of tone, T . We quantify the estimation errors of F and T , which can be easily translated to the estimation errors for $O_{\pm}$.

Theorem 2 (Estimation Errors for Sentiment Topic Vectors). *Suppose Assumptions 1-2 and (16) are satisfied. Let $\hat{\rho}$ be defined as in Lemma 1. As $n, m \rightarrow \infty$, with probability approaching 1, $\|\hat{F} - F\|_1 \leq C\sqrt{\frac{|S|\log(m)}{n\bar{s}}}$, and $\|\hat{T} - \frac{1-\hat{\tau}}{1-\tau}\hat{\rho}T\|_1 \leq C\sqrt{\frac{|S|\log(m)}{n\bar{s}}}$.*

Theorem 2 states that our method in fact estimates $(F, \hat{\rho}T)$ rather than (F, T) . This is the price we pay by not imposing an exact model on the distribution of y_i . Suppose, instead of Model (4), we specify a parametric distribution of y_i , then we could construct more accurate estimates of p_i such that $n^{-1} \sum_{i=1}^n (\hat{p}_i - p_i)^2 \rightarrow 0$, which in turn results in the consistency for the estimation of T . This is confirmed by the next corollary:⁶

Corollary 1. *Suppose conditions of Theorem 2 hold. Let $\hat{p}_i$ be an arbitrary estimate of p_i . We obtain $\hat{O}_{\pm}$ similarly as in (10), with $\widehat{W}$ constructed from these $\hat{p}_i$'s. Suppose $|S| \log(m)/(n\bar{s}) \rightarrow 0$ and $n^{-1} \sum_{i=1}^n (\hat{p}_i - \hat{\tau})^2 \geq a_1$, for a constant $a_1 > 0$. As $n, m \rightarrow \infty$, with probability approaching 1, $\max\{\|\hat{F} - F\|_1, \|\hat{T} - T\|_1\} \leq C \sqrt{\frac{|S| \log(m)}{n\bar{s}}} + C \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{p}_i - p_i)^2}$. Consequently, if $n^{-1} \sum_{i=1}^n (\hat{p}_i - p_i)^2$ converges to 0 in probability, then $\|\hat{F} - F\|_1$ and $\|\hat{T} - T\|_1$ also diminish with probability approaching 1.*

Remark 2. *We compare our results with those for unsupervised topic modeling (e.g., [12, 9, 1]). These methods recover $\hat{O}_{\pm}$ solely from the text data, without using the ordinal responses. While the unsupervised approach can achieve consistency in terms of $\|\hat{T} - T\|_1$, this is inconsequential within our context. As explained in Section 2.3, obtaining an accurate estimate of $\hat{\rho}T$ is sufficient for sentiment ranking. However, the unsupervised approach suffers from a slower rate of convergence. According to Corollary 3.1 in [9], their error rate is at best $\sqrt{\frac{|S| \log(m)}{n\bar{s}}} (1 + \frac{|S|}{\bar{s}})$, where $|S|$ represents the size of the sentiment dictionary, and $\bar{s}$ denotes the average per-document count of sentiment-charged words. In many real-world applications, $|S| \gg \bar{s}$ (for instance, in the application discussed in Section 5, $|S|$ is several hundreds while $\bar{s}$ is typically below 10), so the supervised approach we propose here is preferable.*

3.2 Accuracy of scoring and ranking new documents

Given the word count vector $d \in \mathbb{R}_+^m$ of a new document, we use (11) to obtain an estimate of its true sentiment score p . Recall that $\hat{\rho}$, as defined in Lemma 1, is related to the bias of $\hat{T}$

⁶This corollary applies to any estimate $\hat{p}_i$. For the particular $\hat{p}_i$ in (9), due to its special structure, the rate of convergence for $\|\hat{F} - F\|_1$ is faster without the additional term on the right-hand side; see Theorem 2.

in Theorem 2. As the estimation error propagates, a similar bias occurs to the estimator of p . We thereby define the *rescaled sentiment* score $p^* \in [0, 1]$ as:

$$p^* = \min\{1, \max\{0, \hat{\tau} + (p - \tau)/\hat{\rho}\}\}, \quad \text{where } \tau = n^{-1} \sum_{i=1}^n p_i. \quad (17)$$

This definition slightly differs from the p^* described in Section 2.3, as the p^* mentioned here is a winsorization of $\tau + \hat{\rho}^{-1}(p - \tau)$ within the range of $[0, 1]$. We utilize this winsorized version because the $\hat{p}$ in (11) is always constrained to take values in the interval $[0, 1]$. When p is bounded away from both 0 and 1, and $\hat{\rho}$ is not excessively small, the winsorization has no impact, and the two definitions coincide.

Theorem 3 (Sentiment Accuracy of a New Document). *In Models (1)-(2) and (4), suppose Assumptions 1-2 and (16) are satisfied. Suppose $\rho_{\min} \leq \hat{\rho} \leq \rho_{\max}$, for constants $0 < \rho_{\min} < \rho_{\max}$, and $\rho_{\max}|T_j| \leq (1 - c_1)(1 - \tau)F_j$, for all $j \in S$, where $c_1 \in (0, 1)$ is a constant. Let $d \in \mathbb{R}_+^m$ contain the word counts in a new document with sentiment p , and let s be the total count of sentiment-charged words in this document. Suppose $-\hat{\tau} + c_2 \leq \hat{\rho}^{-1}|p - \tau| \leq 1 - \hat{\tau} - c_2$, where $c_2 > 0$ is a properly small constant. Let $err_n = \frac{\sqrt{|S|\log(m)}}{\Theta\sqrt{ns}} + \frac{1}{\sqrt{s\Theta}}$, with $\Theta := \sum_{j \in S} \frac{(O_{+,j} - O_{-,j})^2}{O_{+,j} + O_{-,j}}$. Suppose $err_n \rightarrow 0$. Let $\hat{p}$ be the penalized MLE in (11). As $n \rightarrow \infty$, for each fixed $\epsilon \in (0, 1)$, there exists a constant $C > 0$, such that, with probability $1 - \epsilon$,*

$$|\hat{p} - p^*| \leq C \min\left\{1, \frac{\Theta}{\lambda}\right\} \times err_n + C \min\left\{1, \frac{\lambda}{\Theta}\right\} \times |p - \tau|. \quad (18)$$

Furthermore, when $\lambda = \Theta|p - \tau|^{-1}err_n$, (18) becomes $|\hat{p} - p^*| \leq C \min\{err_n, |p - \tau|\} \leq Cerr_n$.

The term err_n has two components, $\frac{\sqrt{|S|\log(m)}}{\Theta\sqrt{ns}}$ and $\frac{1}{\sqrt{s\Theta}}$, corresponding to the respective errors from training and testing. Since n is large, it is the second term that will dominate in finite sample. The tuning parameter λ reflects a bias-variance trade-off. In (18), the first term $\min\{1, \frac{\Theta}{\lambda}\} \times err_n$ is related to the “variance”, which decreases with λ ; the second term $\min\{1, \frac{\lambda}{\Theta}\} \times |p - \tau|$ is the “bias”, which increases with λ . In practice, it is often the case that most documents have a neutral tone, so that the bias is negligible relative to the variance.

Besides, text data are very noisy, so imposing a large penalty in MLE significantly reduces the variance. Our estimator shares the same spirit as the James-Stein estimator [8].

Finally, we study the ranking performance of SESTM.

Theorem 4 (Consistency of Rank Correlation). *Under conditions of Theorem 3, suppose we are given L new documents whose $p_1, p_2, \dots, p_L$ are i.i.d. generated from a distribution on $[\tau + \rho_{\min}(-\hat{\tau} + c_2), \tau + \rho_{\min}(1 - \hat{\tau} - c_2)]$ with a continuous probability density. Let s_i be the total count of sentiment-charged words in a new document i . Suppose $\min\{s_1, s_2, \dots, s_L\} \geq c_3 s$, where $s\Theta/\sqrt{\log(L)} \rightarrow \infty$ and $c_3 > 0$ is a constant. We apply (11) with $\lambda \asymp \Theta \text{err}_n$ to estimate sentiments of new documents. Let $SR(\hat{p}, p)$ be the Spearman’s rank correlation between $\{\hat{p}_i\}_{i=1}^L$ and $\{p_i\}_{i=1}^L$. As $n, m, L \rightarrow \infty$, $\mathbb{E}[SR(\hat{p}, p)] \rightarrow 1$.*

Remark 3. Besides Spearman’s rank correlation, there exist other performance metrics for ranking, including mean average precision, mean reciprocal rank, discounted cumulative gain (DCG), and precision/recall of the top- k ranks. They can be studied in a similar manner by applying Theorem 3 and assuming regularity conditions on the true sentiment scores of test documents. In fact, when ranks are obtained by ordering estimated “scores,” the analysis of any performance metric essentially involves quantifying the differences between these estimated scores and a monotone transformation of the true scores (e.g., see [3, 5]). Theorem 3 provides the necessary theoretical foundation for studying any ranking performance metric.

Remark 4. Theorem 4 assumes that the sentiment scores of new documents belong to $[\tau + \rho_{\min}(-\hat{\tau} + c_2), \tau + \rho_{\min}(1 - \hat{\tau} - c_2)]$. However, this assumption can be relaxed by modifying Equation (11) to remove the constraint $p \in [0, 1]$ during optimization. While the estimates $\{\hat{p}_i\}_{i=1}^L$ may fall outside $[0, 1]$, they can still be effectively used to rank the documents.

4 Simulations

In this section, we provide Monte Carlo evidence to illustrate the finite-sample performance of our proposed algorithm. In our simulations, the counts of sentiment-charged and sentiment-

neutral words in each article are generated from Model (1)-(2), where $p_i \sim \text{Unif}(0, 1)$, $s_i \sim \text{Unif}(0, 2\bar{s})$, $n_i \sim \text{Unif}(0, 2\bar{n})$, and for $j = 1, 2, \dots, |S|$, $O_{+,j} = \frac{2}{|S|} \left(1 - \frac{j}{|S|}\right)^2 + \frac{2}{3|S|} \times 1_{\{j < \frac{|S|}{2}\}}$, $O_{-,j} = \frac{2}{|S|} \left(\frac{j}{|S|}\right)^2 + \frac{2}{3|S|} \times 1_{\{j \geq \frac{|S|}{2}\}}$, and Ω_j is drawn from $\frac{1}{m-|S|} \text{Unif}(0, 2)$, for $j = |S| + 1, \dots, m$, then renormalized such that $\sum_j \Omega_j = 1$. As a result, the first $|S|/2$ words are positive, the next $|S|/2$ words are negative, and the remaining $(m - |S|)$ words are neutral with frequencies randomly drawn from a uniform distribution. As such, the difference between $O_{+,j}$ and $O_{-,j}$ reaches the minimum at $j = |S|/2$. This means among all words within the set S , those with j around $|S|/2$ are close to neutral. Next, the sign of the ordinal response is such that $\mathbb{P}(y_i > 0) = p_i$, and its magnitude $|y_i|$ follows a folded Student t-distribution with the degree of freedom parameter set at 4. The standard deviation of the t-distribution has negligible effects on our simulations, since only the ranks of returns matter for our results. We fix the number of Monte Carlo repetitions at $M_c = 200$ and the number of testing articles at $L = 1,000$. In the benchmark case, we set $|S| = 200$, $m = 4,000$, $n = 40,000$, $\bar{s} = 15$, and $\bar{n} = 200$. We vary these choices to examine their effect on the performance of our procedure.

Experiment 1: Finite-sample performance of SESTM. We first evaluate the screening step. Although tuning threshold parameters give better results, for convenience we choose the top $|S|$ words in terms of $|f_j - 0.5| 1_{\{k_j > \kappa\}}$, where κ is set at the 10% quantile of all k_j s. Figure 1 reports the frequency of each word selected in the screening step across all Monte Carlo repetitions. Remarkably, the probability of selecting any word outside S is approximately 0.93% across all repetitions. As expected, the words within set S that are occasionally missed are those associated with j values around $|S|/2$. This observation is not surprising since such words exhibit relatively weak sentiment strength, rendering them nearly neutral in nature.

Next, Figure 2 delineates the precision of our estimation process, taking into consideration potential inaccuracies arising from the screening step. The true values of T and F are represented in black. In this benchmark simulation setting, the scaling constant $\hat{\rho}$ approximates 0.5. The graph clearly shows that the estimators $\hat{F}$ and $\hat{T}$ are centered around their respective

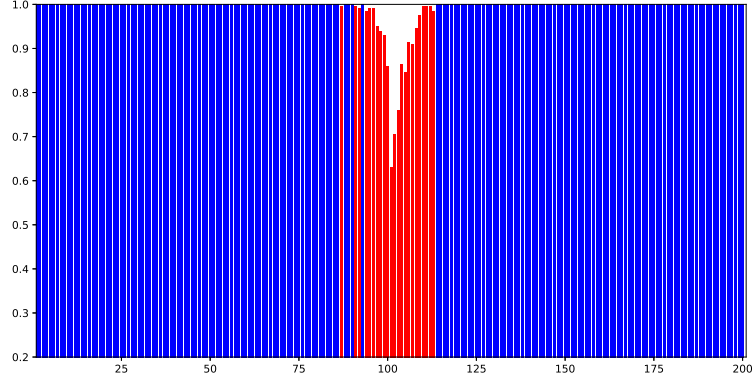


Figure 1: The frequency of each sentiment-charged word selected into $\hat{S}$ in Monte Carlo simulations. The red bars correspond to those words with frequencies less than 100%.

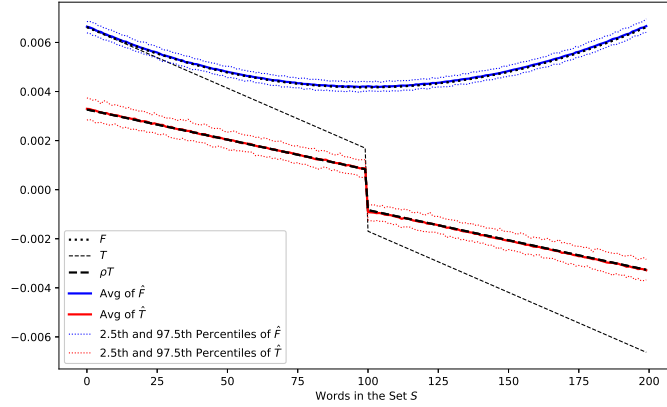


Figure 2: Comparison of the averages of $\hat{F}$ (blue, solid) and $\hat{T}$ (red, solid) across Monte Carlo repetitions with F (black, dotted), T (thin, black, dashed), and $\hat{\rho}T$ (thick, black, dashed), respectively, using the benchmark parameters. The blue and red dotted lines plot the 2.5% and 97.5% quantiles of the Monte Carlo estimates. For each word in S , the averages and percentiles are calculated based on those Monte Carlo samples where this word appears in $\hat{S}$.

targets F and $\hat{\rho}T$ across all words, consistent with our theoretical prediction. Further analysis involves varying the number of training articles (n), the size of the vocabulary (m), the average of the sentiment words per article ($\bar{s}$), and the size of the dictionary of the sentiment words ($|S|$) to scrutinize their influence on the estimates of F and $\hat{\rho}T$, as depicted in Figure 3. The simulation results align with the theoretical implications of Theorem 2. Specifically, the estimation error diminishes with an increase in n or $\bar{s}$, and a decrease in $|S|$ or m .

Next, we evaluate the accuracy of the sentiment ranking step, taking into account any errors propagated from the preceding steps. Notably, data from the test sample are never used in the two previous steps. Figure 4 reports the boxplots of the Spearman’s rank correlation

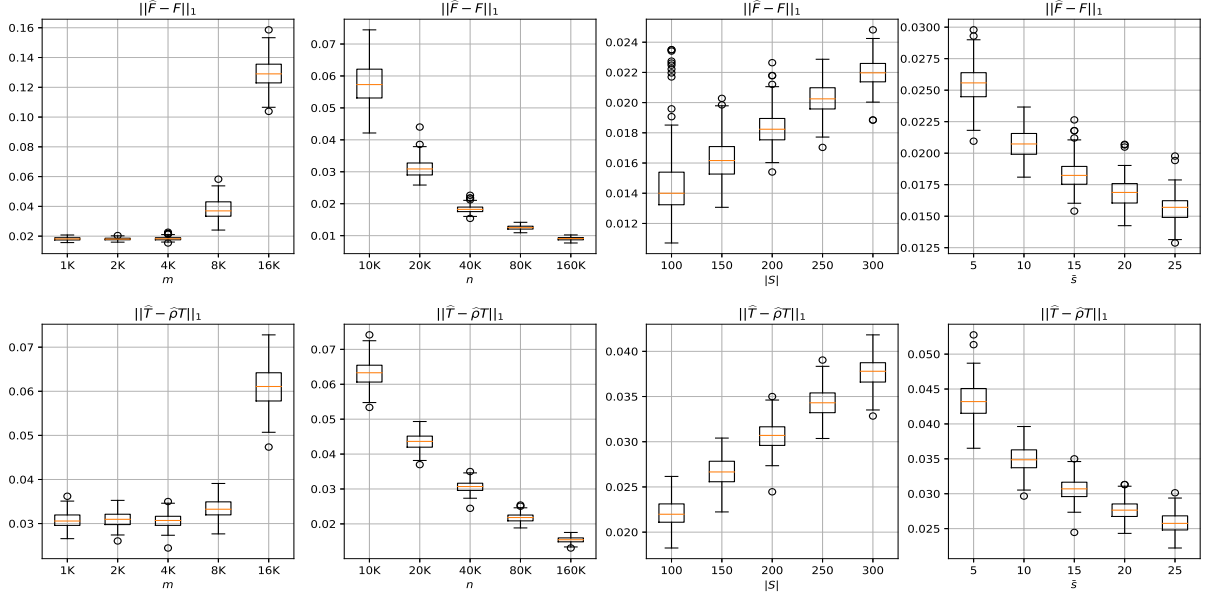


Figure 3: Comparison of F and T estimates across different scenarios. The figure shows the boxplots of $\|\hat{F} - F\|_1$ and $\|\hat{T} - T\|_1$ across Monte Carlo repetitions for different values of $(m, n, |S|, \bar{s})$.

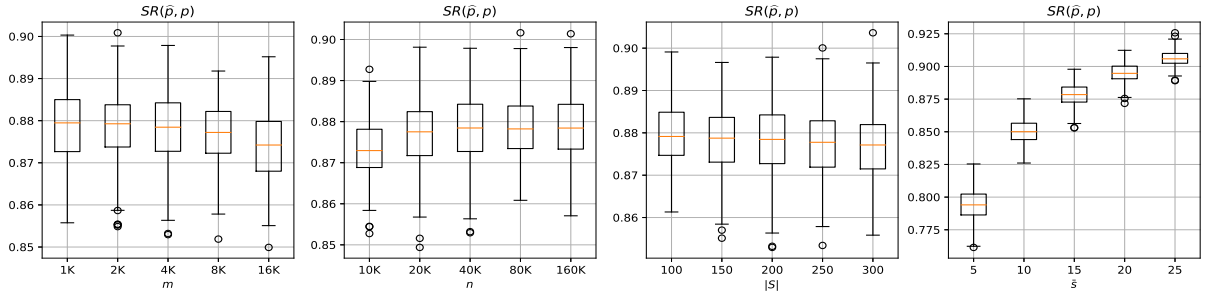


Figure 4: Comparison of ranking performance across different scenarios. The figure shows the boxplots of the Spearman's correlation across Monte Carlo repetitions for various values of $(m, n, |S|, \bar{s})$.

coefficients between the predicted $\hat{p}$ and the true p across 1,000 documents within the test sample in a variety of cases. The rank correlation appears to be most sensitive concerning $\bar{s}$. Elevating it from 5 to 25 results in an improvement in rank correlation from an average of 0.8 to over 0.9, attributable to the larger number of counts from words in S . A rise in n seems to induce a moderately small enhancement, while increasing m and $|S|$ slightly degrades the results. Overall, these observations are congruent with our intuition and the error bounds for estimating F and T , although we do not have any theoretical error bound for $(1 - SR(\hat{p}, p))$.

Experiment 2: Comparison with alternative methods. Finally, we undertake a comparative evaluation of our estimator against alternate methodologies, focusing primarily on

Table 1: Comparison of Spearman’s correlation. This table presents the mean and standard deviation of Spearman’s correlation, gathered across multiple Monte Carlo repetitions in the benchmark scenario using various methodologies, including Naive Bayes, MNIR, and dictionary methods. For the dictionary method, 80%, 90%, and 100% represent the proportion of words in S that are part of the pre-established dictionary utilized for sentiment measurement. The remaining 20% and 10% in the first two cases are randomly selected from the list of sentiment-neutral words.

	SESTM	Naive Bayes	MNIR	Dictionary (80%)	Dictionary (90%)	Dictionary (100%)
Avg SR	0.878	0.833	0.845	0.816	0.837	0.853
Std Dev	0.007	0.010	0.008	0.011	0.009	0.008

three categories: the Naive Bayes, MNIR, and Dictionary methods. The Naive Bayes method is a supervised classification technique that uses word count as input features for sentiment label classification. However, it relies on an assumption that the word count remains independent given the sentiment score, which clearly contradicts the data generating process in our study. Additionally, Naive Bayes focuses solely on classification, disregarding any information pertaining to the (ordered) values of ordinary responses. Similarly, MNIR is also a supervised classification approach, suffering from the same issues due to model misspecification caused by inconsistent assumptions with our data generating process. On the other hand, the Dictionary method involves pre-selecting a sentiment dictionary and determining the sentiment of each test document by simply computing the difference between the respective total counts of positive and negative words. Despite being simple and intuitive, this estimator lacks justifiable asymptotic properties. To further scrutinize the impact of the dictionary, we randomly select a fixed quantity of words, say, 80%, 90%, and 100%, from the true set of sentiment words, and supplement them with remaining words from the sentiment-neutral list. Table 1 delineates the summary statistics of Spearman’s correlation within the context of the benchmark scenario. Our approach, SESTM, as demonstrated by the results, produces a stronger correlation and exhibits smaller variation compared to alternative methods. When equipped with an accurate dictionary, the Dictionary method achieves second place in the ranking, followed by MNIR. Nevertheless, the effectiveness of the Dictionary method rapidly diminishes as the dictionary quality declines. The Naive Bayes approach begins to outperform the Dictionary method once the accuracy of the dictionary drops to 80%.

5 Empirical Analysis of the Dow Jones Newswire

We apply our text-mining framework to extract sentiment information from the Dow Jones Newswire, aiming to utilize sentiment analysis in constructing an investment portfolio. Our null hypothesis of market efficiency predicts that the expected return is dominated by unpredictable news, as this news is rapidly (in its starkest form, immediately) incorporated in prices. The maintained alternative hypothesis of our research is that information in news text is not fully absorbed by market prices instantaneously, for reasons such as limits-to-arbitrage and rationally limited attention. As a result, information contained in news text is predictive of future asset price paths. While this alternative hypothesis is by now uncontroversial, it is hard to overstate its importance, as we have much to learn about the mechanisms through which information enters prices and the frictions that impede these mechanisms.

Our text data set is the *Dow Jones Newswires Machine Text Feed and Archive* database. It contains real-time news feeds from January 1, 1989 to April 30, 2020, amounting to 25,743,474 unique articles (after combining “chained” articles). Approximately 62.2% news articles are tagged with one or more firm identifiers. We restrict our sample to articles tagged with only one firm, reducing the dataset to 11,988,392 articles. Table E.1 in the appendix outlines the step-by-step details of our article filtering process. The appendix also provides a thorough description of the data cleaning procedure, which converts each article into its bag-of-words representation. We match each article with tagged firm’s market capitalization and adjusted daily close-to-close returns from CRSP. To ensure a strong association between text sentiment and returns in our training sample, we label the sentiment of each article published on day t (between 4pm of day $t - 1$ and 4pm of day t) based on the sign of its tagged firm’s three-day return from $t - 1$ to $t + 1$ (from market close on day $t - 2$ to close on day $t + 1$).⁷

⁷For news published on holidays or weekends, we use the next available trading day as the current day t , and the last trading day before the news as day $t - 1$. Using three-day returns does not introduce forward-looking bias, as this labeling procedure applies only to articles in the training sample. Furthermore, since sentiment-driven news events may either cause or be caused by large returns, using a three-day window around the news event may provide a more accurate label for the sentiment in the news text.

Return prediction and portfolio construction. We train the model using rolling window estimation. The rolling window consists of a 15-year interval, the first 10 years for training and the last 5 years for validation/tuning. We then use the subsequent 1-year window for out-of-sample testing. At the end of the testing year, we roll the entire analysis forward by a year and re-train. We iterate this procedure until we exhaust the full sample, which amounts to estimating and validating the model 17 times.

In each training sample, we estimate a collection of SESTM models corresponding to a grid of tuning parameters. There are four tuning parameters in total: $(\alpha_+, \alpha_-, \kappa, \lambda)$. The first two, α_+ and α_- , determine the size of the sentiment dictionary for positive and negative words, respectively, while κ sets the threshold for excluding overly rare words. In practice, it is more convenient (and equivalent) to adjust the number of positive and negative words directly, rather than tuning α_+ and α_- . To reduce computational burden, we select either 50, 100, or 200 words, as the results are not sensitive to this choice. The parameter κ is set to be proportional to the logarithm of the total number of words for articles in the training sample. For the constant term, we explore a grid of values from 50 to 1,000, in increments of 50. Finally, λ is fixed at 10 throughout the analysis, as it does not affect the ranking of sentiment scores but ensures all scores fall within the 0 to 1 range.⁸

For the validation sample, we use all the estimated models to score each news article and select the constellation of tuning parameter values that provides the most desirable model performance. One sensible criterion is the rank correlation with return-based sentiment scores in the validation sample. Alternatively, since the sentiment scores will ultimately be used for portfolio construction, we use the Sharpe ratio of the resulting portfolios as the selection criterion for tuning parameters. The exact portfolio formation procedure follows the one described in the out-of-sample analysis, which we will outline below.

In the out-of-sample test period, we estimate the sentiment scores of articles using the

⁸In total, we tune 60 sets of tuning parameter values, constrained by the available computational resources. In general, increasing the number of tuning parameter sets explored would likely improve model performance. Therefore, the results presented should be considered as conservative estimates, as greater computational resources would allow for a more extensive tuning process and potentially stronger outcomes.

Table 2: Performance of daily news sentiment portfolios, including equal-weighted (EW) and value-weighted (VW) long-short (L-S) portfolios and their long (L) and short (S) legs. The performance measures include (annualized) annual Sharpe ratio, annualized expected returns, risk-adjusted alphas, and R^2 s (in percentage) with respect to the Fama-French three-factor model (“FF3”), the Fama-French five-factor model (“FF5”), and the Fama-French five-factor model augmented to include the momentum factor (“FF5+MOM”). We also report the strategy’s daily turnover, defined as $\frac{1}{2T} \sum_{t=1}^T (\sum_i |w_{i,t+1} - \frac{w_{i,t}(1+y_{i,t+1})}{1+\sum_j w_{j,t}y_{j,t+1}}|)$, where $w_{i,t}$ is the weight of stock i in the portfolio at time t , as well as max drawdown, defined as the peak-to-trough decline relative to the value at peak.

Formation	Sharpe	Turnover	Average	Max	FF3		FF5		FF5+MOM	
	Ratio		Return	Drawdown	α	R^2	α	R^2	α	R^2
EW L-S	5.21	95.55%	36	15.91%	36	1.26	36	1.48	36	1.79
EW L	2.15	95.93%	19	42.21%	18	38.88	18	39.49	18	40.33
EW S	1.52	95.17%	16	71.91%	18	34.50	18	35.43	18	36.73
VW L-S	1.56	93.75%	10	24.38%	10	0.99	10	1.67	10	1.69
VW L	1.04	93.85%	9	35.99%	7	30.96	7	31.02	7	31.50
VW S	0.09	93.66%	1	74.18%	3	25.47	3	25.88	3	26.09

optimally tuned model determined from the validation sample. To evaluate out-of-sample predictive performance in economic terms, we design a trading strategy that leverages sentiment estimates for prediction. Our trading strategy is very simple (the news timeline and our trading activities are summarized in Figure E.3 of the appendix). It is a zero-net-investment portfolio that each day buys 50 stocks with the most positive sentiment scores and shorts 50 stocks with the most negative sentiment scores.⁹ Trading only 100 stocks per day is a reasonable choice in light of the summary statistics of the data, despite it being unfortunately small. On average, we only have 936.5 firms with at least one piece of news on each trading day, among which 181.7 contain sentiment-charged words, for which our algorithm can output non-neutral sentiment scores. In the early part of the sample, there are a handful of days for which fewer than 50 firms have non-neutral scores, in which case we have to trade fewer than 100 stocks but otherwise maintain the zero-cost nature of the portfolio. That said, we also provide results for portfolios with (up to) 200 stocks for robustness check in the appendix.

We consider both equal-weighted and value-weighted schemes when forming the long and short sides of the strategy. Equal weighting is a simple and robust way of assessing predictive power of sentiment over the firm size spectrum, and is anecdotally closer to how hedge funds use news text for portfolio construction. Value weighting heavily overweights large stocks, which may be justifiable for economic reasons (assigning more weight to more productive

⁹In the case a stock is mentioned in multiple news articles on the same day, we use the average sentiment score over the coincident articles.

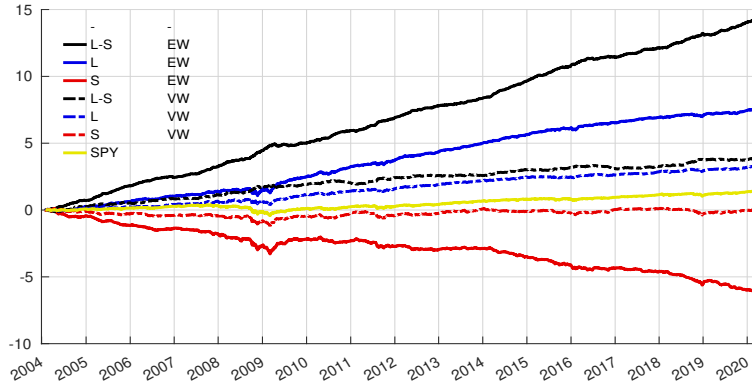


Figure 5: One-day-ahead performance comparison of SESTM (the out-of-sample cumulative log returns). The black, blue, and red colors represent the long-short (L-S), long (L), and short (S) portfolios, respectively. The solid and dashed lines represent equal-weighted (EW) and value-weighted (VW) portfolios, respectively. The yellow solid line is the S&P 500 return (SPY).



Figure 6: Sentiment-charged words. This figure reports the list of words in the sentiment-charged set $\hat{S}$. Font size of a word is proportional to its average sentiment tone over all 17 training samples.

firms) and for practical trade implementation reasons (such as limiting transaction costs). Figure 5 reports the cumulative one-day trading strategy returns (calculated from open-to-open) based on out-of-sample SESTM sentiment forecasts. We report the long (denoted “L”) and short (“S”) sides separately, as well as the overall long-short (“L-S”) strategy performance. We also contrast performance of equal-weighted (“EW”) and value-weighted (“VW”) versions of the strategy. Table 2 reports the corresponding performance statistics of these portfolios.

Three basic facts emerge from the one-day forecast results in Figure 5 and Table 2. First, equal-weighted portfolios substantially outperform their value-weighted counterparts. The long-short strategy with equal weights earns an annualized Sharpe ratio of 5.21, versus 1.56 in the value-weighted case. This indicates that news article sentiment is a stronger predictor of future returns to small stocks, all else equal. There are several potential economic explanations

for this, including the facts that i) small stocks receive less investor attention and thus respond more slowly to news, ii) the underlying fundamentals of small stocks are more uncertain and opaque and thus it requires more effort to process news into actionable price assessments, and iii) small stocks are less liquid and thereby require a longer time for trading to occur to incorporate information into prices. Second, both the long and short sides are significantly profitable. The long side slightly outperforms the short side, with a Sharpe ratio of 2.15 versus 1.52 (in the equal-weighted case). This is due in part to the fact that the long side naturally earns the market equity risk premium while the short side pays it. A potential explanation is that investors face short sales constraints so that the negative news cannot be fully expressed in prices over short horizons. Third, SESTM sentiment trading strategies have little exposure to standard aggregate risk factors. The individual long and short legs of the trade have at most a 40% daily R^2 when regressed on Fama-French factors, while the long-short spread portfolio R^2 is at most 1.8%. In all cases, the average return of the strategy is almost entirely alpha. Note that, by construction, the daily turnover of the portfolio is large. If we completely liquidated the portfolio at the end of each day, we would have a turnover of 100% per day. Actual turnover is slightly lower, on the order of 96% for equal-weighted implementation and 94% for value-weighted, indicating a small amount of persistence in positions. Figure 5 shows that the long-short strategy avoids major drawdowns, and indeed appreciates during the financial crisis while SPY sells off.

Most impactful words: Figure 6 reports the list of sentiment-charged words estimated from our model. These are the words that most strongly correlate with realized returns and thus surpass the correlation screening threshold. Because we re-estimate the model in each of our 17 training samples, the sentiment word lists can change throughout our analysis. The font size of a word in the cloud is proportional to the word’s average sentiment tone ($O_+ - O_-$) over all 17 training samples. Table E.5 in Appendix E.7 provides additional details on selected words, reporting the top 50 positive and negative sentiment words throughout our training

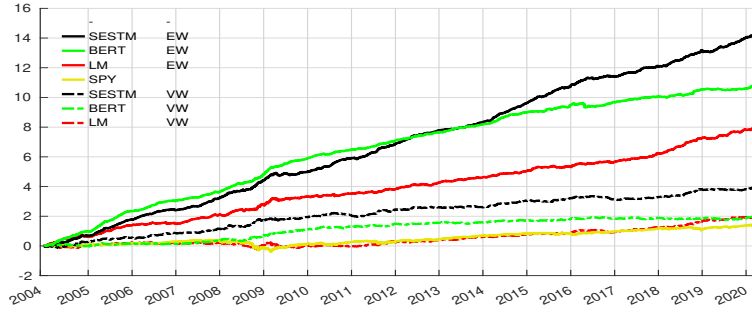


Figure 7: SESTM versus LM, and BERT. This figure compares the daily cumulative returns of long-short portfolios constructed from SESTM, LM, and BERT sentiment scores, separated into equal-weighted (EW, solid lines) and value-weighted (VW, dashed lines) portfolios, respectively. The yellow solid line is the S&P 500 return (SPY).

samples. It also shows if these words overlap with LM and Harvard-IV dictionaries.

The estimated wordlists are remarkably stable over time. Of the top 50 positive sentiment words over all periods, 24 are selected into the positively charged set in at least 12 of the 17 training samples. For the 50 most negative sentiment words, 23 are selected in at least 14 out of 17 samples. The following 14 negative words are selected in *every* training sample: *downgrade, hurt, fell, warn, slowdown, lower, resign, soft, delay, weak, slow, cut, drop, decrease*, and the following nine words are selected into the positive list in 15 or more training samples: *repurchase, declare, upgrade, beat, boost, jump, exceed, raise, strong*. There are interesting distinctions vis-a-vis extant sentiment dictionaries. For example, in comparison to our estimated nine most impactful positive words listed above, only two (*boost, strong*) appear in the LM positive dictionary, and only two (*upgrade, boost*) appear in Harvard-IV. Likewise, five of our 14 most impactful negative words (*fell, lower, soft, drop, decrease*) do not appear in the LM negative dictionary and eight are absent from Harvard-IV. Thus, in addition to the fact that our word lists are accompanied by term-specific sentiment weights (contrasting with the implicit equal weights in extant dictionaries), many words that we find to be most important for understanding realized returns are entirely omitted from pre-existing dictionaries.

Comparison versus dictionary methods, RavenPack, and BERT: We compare SESTM to alternative sentiment scoring methods in terms of return prediction accuracy. The first competitor is dictionary-based sentiment scoring. We construct the LM sentiment score of an

article by aggregating counts of words listed in their positive sentiment dictionary (weighted by tf-idf, as recommended by [10]) and subtracting off weighted counts of words in their negative dictionary. As with SESTM, we average scores from multiple articles for the same firm on the same day. This produces a stock-day signal, $\hat{p}_i^{LM}$, which we use to construct trading strategies in the same manner as the SESTM-based signal, $\hat{p}_i^{SESTM}$, in preceding analyses.

The second competitor is the news sentiment scores from RavenPack News Analytics 4 (RPNA4) (<https://www.ravenpack.com/about/>). RavenPack is a leading provider of big data analytics for financial services, with clients including many of the world’s most prominent hedge funds, banks, and asset managers. We use data from the RPNA4 DJ Edition Equities, released in September 2014, which constructs company-level sentiment scores based on the same Dow Jones sources we use to build SESTM (*Dow Jones Newswires*, *Wall Street Journal*, *Barron’s* and *MarketWatch*). Thus, the underlying news collection available to us is presumably identical to that used by RavenPack. However, the observation count we see in RavenPack is larger than the number of observations we can construct from the underlying Dow Jones news (we discuss this point, along with additional details of the RavenPack data, in Appendix E.6). Following the same procedure, we construct RavenPack daily stock-level sentiment scores ($\hat{p}_i^{RP}$) by averaging all article sentiment scores pertaining to a given firm on a given day (we use RavenPack’s flagship measure, CSS). To avoid any forward-looking bias, we restrict the use of RavenPack sentiment scores in our portfolio analysis to the post-September 2014 period. This ensures that the scores were produced after the release of RPNA4, rather than reflecting earlier data that may have been used to train the model, which could contaminate backfilled sentiment with information not available in real time.

The third alternative we consider is the BERT model, one of the first and most widely used large language models, developed by [4]. For each article, we generate article-level embeddings using the first 512 tokens. While the 512-token limit constrains the length of the text that the BERT model can process, most news articles provide sufficient information

within these first 512 tokens. In practice, including more tokens has not been shown to improve performance. These embeddings act as sufficient statistics, effectively capturing the context of the text. We then use these embeddings to train a logistic regression model with the same labels used in the SESTM approach. Given the high dimensionality of the BERT embeddings—1,024 dimensions—we apply a ridge penalty to mitigate overfitting.¹⁰ The ridge parameter is selected using the same training and validation design as in the previous models. The resulting model outputs sentiment scores $\hat{p}_i^{BERT}$.

We build trading strategies using each of the sentiment scores, $\hat{p}_i^{SESTM}$, $\hat{p}_i^{LM}$, $\hat{p}_i^{RP}$, and $\hat{p}_i^{BERT}$. These strategies follow the same approach as before, where we buy the 50 most positive and short the 50 most negative sentiments. We also evaluate both equal-weighted and value-weighted strategies. Figure 7 focuses on a one-day trading strategy and separately analyzes equal- and value-weighted portfolios. It reports out-of-sample cumulative daily returns to compare average performance and drawdowns across SESTM, LM, and BERT. Table 3 complements this figure by reporting a broader set of performance statistics, including annualized Sharpe ratios, daily turnover, and average returns. Unlike the figure, the table also includes RP in the comparison and restricts the sample to the post-September 2014 period.

The SESTM strategy exhibits clear dominance under equal-weighting, outperforming alternatives in both average returns and Sharpe ratio, while maintaining comparable turnover. In equal-weighted terms, SESTM achieves an annualized Sharpe ratio of 4.85, compared to 3.14 for LM, 4.20 for RavenPack, and 1.89 for BERT. Under value-weighting, however, the performance gap narrows and even reverses: SESTM’s Sharpe ratio drops to 1.43, trailing both RavenPack (1.77) and LM (1.71), though remaining above BERT (0.56).

To more carefully assess the differences in performance across methods, Table 3 also reports a series of portfolio spanning tests. For each trading strategy, we regress its returns on the returns of each of the competing strategies, while also controlling for daily returns to the five

¹⁰Due to space constraints, we only report results from the BERT-large model, which has 340 million parameters. The BERT-base model, with 110 million parameters and 768-dimensional embeddings, performs worse empirically.

Table 3: Comparison of SESTM with LM, RavenPack, and BERT. This table replicates the analysis from Table 2 for equal-weighted and value-weighted long-short portfolios shown in Figure 7, but focuses only on the out-of-sample period beginning in October 2014. It also incorporates additional covariates in a series of portfolio spanning tests. Sharpe ratios are annualized; returns and alphas are reported in basis points per day.

	Sharpe Ratio	Turnover	Average Return	FF6+SESTM			FF6+LM			FF6+RP			FF6+BERT		
				α	$t(\alpha)$	R^2	α	$t(\alpha)$	R^2	α	$t(\alpha)$	R^2	α	$t(\alpha)$	R^2
SESTM															
EW	4.85	95.10%	37				33	10.27	4.50	29	9.16	7.10	35	10.97	3.00
VW	1.43	91.60%	9				6	2.32	8.30	6	2.49	6.70	7	2.93	5.30
LM															
EW	3.14	95.40%	23	15	5.05	5.10				18	5.84	3.60	21	6.94	2.40
VW	1.71	93.40%	10	8	3.36	5.90				9	3.78	1.90	9	3.79	2.80
RavenPack															
EW	4.20	97.00%	30	21	7.02	7.00	27	8.91	2.90				28	9.45	2.10
VW	1.77	97.20%	11	8	3.47	6.70	9	3.77	4.40				9	3.85	4.50
BERT															
EW	1.89	93.70%	15	11	3.17	2.70	14	4.31	1.50	13	3.78	1.90			
VW	0.56	90.70%	4	4	1.29	3.50	4	1.23	3.40	4	1.44	2.60			

Fama-French factors plus the UMD momentum factor (denoted FF6 in the table). We evaluate R^2 and regression intercept (α). If a trading strategy has a significant α after controlling for an alternative, it indicates that the underlying sentiment measure isolates predictive information not fully subsumed by the alternative. Likewise, the R^2 measures the extent to which trading strategies duplicate each other. An interesting result of the spanning tests is the overall low correlation among strategies and with Fama-French factors. The highest R^2 is 8.30% for SESTM regressed on FF6 and LM. The SESTM α 's are in each case almost as large as its raw return. At most, 7.10% of the SESTM strategy performance is explained by the controls (i.e., an equal-weighted α of 29 basis points versus the raw average return of 37 basis points). We also see significant positive alphas for the alternative strategies after controlling for SESTM, indicating not only that they achieve significant positive returns, but also that a component of those excess returns are uncorrelated with SESTM and FF6. In short, SESTM, RavenPack, LM, and BERT capture different varieties of information content in news.

6 Conclusion

We introduce a new framework for extracting sentiment information from text that offers several key advantages over existing methods. Unlike pre-defined sentiment dictionaries, our method learns a domain-specific dictionary directly from the data, allowing it to adapt to various domains and capture nuanced sentiment expressions. It is also highly versatile, han-

dling both discrete and continuous ordinal responses, unlike many existing methods limited to discrete data. Furthermore, our approach operates as a “white box” model, offering clear interpretability of all parameters while being computationally efficient and requiring only basic statistical tools.

Compared to DNN approaches, our framework is easier to train, interpret, and diagnose. Additionally, it is backed by theoretical guarantees that provide valuable insights into how sentiment estimation and ranking accuracy depend on factors such as the number of documents and the size of the sentiment dictionary, a feature lacking in many alternative methods.

While our current model relies solely on word counts and functions as a uni-gram topic model, there are important avenues for future research that could enhance its explanatory power. Extending the model to incorporate multi-gram topic models or word-embedding techniques would allow it to capture additional information from word order and context. Moreover, sentiment analysis likely involves multiple dimensions that drive market behavior in distinct ways. For example, distinguishing between long-term and short-term sentiment, or differentiating between macroeconomic and company-specific sentiment, could provide deeper insights into market reactions. These extensions would move beyond a one-dimensional sentiment model and offer a richer understanding of the relationship between sentiment and financial outcomes.

Additionally, we observed in Section 5, that the most influential sentiment words varied across different time windows, suggesting the potential benefits of allowing sentiment topic vectors to vary over time. Re-estimating sentiment dictionaries periodically or modeling sentiment topics as smooth functions of time—potentially using change-point detection and kernel smoothing techniques—could improve the model’s ability to capture these temporal dynamics. Document-level covariates, such as author attributes, could also be incorporated to further refine how sentiment is expressed in the text.

Finally, while we demonstrate the practical application of our methodology in enhancing

sentiment-guided portfolio construction and compare it favorably with other sentiment analysis methods, the potential extensions discussed provide exciting directions for future research. These refinements could lead to more nuanced and accurate insights, contributing to a better understanding of how sentiment influences price formation in financial markets.

7 Conflict of Interest Statement

The authors report there are no competing interests to declare.

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Supplementary Material of “Enhancing Investment Decisions with Sentiment Analysis: A Probabilistic Ranking Framework”

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Abstract

This supplementary material comprises mathematical proofs and additional empirical results that accompany the main article.

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A Identification

The model in Section 2.1 is not identifiable without additional constraints. This can be seen from the following proposition (whose proof is elementary, thus omitted).

Proposition A.1 (Non-identifiability). *For any $\alpha \in \mathbb{R}$, let $\tilde{O}_+ = (1 - \alpha)O_+ + \alpha O_-$, $\tilde{O}_- = (1 + \alpha)O_- - \alpha O_+$, and $\tilde{p}_i = p_i + \alpha$, $1 \leq i \leq n$. Let (τ, F, T) be as in (3), and similarly define $\tilde{\tau} = n^{-1} \sum_{i=1}^n \tilde{p}_i$, $\tilde{F} = \tilde{\tau}\tilde{O}_+ + (1 - \tilde{\tau})\tilde{O}_-$, and $\tilde{T} = (1 - \tilde{\tau})(\tilde{O}_+ - \tilde{O}_-)$. The following statements are true:*

- (1) $\sum_{j=1}^m \tilde{O}_{+,j} = 1$, and $\sum_{j=1}^m \tilde{O}_{-,j} = 1$. Consequently, if $\tilde{O}_{\pm}$ are nonnegative vectors, then they are probability mass functions (PMFs).
- (2) $\tilde{p}_i \tilde{O}_+ + (1 - \tilde{p}_i) \tilde{O}_- = p_i O_+ + (1 - p_i) O_-$, for all $1 \leq i \leq n$.

(3) $\tilde{F} = F$, and $\tilde{T} = (1 - \tau)^{-1}(1 - \tau - \alpha)T$.

We consider a perturbation from $(O_+, O_-, p_1, p_2, \dots, p_n)$ to $(\tilde{O}_+, \tilde{O}_-, \tilde{p}_1, \tilde{p}_2, \dots, \tilde{p}_n)$. When $O_\pm$ are strictly positive vectors and $0 < p_i < 1$ for all $1 \leq i \leq n$, there exists a sufficiently small α (in magnitude) such that $\tilde{O}_\pm$ are nonnegative vectors and $0 \leq \tilde{p}_i \leq 1$ for all $1 \leq i \leq n$. It follows from (1) of Proposition A.1 that $(\tilde{O}_+, \tilde{O}_-, \tilde{p}_1, \tilde{p}_2, \dots, \tilde{p}_n)$ are eligible parameters for Model (2). We recall that $d_{[S],i} \sim \text{Multinomial}(N_i, Q_i)$, with $Q_i = p_i O_+ + (1 - p_i) O_-$. By (2) of Proposition A.1, the distributions on $d_{[S],i}$ are exactly the same after perturbation. This shows that the model is not identifiable. Result (3) of Proposition A.1 connects the post-perturbation (F, T) and the pre-perturbation (F, T) .

Proposition A.1 implies that there is one extra degree of freedom, α , in the parameters of Model (2). We need to impose an identifiability condition. One option is to impose a constraint on the average true sentiment scores of training documents:

$$n^{-1} \sum_{i=1}^n p_i = \tau_0, \quad \text{for some pre-specified } \tau_0 \in (0, 1) \text{ (e.g., } \tau_0 = 1/2\text{)}. \quad (\text{A.1})$$

This constraint removes the extra degree of freedom and makes the model identifiable. Another option is to impose a constraint on the topic vectors $O_\pm$. We can assume:

$$\text{there exist two words } j_1^* \text{ and } j_2^* \text{ such that } O_{-,j_1^*} = 0 \text{ and } O_{+,j_2^*} = 0. \quad (\text{A.2})$$

This is actually the anchor-word condition [2, 1] for a standard Hofmann's topic model, where j_1^* and j_2^* are the respective anchor words of the positive and negative sentiment topics. Under (A.2), $\tilde{O}_\pm$ are both nonnegative vectors only when $\alpha = 0$, see [2]; hence, the model is identifiable.

It is important to note that the choice of the identifiability condition does not affect our algorithm. For this reason, we prefer not to pick a particular identifiability condition but state our statistical theory in a way applicable to all choices of the identifiability condition. To this end, we introduce

$$\tau := n^{-1} \sum_{i=1}^n p_i, \quad (\text{A.3})$$

and let the definition of (F, T) and the statements of all theorems and lemmas depend on τ . If one uses the identifiability condition (A.1), then τ can be set at a pre-specified value τ_0 (e.g., $\tau_0 = 1/2$), and the statements of main theorems and lemmas would be simplified.

B Normalized Ranks with Ties

When y_i takes discrete values, there will be many ties in the ranking of y_i . There are several ways to rank those documents associated with the same value of y_i . Our method and statistical theory is flexible to accommodate different choices of ranks.

We illustrate this point using an example where y_i 's are the star ratings associated with customer reviews. Suppose $y_i \in \{1, 2, 3, 4, 5\}$. We denote ϵ_k as the fraction of k -star reviews, and set $\epsilon_0 = 0$. Let $\delta_k = \sum_{j=0}^k \epsilon_j$, for $1 \leq k \leq K$. It is obvious that

The ranks of all k -star reviews should be between $n\delta_{k-1}$ and $n\delta_k$.

Equivalently speaking, the normalized ranks should satisfy that

$$\delta_{k-1} \leq \hat{p}_i \leq \delta_k, \quad \text{for all documents with } y_i = k. \quad (\text{B.4})$$

As long as (B.4) is satisfied, we are flexible to assign ranks to these documents. Two common approaches are as follows. The first approach assigns equal normalized ranks to all tied documents:

$$\hat{p}_i = (\delta_{k-1} + \delta_k)/2, \quad \text{for all documents with } y_i = k. \quad (\text{B.5})$$

The second approach assigns continuous normalized ranks, for example:

$$\hat{p}_i \text{ is uniformly distributed in } [\delta_{k-1}, \delta_k] \text{ for documents with } y_i = k. \quad (\text{B.6})$$

Our statistical theory applies to both choices, except that the quantify $\hat{\rho}$ defined in Lemma 1 takes different values depending on the choice of ranking method.

For illustration, we calculate $\hat{\rho}$ in a special case where $n^{-1} \sum_{i=1}^n p_i = 1/2$ and $\epsilon_k = 1/5$ for all $1 \leq k \leq K$. We shall denote the $\hat{\rho}$ associated with two options by $\hat{\rho}_1$ and $\hat{\rho}_2$, respectively. For the first option in (B.5), when $y_i = k$, $\hat{p}_i = (2k - 1)/10$. By direct calculations,

$$\hat{\rho}_1 = \frac{1}{2} \sum_{k=1}^5 (k - 3)(\bar{p}_k - 1/2), \quad (\text{B.7})$$

where $\bar{p}_k$ is the average sentiment score of k -star reviews. For the second option in (B.6), note that $\sum_{i=1}^n (\frac{i}{n} - \frac{1}{2})^2 = \sum_{i=1}^n \frac{i^2}{n^2} - \sum_{i=1}^n \frac{i}{n} + \frac{n}{4} = \frac{n}{3} - \frac{n}{2} + \frac{n}{4} + O(1) = \frac{n}{12} + O(1)$. It follows that

$$\hat{\rho}_2 = \frac{12}{n} \sum_{i=1}^n (p_{(i)} - 1/2)(i/n - 1/2) + O(n^{-1}), \quad (\text{B.8})$$

where $p_{(i)}$ is the true sentiment score of the review with a normalized rank i/n . We can further simplify (B.7) and (B.8) in an *idealized* setting where *for any two documents, if $y_i \leq y_j$, then $p_i \leq p_j$* (Model (4) only guarantees that if $\mathbb{E}y_i \leq \mathbb{E}y_j$, then $p_i \leq p_j$; hence, what we consider here is indeed an idealized setting, which is only for illustration of insights). In this idealized setting, we can obtain explicit values of $\hat{\rho}_1$ and $\hat{\rho}_2$ for some particular models of p_i :

- Model 1: $p_i = (\delta_{k-1} + \delta_k)/2$ for all k -star reviews. For this model of p_i , it can be shown that $\hat{\rho}_1 = 1$ and $\hat{\rho}_2 = 24/25 + O(n^{-1})$.
- Model 2: The p_i 's of k -star reviews are uniformly distributed in $[\delta_{k-1}, \delta_k]$. For this model of p_i , it can be shown that $\hat{\rho}_1 = 1$, and $\hat{\rho}_2 = 1 + O(n^{-1})$.

We aim for $\hat{\rho}$ to be close to 1, which makes the first option slightly preferable to the second (at least in this idealized setting, based on either of the models for p_i described above).

As explained in the main article, the value of $\hat{\rho}$ does not affect the ranking consistency of SESTM. Hence, it makes no difference whether we choose option (B.5) or option (B.6).

Furthermore, (B.5) and (B.6) are not the only options. For example, in (B.5), we can set $\hat{p}_i = a_k$ for any arbitrary number $a_k \in (\delta_{k-1}, \delta_k)$, or allow $\hat{p}_i$ to take several distinct values within (δ_{k-1}, δ_k) . In these cases, $\hat{\tau} = 1/2$ is no longer true, but the entire statistical theory remains valid (see Lemma 1 and Section 3).

C Mathematical Proofs

C.1 General Setup

Our model in the main article is summarized in (1), (2), and (4), in which y_i 's are independent of each other. In fact, the SESTM algorithm is also applicable to settings with dependent y_i 's. In this supplementary material, we prove the theoretical results of SESTM in a more general model allowing dependent y_i 's. It includes the model in the main article as a special case.

Recall that Model (4) only concerns the marginal distribution of each y_i . To describe the dependence among y_i 's, we borrow the notion of *dependency graph* [4].

Definition C.1. *An undirected graph Γ with nodes $\{1, 2, \dots, n\}$ is called a dependency graph if, for any $i \in \{1, 2, \dots, n\}$ and $V \subset \{1, 2, \dots, n\} \setminus \{i\}$, there is no edge between i and nodes*

in V implies that y_i is independent of $\{y_j\}_{j \in V}$.

The dependence graph is a way to measure the complexity of dependence among y_i 's. Given the joint distribution of $y_1, y_2, \dots, y_n$, if there exists a valid dependence graph whose maximum degree is properly small, then we think the dependence structure is not too complicated. For example, when y_i 's are independent, the “empty” graph (with no edge) is a valid dependence graph. When y_i 's jointly follow a multivariate Gaussian distribution with a covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$, a valid dependence graph is such that there is an edge between two nodes i and j if and only if $\Sigma(i, j) \neq 0$, and the maximum degree of this graph is determined by the row-wise sparsity of Σ . Using this definition of dependence graph, we shall not specify the joint distribution of y_i 's explicitly but only make a mild assumption as follows: Suppose there exists a valid dependency graph for $\{y_i\}_{1 \leq i \leq n}$ whose maximum degree, K_n , satisfies that

$$K_n + 1 = o(n / \log^2(m)). \quad (\text{C.9})$$

In addition to replacing the independence assumption on y_i 's by the assumption in (C.9), we also slightly modify the regularity condition (e) in Assumption 2 to the following one:

$$\left| \sum_{i=1}^n (g(p_i) - \bar{g})(s_i - \bar{s}) \right| \leq C \bar{s} \sqrt{n(K_n + 1) \log(m)}. \quad (\text{C.10})$$

Assumptions used in this supplementary material: We assume that (1)-(2) hold for the joint distribution of d_i 's, (4) and (C.9) hold for the joint distribution of y_i 's, and Assumption 1, Assumption 2(a)-(d), and (C.10) are satisfied. The claims of some theorems and lemmas will also change slightly. Letting $K_n = 0$ reduces to the claims in the main article.

C.2 Proof of Lemma 1

Proof. Recall that the matrix M is defined by

$$M = \begin{bmatrix} 1 & 1 \\ 1 & \tau/(\tau - 1) \end{bmatrix} (W\widehat{W}') \cdot (\widehat{W}\widehat{W}')^{-1} \begin{bmatrix} \widehat{\tau} & 1 - \widehat{\tau} \\ 1 - \widehat{\tau} & \widehat{\tau} - 1 \end{bmatrix} \equiv M_1 \cdot M_2. \quad (\text{C.11})$$

In what follows, we calculate M_1 and M_2 separately.

First, consider M_1 . From the definitions of W and $\widehat{W}$ in (8) and (10), we obtain:

$$W\widehat{W}' = \begin{bmatrix} \sum_{i=1}^n p_i \widehat{p}_i & \sum_{i=1}^n p_i (1 - \widehat{p}_i) \\ \sum_{i=1}^n (1 - p_i) \widehat{p}_i & \sum_{i=1}^n (1 - p_i) (1 - \widehat{p}_i) \end{bmatrix}.$$

Write $\mu = \tau/(1 - \tau)$ for short. It follows that

$$\begin{aligned} M_1 &= \begin{bmatrix} 1 & 1 \\ 1 & -\mu \end{bmatrix} \begin{bmatrix} \sum_{i=1}^n p_i \hat{p}_i & \sum_{i=1}^n p_i (1 - \hat{p}_i) \\ \sum_{i=1}^n (1 - p_i) \hat{p}_i & \sum_{i=1}^n (1 - p_i) (1 - \hat{p}_i) \end{bmatrix} \\ &= \begin{bmatrix} \sum_{i=1}^n \hat{p}_i & \sum_{i=1}^n (1 - \hat{p}_i) \\ \sum_{i=1}^n [p_i - \mu(1 - p_i)] \hat{p}_i & \sum_{i=1}^n [p_i - \mu(1 - p_i)] (1 - \hat{p}_i) \end{bmatrix}. \end{aligned} \quad (\text{C.12})$$

Since $\hat{\tau}$ is the average of $\hat{p}_i$'s, we immediately have $\sum_i \hat{p}_i = n\hat{\tau}$ and $\sum_i (1 - \hat{p}_i) = n(1 - \hat{\tau})$. In addition, since $\mu = \tau/(1 - \tau)$, it follows that $p_i - \mu(1 - p_i) = (1 + \mu)p_i - \mu = (p_i - \tau)/(1 - \tau)$. Plugging these equalities into (C.12) gives

$$M_1 = \begin{bmatrix} n\hat{\tau} & n(1 - \hat{\tau}) \\ \frac{1}{1 - \tau} \sum_{i=1}^n (p_i - \tau) \hat{p}_i & \frac{1}{1 - \tau} \sum_{i=1}^n (p_i - \tau) (1 - \hat{p}_i) \end{bmatrix}. \quad (\text{C.13})$$

Next, consider M_2 . We start from calculating $(\widehat{W}\widehat{W}')^{-1}$. By direct calculations,

$$\widehat{W}\widehat{W}' = \begin{bmatrix} \sum_{i=1}^n \hat{p}_i^2 & \sum_{i=1}^n \hat{p}_i (1 - \hat{p}_i) \\ \sum_{i=1}^n \hat{p}_i (1 - \hat{p}_i) & \sum_{i=1}^n (1 - \hat{p}_i)^2 \end{bmatrix}. \quad (\text{C.14})$$

The inverse of any 2×2 matrix has an explicit formular as follows:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Therefore, it follows from (C.14) that

$$(\widehat{W}\widehat{W}')^{-1} = \frac{1}{\Phi_n} \begin{bmatrix} \sum_{i=1}^n (1 - \hat{p}_i)^2 & -\sum_{i=1}^n \hat{p}_i (1 - \hat{p}_i) \\ -\sum_{i=1}^n \hat{p}_i (1 - \hat{p}_i) & \sum_{i=1}^n \hat{p}_i^2 \end{bmatrix}, \quad (\text{C.15})$$

where $\Phi_n := (\sum_i \hat{p}_i^2) [\sum_i (1 - \hat{p}_i)^2] - [\sum_i \hat{p}_i (1 - \hat{p}_i)]^2$. By direct calculations, we have

$$\begin{aligned} \Phi_n &= \sum_i \sum_j \hat{p}_i^2 (1 - \hat{p}_j)^2 - \sum_i \sum_j \hat{p}_i (1 - \hat{p}_i) \hat{p}_j (1 - \hat{p}_j) \\ &= \sum_i \sum_j \hat{p}_i (1 - \hat{p}_j) \underbrace{[\hat{p}_i (1 - \hat{p}_j) - (1 - \hat{p}_i) \hat{p}_j]}_{=\hat{p}_i - \hat{p}_j} \\ &= \sum_i \hat{p}_i^2 \sum_j (1 - \hat{p}_j) - \sum_i \hat{p}_i \sum_j \hat{p}_j (1 - \hat{p}_j) \\ &= n(1 - \hat{\tau}) \sum_i \hat{p}_i^2 - n\hat{\tau} \sum_j \hat{p}_j (1 - \hat{p}_j) \quad (\text{because } \hat{\tau} \text{ is the average of } \hat{p}_i) \end{aligned}$$

$$\begin{aligned}
&= n \sum_i [(1 - \hat{\tau})\hat{p}_i^2 - \hat{\tau}\hat{p}_i(1 - \hat{p}_i)] \\
&= n \sum_i \hat{p}_i(\hat{p}_i - \hat{\tau}) - n \sum_i \hat{\tau}(\hat{p}_i - \hat{\tau}) \quad (\text{note that } \sum_i (\hat{p}_i - \hat{\tau}) = 0) \\
&= n \sum_i (\hat{p}_i - \hat{\tau})^2.
\end{aligned}$$

Plugging it into (C.15) gives

$$(\widehat{W}\widehat{W}')^{-1} = \frac{1}{n \sum_{i=1}^n (\hat{p}_i - \hat{\tau})^2} \begin{bmatrix} \sum_{i=1}^n (1 - \hat{p}_i)^2 & -\sum_{i=1}^n \hat{p}_i(1 - \hat{p}_i) \\ -\sum_{i=1}^n \hat{p}_i(1 - \hat{p}_i) & \sum_{i=1}^n \hat{p}_i^2 \end{bmatrix}. \quad (\text{C.16})$$

With the expression of $(\widehat{W}\widehat{W}')^{-1}$, we now calculate M_2 :

$$\begin{aligned}
M_2 &= \frac{1}{n \sum_{i=1}^n (\hat{p}_i - \hat{\tau})^2} \begin{bmatrix} \sum_{i=1}^n (1 - \hat{p}_i)^2 & -\sum_{i=1}^n \hat{p}_i(1 - \hat{p}_i) \\ -\sum_{i=1}^n \hat{p}_i(1 - \hat{p}_i) & \sum_{i=1}^n \hat{p}_i^2 \end{bmatrix} \begin{bmatrix} \hat{\tau} & 1 - \hat{\tau} \\ 1 - \hat{\tau} & \hat{\tau} - 1 \end{bmatrix} \\
&= \frac{1}{n \sum_{i=1}^n (\hat{p}_i - \hat{\tau})^2} \begin{bmatrix} \sum_{i=1}^n (1 - \hat{p}_i)[\hat{\tau}(1 - \hat{p}_i) - (1 - \hat{\tau})\hat{p}_i] & (1 - \hat{\tau}) \sum_{i=1}^n [(1 - \hat{p}_i)^2 + \hat{p}_i(1 - \hat{p}_i)] \\ \sum_{i=1}^n \hat{p}_i[-\hat{\tau}(1 - \hat{p}_i) + (1 - \hat{\tau})\hat{p}_i] & -(1 - \hat{\tau}) \sum_{i=1}^n [\hat{p}_i(1 - \hat{p}_i) + \hat{p}_i^2] \end{bmatrix} \\
&= \frac{1}{n \sum_{i=1}^n (\hat{p}_i - \hat{\tau})^2} \begin{bmatrix} -\sum_{i=1}^n (1 - \hat{p}_i)(\hat{p}_i - \hat{\tau}) & (1 - \hat{\tau}) \sum_{i=1}^n (1 - \hat{p}_i) \\ \sum_{i=1}^n \hat{p}_i(\hat{p}_i - \hat{\tau}) & -(1 - \hat{\tau}) \sum_{i=1}^n \hat{p}_i \end{bmatrix} \\
&= \frac{1}{n \sum_{i=1}^n (\hat{p}_i - \hat{\tau})^2} \begin{bmatrix} -\sum_{i=1}^n (1 - \hat{p}_i)(\hat{p}_i - \hat{\tau}) & n(1 - \hat{\tau})^2 \\ \sum_{i=1}^n \hat{p}_i(\hat{p}_i - \hat{\tau}) & -n(1 - \hat{\tau})\hat{\tau} \end{bmatrix}. \quad (\text{C.17})
\end{aligned}$$

Finally, we combine the respective expression of M_1 and M_2 in (C.13) and (C.17) to get

$$\begin{aligned}
M = M_1 M_2 &= \frac{1}{n \sum_{i=1}^n (\hat{p}_i - \hat{\tau})^2} \begin{bmatrix} n\hat{\tau} & n(1 - \hat{\tau}) \\ \frac{1}{1 - \hat{\tau}} \sum_{i=1}^n (p_i - \tau)\hat{p}_i & \frac{1}{1 - \hat{\tau}} \sum_{i=1}^n (p_i - \tau)(1 - \hat{p}_i) \end{bmatrix} \\
&\quad \times \begin{bmatrix} -\sum_{i=1}^n (1 - \hat{p}_i)(\hat{p}_i - \hat{\tau}) & n(1 - \hat{\tau})^2 \\ \sum_{i=1}^n \hat{p}_i(\hat{p}_i - \hat{\tau}) & -n(1 - \hat{\tau})\hat{\tau} \end{bmatrix}.
\end{aligned}$$

By direct calculations,

$$\begin{aligned}
M(1, 2) &= \frac{1}{n \sum_i (\hat{p}_i - \hat{\tau})^2} [n\hat{\tau} \cdot n(1 - \hat{\tau})^2 - n(1 - \hat{\tau}) \cdot n(1 - \hat{\tau})\hat{\tau}] = 0, \\
M(1, 1) &= \frac{1}{n \sum_i (\hat{p}_i - \hat{\tau})^2} \sum_i [-n\hat{\tau}(1 - \hat{p}_i) + n(1 - \hat{\tau})\hat{p}_i](\hat{p}_i - \hat{\tau}) \\
&= \frac{1}{n \sum_i (\hat{p}_i - \hat{\tau})^2} \sum_i (n\hat{p}_i - n\hat{\tau})(\hat{p}_i - \hat{\tau}) = 1,
\end{aligned}$$

$$\begin{aligned}
M(2, 2) &= \frac{1}{n(1-\tau) \sum_i (\hat{p}_i - \hat{\tau})^2} \sum_i (p_i - \tau) [n(1-\hat{\tau})^2 \hat{p}_i - n(1-\hat{\tau}) \hat{\tau} (1 - \hat{p}_i)] \\
&= \frac{1 - \hat{\tau}}{(1 - \tau) \sum_i (\hat{p}_i - \hat{\tau})^2} \sum_i (p_i - \tau) (\hat{p}_i - \hat{\tau}) = \frac{1 - \hat{\tau}}{1 - \tau} \hat{\rho}, \\
M(2, 1) &= \frac{-[\sum_i (p_i - \tau) \hat{p}_i] [\sum_i (1 - \hat{p}_i) (\hat{p}_i - \hat{\tau})] + [\sum_i (p_i - \tau) (1 - \hat{p}_i)] [\sum_i \hat{p}_i (\hat{p}_i - \hat{\tau})]}{n(1-\tau) \sum_i (\hat{p}_i - \hat{\tau})^2} \\
&= \frac{-\sum_i \sum_j (p_i - \tau) \hat{p}_i (1 - \hat{p}_j) (\hat{p}_j - \hat{\tau}) + \sum_i \sum_j (p_i - \tau) (1 - \hat{p}_i) \hat{p}_j (\hat{p}_j - \hat{\tau})}{n(1-\tau) \sum_i (\hat{p}_i - \hat{\tau})^2} \\
&= \frac{\sum_i \sum_j (p_i - \tau) (\hat{p}_j - \hat{\tau}) (\hat{p}_j - \hat{p}_i)}{n(1-\tau) \sum_i (\hat{p}_i - \hat{\tau})^2} \\
&= \frac{1}{n(1-\tau) \sum_i (\hat{p}_i - \hat{\tau})^2} \left[\sum_i (p_i - \tau) \sum_j \hat{p}_j (\hat{p}_j - \hat{\tau}) - \sum_i \hat{p}_i (p_i - \tau) \sum_j (\hat{p}_j - \hat{\tau}) \right] \\
&= 0,
\end{aligned}$$

where the last line is because $\sum_i (p_i - \tau) = 0$ and $\sum_j (\hat{p}_j - \hat{\tau}) = 0$. It follows immediately that

$$M = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1-\hat{\tau}}{1-\tau} \hat{\rho} \end{bmatrix}.$$

This proves the claim. $\square$

C.3 Proof of Lemma 2

Proof. For each word j , let L_j^+ denote the total count of this word in all articles with positive returns; define L_j^- similarly. Write $t_i = \text{sgn}(y_i) \in \{\pm 1\}$, for $1 \leq i \leq n$, and let $\bar{t} = n^{-1} \sum_{i=1}^n t_i$. It is seen that

$$L_j^\pm = \sum_{i=1}^n d_{j,i} \cdot (1 \pm t_i)/2, \quad \text{and} \quad \hat{\pi} = (1 + \bar{t})/2.$$

By definition, $f_j = L_j^+ / (L_j^+ + L_j^-) = [1 + (L_j^+ - L_j^-) / (L_j^+ + L_j^-)] / 2$. As a result,

$$f_j - \hat{\pi} = \frac{1}{2} \left(\frac{L_j^+ - L_j^-}{L_j^+ + L_j^-} - \bar{t} \right) = \frac{\sum_{i=1}^n (t_i - \bar{t}) \cdot d_{j,i}}{\sum_{i=1}^n d_{j,i}}. \quad (\text{C.18})$$

Below, we study f_j for $j \in S$ and $j \in N$, separately.

In the first part of the proof, we consider $j \in S$. As in Section 2.1, let $\tau = n^{-1} \sum_i p_i$, $F = \tau O_+ + (1 - \tau) O_-$, and $T = (1 - \tau)(O_+ - O_-)$. We re-parametrization of p_i to $\eta_i = p_i - \tau$. Then, $p_i O_+ + (1 - p_i) O_- = F + (1 - \tau)^{-1} \eta_i T$. Combining this with Model (2) gives

$$d_{j,i} \sim \text{Binomial} \left(s_i, F_j + (1 - \tau)^{-1} \eta_i T_j \right).$$

Let $\{b_{j,i,\ell}\}_{\ell=1}^{s_i}$ be a collection of *iid* Bernoulli variables with a success probability $F_j + \frac{\eta_i}{1-\tau}T_j$. Then, $d_{j,i} \stackrel{(d)}{=} \sum_{\ell=1}^{s_i} b_{j,i,\ell}$, where $\stackrel{(d)}{=}$ means “having the same distribution.” It follows that

$$f_j - \hat{\pi} \stackrel{(d)}{=} \frac{\sum_{i=1}^n \sum_{\ell=1}^{s_i} (t_i - \bar{t}) \cdot b_{j,i,\ell}}{\sum_{i=1}^n \sum_{\ell=1}^{s_i} b_{j,i,\ell}}, \quad \text{where } b_{j,i,\ell} \sim \text{Bernoulli}\left(F_j + \frac{\eta_i}{1-\tau}T_j\right). \quad (\text{C.19})$$

Define $\Delta_{1j} = \sum_{i=1}^n \sum_{\ell=1}^{s_i} (t_i - \bar{t})(b_{j,i,\ell} - \mathbb{E}b_{j,i,\ell})$ and $\Delta_{2j} = \sum_{i=1}^n \sum_{\ell=1}^{s_i} (b_{j,i,\ell} - \mathbb{E}b_{j,i,\ell})$. Since $\mathbb{E}b_{j,i,\ell} = F_j + \frac{\eta_i}{1-\tau}T_j$, we can re-write (C.19) as

$$f_j - \hat{\pi} = \frac{\sum_{i=1}^n s_i(t_i - \bar{t})(F_j + \frac{\eta_i}{1-\tau}T_j) + \Delta_{1j}}{\sum_{i=1}^n s_i(F_j + \frac{\eta_i}{1-\tau}T_j) + \Delta_{2j}}.$$

Note that $\bar{t}$ is the average of t_i . It yields that $\sum_{i=1}^n s_i(t_i - \bar{t})F_j = F_j \sum_{i=1}^n (s_i - \bar{s})(t_i - \bar{t})$. As a result,

$$f_j - \hat{\pi} = \frac{F_j \sum_{i=1}^n (s_i - \bar{s})(t_i - \bar{t}) + \frac{T_j}{1-\tau} \sum_{i=1}^n s_i(t_i - \bar{t})\eta_i + \Delta_{1j}}{n\bar{s}F_j + \frac{T_j}{1-\tau} \sum_{i=1}^n s_i\eta_i + \Delta_{2j}}. \quad (\text{C.20})$$

To proceed the proof, we need two technical lemmas. One is the classical Bernstein’s inequality [6] for independent variables:

Lemma C.1 (Bernstein inequality). *Suppose $X_1, \dots, X_n$ are independent random variables such that $\mathbb{E}X_i = 0$, $|X_i| \leq b$ and $\text{Var}(X_i) \leq \sigma_i^2$ for all i . Let $\sigma^2 = \sum_{i=1}^n \sigma_i^2$. For any $t > 0$,*

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i\right| \geq t\right) \leq 2 \exp\left(-\frac{t^2/2}{\sigma^2 + bt/3}\right).$$

The other is a Hoeffding-type inequality for dependent variables. In Section C.1, we have introduced the notion of dependency graph for describing the dependency structure for a set of variables. The following lemma comes from Theorem 2.1 and Equation (2.2) in [4].

Lemma C.2 (Hoeffding inequality for weakly dependent variables). *Suppose $X_1, \dots, X_n$ are random variables such that $\mathbb{E}X_i = 0$ and $a_i \leq X_i \leq b_i$. Suppose Γ is a valid dependency graph for the joint distribution of $\{X_i\}_{i=1}^n$. The maximum degree of Γ is denoted by $d^*(\Gamma)$. Then, for any $t > 0$,*

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i\right| \geq t\right) \leq 2 \exp\left(-\frac{2t^2}{[d^*(\Gamma) + 1] \sum_{i=1}^n (b_i - a_i)^2}\right).$$

When applying Lemmas C.1-C.2, we often use particular choices of t . In Lemma C.1, for any $\epsilon \in (0, 1)$, we can let $t = \sigma\sqrt{2\log(2/\epsilon)} + (2b/3)\log(2/\epsilon)$ to make $\mathbb{P}(|\sum_{i=1}^n X_i| \geq t)$ upper

bounded by ϵ . In particular, as $n \rightarrow \infty$, for any $\epsilon_n \rightarrow 0$, there exists a constant $C > 0$ such that simultaneously for all sufficiently large n , with probability $1 - \epsilon_n$,

$$\left| \sum_{i=1}^n X_i \right| \leq C \left[\sigma \sqrt{\log(1/\epsilon_n)} + b \log(1/\epsilon_n) \right]. \quad (\text{C.21})$$

Similarly, in Lemma C.2, for any sequence $\epsilon_n \rightarrow 0$, we can choose an appropriate t and show that, with probability $1 - \epsilon_n$,

$$\left| \sum_{i=1}^n X_i \right| \leq C \sqrt{[d^*(\Gamma) + 1] \sum_{i=1}^n (b_i - a_i)^2 \log(1/\epsilon_n)}. \quad (\text{C.22})$$

In the proof below, we will use (C.21) and (C.22), which are more convenient than the original statements in Lemmas C.1-C.2.

With these technical preparations, we now proceed to study the right hand side of (C.20). First, we bound $|\Delta_{1j}|$ and $|\Delta_{2j}|$. The random variables $\{b_{j,i,\ell} - \mathbb{E}b_{j,i,\ell}\}$ are independent, mean-zero, and they satisfy that $\sum_{i=1}^n \sum_{\ell=1}^{s_i} \text{var}(b_{j,i,\ell}) \leq \sum_{i=1}^n \sum_{\ell=1}^{s_i} (F_j + \frac{\eta_i}{1-\tau} T_j)$. Note that $|\eta_i| \leq 1$, $F_j = \tau O_{+,j} + (1-\tau)O_{-,j}$, and $T_j = (1-\tau)(O_{+,j} - O_{-,j})$. Since $T_j \leq \tau^{-1}(1-\tau)F_j$, we have $\sum_{i=1}^n \sum_{\ell=1}^{s_i} \text{var}(b_{j,i,\ell}) \leq \sum_{i=1}^n \sum_{\ell=1}^{s_i} (1 + \tau^{-1})F_j = (1 + \tau^{-1})n\bar{s}F_j \leq Cn\bar{s}F_j$, where in the last inequality we have used Assumption 2(a). By (C.21), with probability $1 - m^{-1}\epsilon_n$,

$$\begin{aligned} |\Delta_{2j}| &= \left| \sum_{i=1}^n \sum_{\ell=1}^{s_i} (b_{j,i,\ell} - \mathbb{E}b_{j,i,\ell}) \right| \leq C \sqrt{n\bar{s}F_j \log(m/\epsilon_n)} + C \log(m/\epsilon_n) \\ &\leq C \sqrt{n\bar{s}F_j \log(m/\epsilon_n)}, \end{aligned} \quad (\text{C.23})$$

where the last line follows from the assumption $n\bar{s}F_j / \log(m/\epsilon_n) \rightarrow \infty$. We recall that $\Delta_{1j} = \sum_{i=1}^n \sum_{\ell=1}^{s_i} (t_i - \bar{t})(b_{j,i,\ell} - \mathbb{E}b_{j,i,\ell})$ and that $t_i = \text{sgn}(y_i)$ are independent of d_i 's. Conditioning on $\{y_i\}_{i=1}^n$, by similar arguments as above, with probability $1 - m^{-1}\epsilon_n$,

$$\begin{aligned} |\Delta_{1j}| &\leq C \sqrt{F_j \sum_{i=1}^n s_i (t_i - \bar{t})^2 \log(m/\epsilon_n)} + C(\max_i |t_i - \bar{t}|) \log(m/\epsilon_n) \\ &\leq C \sqrt{n\bar{s}F_j \log(m/\epsilon_n)}, \end{aligned} \quad (\text{C.24})$$

where the last inequality is due to $|t_i - \bar{t}| \leq 1$. Since the bound in (C.24) does not depend on y_i 's, this inequality also holds unconditionally with the same probability.

We then insert (C.23)-(C.24) into (C.20). Consider the term $(n\bar{s}F_j + \frac{T_j}{1-\tau} \sum_{i=1}^n s_i \eta_i)$ in the denominator of (C.20). It can be re-written as $\sum_{i=1}^n s_i [p_i O_{+,j} + (1-p_i)O_{-,j}]$, which, by

Assumption 2(d), is lower bounded by $c_0 \sum_{i=1}^n s_i F_j \geq c_0 \bar{s} F_j$, for a constant $c_0 > 0$. Meanwhile, since $|\eta_i| \leq 1$ and $T_j \leq \tau^{-1}(1-\tau)F_j$, this term also has an upper bound: $n\bar{s}F_j + \frac{T_j}{1-\tau} \sum_{i=1}^n s_i \eta_i \leq (1 + \tau^{-1})n\bar{s}F_j$. Combining the above results with (C.20) gives

$$\begin{aligned} |f_j - \hat{\pi}| &\geq \frac{|\frac{T_j}{1-\tau} \sum_{i=1}^n s_i(t_i - \bar{t})\eta_i| - |F_j \sum_{i=1}^n (s_i - \bar{s})(t_i - \bar{t})| - |\Delta_{1j}|}{n\bar{s}F_j + \frac{T_j}{1-\tau} \sum_{i=1}^n s_i \eta_i + \Delta_{2j}} \\ &\geq \frac{|\frac{T_j}{1-\tau} \sum_{i=1}^n s_i(t_i - \bar{t})\eta_i|}{(1 + \tau^{-1})n\bar{s}F_j + |\Delta_{2j}|} - \frac{F_j |\sum_{i=1}^n (s_i - \bar{s})(t_i - \bar{t})|}{c_0 n\bar{s}F_j - |\Delta_{2j}|} - \frac{|\Delta_{1j}|}{c_0 n\bar{s}F_j - |\Delta_{2j}|} \\ &\gtrsim \frac{|T_j \sum_{i=1}^n s_i(t_i - \bar{t})\eta_i|}{(1 + \tau^{-1})(1 + \tau^{-1})n\bar{s}F_j} - O\left(\frac{|\sum_{i=1}^n (s_i - \bar{s})(t_i - \bar{t})|}{n\bar{s}} + \frac{\sqrt{\log(m/\epsilon_n)}}{\sqrt{n\bar{s}F_j}}\right). \end{aligned} \quad (\text{C.25})$$

In the third line, we have plugged in the upper bounds in (C.23)-(C.24); additionally, we have used the fact that $|\Delta_{1j}| = o(n\bar{s}F_j)$ and $|\Delta_{2j}| = o(n\bar{s}F_j)$, due to $n\bar{s}F_j / \log(m/\epsilon_n) \rightarrow \infty$.

Last, we deal with the randomness of $\{t_i\}_{i=1}^n$ in (C.25). Recall that $t_i = \text{sgn}(y_i)$. By (C.9), y_i 's are dependent, and the dependence structure is captured by a dependency graph with maximum degree K_n . It follows immediately that the same graph can be used as a dependency graph for t_i 's. Hence, we can apply (C.22) with $d^*(\Gamma) = K_n$.

In (C.25), we first study the term $|\sum_{i=1}^n s_i(t_i - \bar{t})\eta_i|$. Since $\mathbb{E}t_i = 2g(p_i) - 1$ and $\mathbb{E}\bar{t} = 2\bar{g} - 1$, we have the decomposition: $t_i - \bar{t} = 2[g(p_i) - \bar{g}] + (t_i - \mathbb{E}t_i) - (\bar{t} - \mathbb{E}\bar{t})$. It follows that

$$\begin{aligned} \left| \sum_{i=1}^n s_i(t_i - \bar{t})\eta_i \right| &\geq 2 \left| \sum_{i=1}^n s_i \eta_i [g(p_i) - \bar{g}] \right| - \left| \sum_{i=1}^n s_i \eta_i (t_i - \mathbb{E}t_i) \right| - \left| \sum_{i=1}^n s_i \eta_i (\bar{t} - \mathbb{E}\bar{t}) \right| \\ &= 4n\bar{s}\theta - \left| \sum_{i=1}^n s_i \eta_i (t_i - \mathbb{E}t_i) \right| - \left| n^{-1} \sum_{i=1}^n s_i \eta_i \right| \left| \sum_{i=1}^n (t_i - \mathbb{E}t_i) \right|, \end{aligned} \quad (\text{C.26})$$

where the last line is from the definition of θ in (15) and the equality $\bar{t} - \mathbb{E}\bar{t} = n^{-1} \sum_{i=1}^n (t_i - \mathbb{E}t_i)$. We apply (C.22) to $X_i = t_i - \mathbb{E}t_i$. Since $|t_i - \mathbb{E}t_i| \leq 1$, it gives that, with probability $1 - m^{-1}\epsilon_n$,

$$\left| \sum_{i=1}^n (t_i - \mathbb{E}t_i) \right| \leq C\sqrt{n(K_n + 1)\log(m/\epsilon_n)}.$$

Similarly, we apply (C.22) to $X_i = s_i \eta_i (t_i - \mathbb{E}t_i)$. Since $|\eta_i| = |2p_i - 1| \leq 1$ and $\sum_{i=1}^n s_i^2 \leq Cn\bar{s}^2$ (by Assumption 2(c)), it follows that, with probability $1 - m^{-1}\epsilon_n$,

$$\left| \sum_{i=1}^n s_i \eta_i (t_i - \mathbb{E}t_i) \right| \leq C\bar{s}\sqrt{n(K_n + 1)\log(m/\epsilon_n)}.$$

Plugging the above results into (C.26) and noting that $|n^{-1} \sum_{i=1}^n s_i \eta_i| \leq C\bar{s}$, we immediately have: With probability $1 - m^{-1}\epsilon_n$,

$$\left| \sum_{i=1}^n s_i(t_i - \bar{t})\eta_i \right| \geq 4n\bar{s}\theta - O\left(\bar{s}\sqrt{n(K_n + 1)\log(m/\epsilon_n)}\right). \quad (\text{C.27})$$

Next, we study the term $|\sum_{i=1}^n (s_i - \bar{s})(t_i - \bar{t})|$ in (C.25). Using the decomposition $t_i - \bar{t} = 2[g(p_i) - \bar{g}] + (t_i - \mathbb{E}t_i) - (\bar{t} - \mathbb{E}\bar{t})$ again, we get:

$$\left| \sum_{i=1}^n (s_i - \bar{s})(t_i - \bar{t}) \right| \leq 2 \left| \sum_{i=1}^n (s_i - \bar{s})[g(p_i) - \bar{g}] \right| + \left| \sum_{i=1}^n (s_i - \bar{s})(t_i - \mathbb{E}t_i) \right| + \left| \sum_{i=1}^n (s_i - \bar{s})(\bar{t} - \mathbb{E}\bar{t}) \right|.$$

The last term is zero. The first term is bounded by $C\bar{s}\sqrt{n(K_n + 1)\log(m)}$, due to (C.10) (it is the assumption we have used to replace Assumption 2(d) in the general case of dependent y_i). The second term can be bounded using (C.22) again. We omit the details and state the results directly: With probability $1 - m^{-1}\epsilon_n$, $|\sum_{i=1}^n (s_i - \bar{s})(t_i - \mathbb{E}t_i)| \leq C\bar{s}\sqrt{n(K_n + 1)\log(m/\epsilon_n)}$. Combining these results gives

$$\left| \sum_{i=1}^n (s_i - \bar{s})(t_i - \bar{t}) \right| \leq C\bar{s}\sqrt{n(K_n + 1)\log(m/\epsilon_n)}. \quad (\text{C.28})$$

We plug (C.27) and (C.28) into (C.25). It follows that, with probability $1 - m^{-1}\epsilon_n$,

$$|f_j - \hat{\pi}| \gtrsim \frac{4\tau\theta|T_j|}{(1 - \tau^2)F_j} - O\left(\frac{\sqrt{(K_n + 1)\log(m/\epsilon_n)}}{\sqrt{n}} + \frac{\sqrt{\log(m/\epsilon_n)}}{\sqrt{n\bar{s}F_j}}\right). \quad (\text{C.29})$$

Combined with the probability union bound, it yields that the above inequality holds simultaneously for all $j \in S$ with probability $1 - \epsilon_n$. By definition, $|T_j| = (1 - \tau)|O_{+,j} - O_{-,j}|$, and $F_j = \tau O_{+,j} + (1 - \tau)O_{-,j} \leq 2(O_{+,j} + O_{-,j})$. Therefore, the first term on the right hand side of (C.29) is lower bounded by $\frac{2\tau\theta}{1+\tau} \cdot \frac{|O_{+,j} - O_{-,j}|}{O_{+,j} + O_{-,j}}$. In addition, $F_j = \tau O_{+,j} + (1 - \tau)O_{-,j} \geq C^{-1}(O_{+,j} + O_{-,j})$, because τ is bounded away from 0 and 1 (by Assumption 2(a)). We plug these lower bounds into (C.29) to obtain: With probability $1 - \epsilon_n$,

$$|f_j - \hat{\pi}| \gtrsim \frac{2\tau\theta|O_{+,j} - O_{-,j}|}{(1 + \tau)F_j} - O\left(\frac{\sqrt{\log(m/\epsilon_n)}}{\sqrt{n}} + \frac{\sqrt{\log(m/\epsilon_n)}}{\sqrt{n\bar{s}(O_{+,j} + O_{-,j})}}\right).$$

This proves the first claim.

In the second part of the proof, we consider $j \in N$. Now, $f_j - \hat{\pi}$ has the same expression as in (C.18), and we aim to study its numerator and denominator. By Model (1),

$$d_{j,i} \sim \text{Binomial}(n_i, \Omega_{j,i}), \quad 1 \leq i \leq n.$$

Similarly as before, we introduce independent random variables $\{h_{j,i,\ell}\}_{1 \leq i \leq n, 1 \leq \ell \leq n_i}$, with $h_{j,i,\ell} \sim \text{Bernoulli}(\Omega_{j,i})$. Then, $d_{j,i}$ can be replaced by $\sum_{\ell=1}^{n_i} h_{j,i,\ell}$. We thus re-write the numerator and denominator in (C.18) as

$$U_j \equiv \sum_{i=1}^n (t_i - \bar{t})d_{j,i} = \sum_{i=1}^n \sum_{\ell=1}^{n_i} (t_i - \bar{t})h_{j,i,\ell}, \quad \text{and} \quad V_j \equiv \sum_{i=1}^n d_{j,i} = \sum_{i=1}^n \sum_{\ell=1}^{n_i} h_{j,i,\ell}.$$

We first study U_j . Re-write

$$U_j = \sum_{i=1}^n \sum_{\ell=1}^{n_i} (t_i - \bar{t})(h_{j,i,\ell} - \Omega_{j,i}) + \sum_{i=1}^n n_i(t_i - \bar{t})\Omega_{j,i} \equiv (I_1) + (I_2).$$

Conditioning on $\{y_i\}_{i=1}^n$, the variables $\{(t_i - \bar{t})(h_{j,i,\ell} - \Omega_{j,i})\}_{1 \leq i \leq n, 1 \leq \ell \leq n_i}$ are independent, $|h_{j,i,\ell} - \Omega_{j,i}| \leq 1$, and $\sum_{i=1}^n \sum_{\ell=1}^{n_i} \text{var}(h_{j,i,\ell}) \leq \sum_{i=1}^n n_i \Omega_{j,i} = n\bar{n}q_j$. We apply (C.21) to obtain the following: Conditioning on $\{y_i\}_{i=1}^n$, with probability $1 - m^{-1}\epsilon_n$,

$$|(I_1)| \leq C\sqrt{n\bar{n}q_j \log(m/\epsilon_n)} + C\log(m/\epsilon_n) \leq C\sqrt{n\bar{n}q_j \log(m/\epsilon_n)}, \quad (\text{C.30})$$

where the last inequality comes from $n\bar{n}q_j/\log(m/\epsilon_n) \rightarrow \infty$. This bound does not depend on y_i 's, therefore, it also holds marginally without conditioning. We then bound $|(I_2)|$. Write

$$(I_2) = Z - \Delta, \quad \text{where} \quad Z = \sum_{i=1}^n n_i(t_i - \mathbb{E}\bar{t})\Omega_{j,i}, \quad \Delta = \sum_{i=1}^n n_i(\bar{t} - \mathbb{E}\bar{t})\Omega_{j,i}.$$

We use Assumption 1. Since $\mathbb{E}[(t_i - \bar{t})\Omega_{j,i}] = \mathbb{E}[(t_i - \bar{t}) \cdot \mathbb{E}(n_i^{-1}d_{j,i}|y_1, \dots, y_n)] = \mathbb{E}[n_i^{-1}(t_i - \bar{t})d_{j,i}]$, it holds that

$$\mathbb{E}[(I_2)] = \sum_{i=1}^n \mathbb{E}[(\text{sgn}(y_i) - \overline{\text{sgn}}_y) d_{j,i}] = 0. \quad (\text{C.31})$$

As a result,

$$|(I_2)| = |(I_2) - \mathbb{E}[(I_2)]| \leq |Z - \mathbb{E}Z| + |\Delta| + \mathbb{E}|\Delta|. \quad (\text{C.32})$$

We bound the three terms on the right hand separately. Note that $|\Delta| = |\bar{t} - \mathbb{E}\bar{t}| \cdot |\sum_{i=1}^n n_i \Omega_{j,i}| \leq |\bar{t} - \mathbb{E}\bar{t}| \cdot n\bar{n}q_j \leq \bar{n}q_j |\sum_{i=1}^n (t_i - \mathbb{E}t_i)|$. In the equation below (C.26), we have already proved that $|\sum_{i=1}^n (t_i - \mathbb{E}t_i)| \leq C\sqrt{n(K_n + 1)\log(m/\epsilon_n)}$, with probability $1 - m^{-1}\epsilon_n$. Combining these results gives

$$|\Delta| \leq C\bar{n}q_j \sqrt{n(K_n + 1)\log(m/\epsilon_n)}.$$

Furthermore, if we apply Lemma C.2, instead of (C.22), to $\sum_{i=1}^n (t_i - \mathbb{E}t_i)$, we will find out that this is a sub-Gaussian variable with a sub-Gaussian norm $O(n(K_n + 1))$. In particular, its first absolute moment is $O(\sqrt{n(K_n + 1)})$. It follows that

$$\mathbb{E}|\Delta| \leq C\bar{n}q_j \mathbb{E}\left(\left|\sum_{i=1}^n (t_i - \mathbb{E}t_i)\right|\right) \leq C\bar{n}q_j \sqrt{n(K_n + 1)}.$$

Write $Z - \mathbb{E}Z = \sum_{i=1}^n n_i \{(t_i - \mathbb{E}\bar{t})\Omega_{j,i} - \mathbb{E}[(t_i - \mathbb{E}\bar{t})\Omega_{j,i}]\}$. We shall apply (C.22). Note that each summand only depends on y_i . Hence, the dependency graph in (C.9) is still a valid

dependency graph here. Additionally, each summand is bounded by $2n_i\Omega_{j,i}$ in absolute value, and $\sum_{i=1}^n (2n_i\Omega_{j,i})^2 = 4 \sum_{i=1}^n n_i^2 \Omega_{j,i}^2 \leq Cn\bar{n}^2 q_j^2$ (by Assumption 2(c)). It follows from (C.22) that, with probability $1 - m^{-1}\epsilon_n$,

$$|Z - \mathbb{E}Z| \leq C\bar{n}q_j \sqrt{n(K_n + 1) \log(m/\epsilon_n)}.$$

We plug the above inequalities into (C.32) and find out that

$$|(I_2)| \leq C\bar{n}q_j \sqrt{n(K_n + 1) \log(m/\epsilon_n)}. \quad (\text{C.33})$$

Combining (C.30) and (C.33) gives

$$|U_j| \leq C \left(\sqrt{n\bar{n}q_j \log(m/\epsilon_n)} + \bar{n}q_j \sqrt{n(K_n + 1) \log(m/\epsilon_n)} \right). \quad (\text{C.34})$$

We then study V_j . Rewrite

$$V_j = \sum_{i=1}^n n_i \mathbb{E}[\Omega_{j,i}] + \sum_{i=1}^n n_i \{\Omega_{j,i} - \mathbb{E}[\Omega_{j,i}]\} + \sum_{i=1}^n \sum_{\ell=1}^{n_i} (h_{j,i,\ell} - \Omega_{j,i}).$$

The last two terms can be bounded in similar ways as we bound $|Z - \mathbb{E}Z|$ and $|(I_1)|$. Moreover, $\sum_{i=1}^n n_i \mathbb{E}[\Omega_{j,i}] = \sum_{i=1}^n n_i \Omega_{j,i} = n\bar{n}q_j$. It follows that

$$V_j \geq n\bar{n}q_j - C \left(\sqrt{n\bar{n}q_j \log(m/\epsilon_n)} + \bar{n}q_j \sqrt{n(K_n + 1) \log(m/\epsilon_n)} \right) \gtrsim n\bar{n}q_j, \quad (\text{C.35})$$

where the last inequality is because $n\bar{n}q_j / \log(m/\epsilon_n) \rightarrow \infty$ and $nK_n^{-1} / \log(m/\epsilon_n) \rightarrow \infty$. We put (C.34) and (C.35) together to get: With probability $1 - m^{-1}\epsilon_n$,

$$|f_j - \hat{\pi}| = \frac{|U_j|}{V_j} = O \left(\frac{\sqrt{\log(m/\epsilon_n)}}{\sqrt{n\bar{n}q_j}} + \frac{\sqrt{(K_n + 1) \log(m/\epsilon_n)}}{\sqrt{n}} \right).$$

This proves the second claim. $\square$

C.4 Proof of Theorem 1

Proof. Since we consider a more general model (see Section C.1) than the model in the main article, we also slightly modify the assumption (16) to the following one:

$$n\theta^2 \min_{j \in S} \frac{(O_{+,j} - O_{-,j})^2}{(O_{+,j} + O_{-,j})^2} \geq \frac{\log^2(m)}{\min\{(K_n + 1)^{-1}, \bar{s} \min_{j \in S} (O_{+,j} + O_{-,j}), \bar{n} \min_{j \in N} q_j\}}. \quad (\text{C.36})$$

We note that (16) is a special case of (C.36) with $K_n = 0$.

We now show the claim. Recall that in the proof of Lemma 2, letting $\epsilon_n = 1/m$, we have shown (see (C.29)): With probability $1 - O(m^{-1})$, simultaneously for all $1 \leq j \leq m$,

$$|f_j - \hat{\pi}| \begin{cases} \geq \frac{4\tau}{1-\tau^2} \theta F_j^{-1} |T_j| + O(e_n), & j \in S, \\ \leq O(e_n), & j \in N, \end{cases}$$

where $e_n^2 = (\min\{(K_n + 1)^{-1}, \bar{s} \min_{j \in S} F_j, \bar{n} \min_{j \in N} q_j\})^{-1} \frac{\log(m)}{n}$. The assumption (C.36), together with $a_0 \leq \tau \leq 1 - a_0$ (see Assumption 2(a)), ensures $\theta F_j^{-1} |T_j| \gg e_n \sqrt{\log(m)}$. By setting the threshold at $e_n \sqrt{\log(\log(m))}$, all words in S will retain, and all words in N will be screened out. This proves the claim. $\square$

C.5 Proof of Theorem 2

Proof. By Theorem 1, $\mathbb{P}(\hat{S} = S) = 1 - o(1)$. Hence, it suffices to prove the claim by replacing $\hat{S}$ with S in the estimation step. In particular, we replace $\hat{h}_i$ in (7) by $h_i = d_{[S],i}/s_i$. Write $H = [h_1, \dots, h_n]$. The estimate $\hat{O}$ is obtained by setting the negative entries of $\tilde{O} = H\hat{W}'(\hat{W}\hat{W}')^{-1}$ to zero and renormalizing it to have a unit ℓ^1 -norm. By Model (2), $\mathbb{E}H = OW$. It motivates us to define an intermediate matrix

$$O^* = OW\hat{W}'(\hat{W}\hat{W}')^{-1}. \quad (\text{C.37})$$

Let $F^* = \hat{\tau}O_+^* + (1 - \hat{\tau})O_-^*$ and $T^* = (1 - \hat{\tau})(O_+^* - O_-^*)$. By definition,

$$\begin{aligned} [F^*, T^*] &= O^* \begin{bmatrix} \hat{\tau} & 1 - \hat{\tau} \\ 1 - \hat{\tau} & \hat{\tau} - 1 \end{bmatrix} = O(W\hat{W}')(\hat{W}\hat{W}')^{-1} \begin{bmatrix} \hat{\tau} & 1 - \hat{\tau} \\ 1 - \hat{\tau} & \hat{\tau} - 1 \end{bmatrix} \\ &= [F, T] \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & \tau/(\tau - 1) \end{bmatrix} (W\hat{W}')(\hat{W}\hat{W}')^{-1}}_{\equiv M} \begin{bmatrix} \hat{\tau} & 1 - \hat{\tau} \\ 1 - \hat{\tau} & \hat{\tau} - 1 \end{bmatrix}. \end{aligned}$$

We have seen in Lemma 1 that $M = \text{diag}(1, \frac{1-\hat{\tau}}{1-\tau}\hat{\rho})$. It follows that

$$F^* = F, \quad T^* = \frac{1 - \hat{\tau}}{1 - \tau} \hat{\rho} T. \quad (\text{C.38})$$

Hence, to show the claim, we only need to bound $\|\hat{F} - F^*\|_1$ and $\|\hat{T} - T^*\|_1$. Note that $\hat{F} - F^* = \hat{\tau}(\hat{O}_+ - O_+^*) + (1 - \hat{\tau})(\hat{O}_- - O_-^*)$. It follows that $\|\hat{F} - F^*\|_1 \leq \max\{\|\hat{O}_+ - O_+^*\|_1, \|\hat{O}_- - O_-^*\|_1\}$. A similar inequality holds for $\|\hat{T} - T^*\|_1$. It reduces to showing that with probability $1 - o(1)$,

$$\|\hat{O}_{\pm} - O_{\pm}^*\|_1 \leq C\sqrt{|S|\log(m)/(n\bar{s})}. \quad (\text{C.39})$$

We now show (C.39). Let $\bar{O} = [\bar{O}_+, \bar{O}_-]$ be the matrix obtained by setting negative entries of $\tilde{O} = H\widehat{W}'(\widehat{W}\widehat{W}')^{-1}$ to zero. Then, $\hat{O}_\pm = (1/\|\bar{O}_\pm\|_1)\bar{O}_\pm$. It follows that, for $j \in S$,

$$|\hat{O}_{\pm,j} - O_{\pm,j}^*| \leq |\bar{O}_{\pm,j} - O_{\pm,j}^*| + |\bar{O}_{\pm,j}| \cdot \left| \frac{1}{\|\bar{O}_\pm\|_1} - 1 \right|.$$

From the definitions of (F^*, T^*) , we deduce: $O_+^* = F^* + T^*$ and $O_-^* = F^* - \frac{\hat{\tau}}{1-\hat{\tau}}T^*$. Combining it with (C.38) gives $O_+^* = F + \frac{1-\hat{\tau}}{1-\tau}\hat{\rho}T$ and $O_-^* = F - \frac{\hat{\tau}}{1-\tau}\hat{\rho}T$. Note that F is a nonnegative vector whose entries sum to 1, T is a vector whose entries sum to 0, and $\frac{1}{1-\tau}|T_j| \leq \min\{\frac{1}{\tau}F_j, \frac{1}{1-\tau}F_j\}$ for each j . As a result, when $0 < \hat{\rho} \leq 1$, the two vectors $O_\pm^*$ are nonnegative, with $\|O_\pm^*\|_1 = 1$. Since $\|O_\pm^*\|_1 = 1$, it holds that $|\|\bar{O}_\pm\|_1^{-1} - 1| = \|\bar{O}_\pm\|_1^{-1}|\|\bar{O}_\pm\|_1 - \|O_\pm^*\|_1| \leq \|\bar{O}_\pm\|_1^{-1}\|\bar{O}_\pm - O_\pm^*\|_1$. Hence,

$$|\hat{O}_{\pm,j} - O_{\pm,j}^*| \leq |\bar{O}_{\pm,j} - O_{\pm,j}^*| + \frac{|\bar{O}_{\pm,j}|}{\|\bar{O}_\pm\|_1} \|\bar{O}_\pm - O_\pm^*\|_1. \quad (\text{C.40})$$

Summing over j on both sides gives

$$\|\hat{O}_\pm - O_\pm^*\|_1 \leq 2\|\bar{O}_\pm - O_\pm^*\|_1.$$

Moreover, since $O_\pm^*$ are nonnegative vectors, the operation of truncating out negative entries in $\tilde{O}_\pm$ (to obtain $\bar{O}_\pm$) always makes it closer to $O_\pm^*$. It implies $\|\bar{O}_\pm - O_\pm^*\|_1 \leq \|\tilde{O}_\pm - O_\pm^*\|_1$. Combining the above gives

$$\|\hat{O}_\pm - O_\pm^*\|_1 \leq 2\|\tilde{O}_\pm - O_\pm^*\|_1. \quad (\text{C.41})$$

Therefore, to show (C.39), it suffices to show that

$$\|\tilde{O}_\pm - O_\pm^*\|_1 \leq C\sqrt{|S|\log(m)/(n\bar{s})}. \quad (\text{C.42})$$

It remains to show (C.42). Recall that $\tilde{O} = H\widehat{W}'(\widehat{W}\widehat{W}')^{-1}$ and $O^* = H\widehat{W}'(\widehat{W}\widehat{W}')^{-1}$. As mentioned, $\mathbb{E}H = OW$. Define $Z = H - OW$ and write

$$\tilde{O} = (OW + Z)\widehat{W}'(\widehat{W}\widehat{W}')^{-1} = O^* + Z\widehat{W}'(\widehat{W}\widehat{W}')^{-1}.$$

Let $z_i \in \mathbb{R}^n$ be the i -th column of Z , $1 \leq i \leq n$. Plugging in the form of $\widehat{W}$, we have

$$Z\widehat{W}'(\widehat{W}\widehat{W}')^{-1} = \left[\sum_{i=1}^n \hat{p}_i z_i \quad \sum_{i=1}^n (1 - \hat{p}_i) z_i \right] (\widehat{W}\widehat{W}')^{-1}.$$

It follows that

$$|\tilde{O}_{\pm,j} - O_{\pm,j}^*| \leq \|[Z\widehat{W}'(\widehat{W}\widehat{W}')^{-1}]_{j,\cdot}\|_\infty$$

$$\begin{aligned}
&\leq \left(\left| \frac{1}{n} \sum_{i=1}^n \widehat{p}_i Z_{j,i} \right| + \left| \frac{1}{n} \sum_{i=1}^n (1 - \widehat{p}_i) Z_{j,i} \right| \right) \|n(\widehat{W}\widehat{W}')^{-1}\|_{\infty} \\
&\leq C \left(\left| \frac{1}{n} \sum_{i=1}^n \widehat{p}_i Z_{j,i} \right| + \left| \frac{1}{n} \sum_{i=1}^n (1 - \widehat{p}_i) Z_{j,i} \right| \right), \tag{C.43}
\end{aligned}$$

where in the last line we have used the explicit expression of $(\widehat{W}\widehat{W}')^{-1}$ in (C.16). We now bound $|\frac{1}{n} \sum_{i=1}^n \widehat{p}_i Z_{j,i}|$. The bound for $|\frac{1}{n} \sum_{i=1}^n (1 - \widehat{p}_i) Z_{j,i}|$ can be obtained similarly, so the proof is omitted. By definition, $Z_{j,i} = H_{j,i} - \mathbb{E}H_{j,i}$, where $H_{j,i} = s_i^{-1}d_{j,i}$. By Models (1)-(2),

$$d_{j,i} \sim \text{Binomial}(s_i, p_i O_{+,j} + (1 - p_i) O_{-,j}).$$

Let $\{b_{j,i,\ell}\}_{\ell=1}^{s_i}$ be a collection of *iid* Bernoulli variables with a success probability $[p_i O_{+,j} + (1 - p_i) O_{-,j}]$. Then, $d_{j,i}$ has the same distribution as $\sum_{\ell=1}^{s_i} b_{j,i,\ell}$. As a result, $Z_{j,i} \stackrel{(d)}{=} \sum_{\ell=1}^{s_i} s_i^{-1} (b_{j,i,\ell} - \mathbb{E}b_{j,i,\ell})$, and we can write

$$\sum_{i=1}^n \widehat{p}_i Z_{j,i} = \sum_{i=1}^n \sum_{\ell=1}^{s_i} \widehat{p}_i s_i^{-1} (b_{j,i,\ell} - \mathbb{E}b_{j,i,\ell}).$$

Conditioning on $\{y_i\}_{i=1}^n$, $\widehat{p}_i$'s become non-random, and we shall apply (C.21). The variables $\widehat{p}_i s_i^{-1} (b_{j,i,\ell} - \mathbb{E}b_{j,i,\ell})$ are mutually independent, upper bounded by $2s_{\min}^{-1}$, each with mean 0 and a variance $\leq s_i^{-2} (O_{+,j} + O_{-,j}) = 4\tau^{-1} s_i^{-2} F_j \leq C s_i^{-2} F_j$. The sum of variances of these variables is bounded by $\sum_{i=1}^n \sum_{\ell=1}^{s_i} C s_i^{-2} F_j \leq C (\sum_i s_i^{-1}) F_j \leq C n \bar{s}^{-1} F_j$, where the last inequality follows from Assumption 2(c). By (C.21), with probability $1 - O(m^{-2})$,

$$\left| \sum_{i=1}^n \widehat{p}_i Z_{j,i} \right| \leq C \sqrt{n \bar{s}^{-1} F_j \log(m)} + C s_{\min}^{-1} \log(m) \leq C \sqrt{n \bar{s}^{-1} F_j \log(m)}, \tag{C.44}$$

where the last line is due to $\bar{s}/s_{\min} \leq C n \bar{s} F_j / \log(m)$ (see Assumption 2(c)). The bound for $|\sum_{i=1}^n (1 - \widehat{p}_i) Z_{j,i}|$ is similar. Plugging them into (C.43) gives

$$|\widetilde{O}_{\pm,j} - O_{\pm,j}^*| \leq C \frac{\sqrt{F_j \log(m)}}{\sqrt{n \bar{s}}}. \tag{C.45}$$

It follows from Cauchy-Schwarz inequality that

$$\|\widetilde{O}_{\pm} - O_{\pm}^*\|_1 \leq C \sqrt{\frac{\log(m)}{n \bar{s}}} \sum_{j \in S} \sqrt{F_j} \leq C \sqrt{\frac{\log(m)}{n \bar{s}}} \cdot |S|^{\frac{1}{2}} \left(\sum_{j \in S} F_j \right)^{\frac{1}{2}} \leq C \sqrt{\frac{|S| \log(m)}{n \bar{s}}}.$$

This proves (C.39). The proof is now complete. $\square$

C.6 Proof of Corollary 1

Proof. Let $(O_\pm^*, F^*, T^*)$ be the same as in the proof of Theorem 2. In (C.39) and the paragraph above it, we have shown: (i) $\max\{\|\hat{F} - F^*\|_1, \|\hat{T} - T^*\|_1\} \leq \max\{\|\hat{O}_+ - O_+^*\|_1, \|\hat{O}_- - O_-^*\|_1\}$; (ii) with probability $1 - o(1)$, $\|\hat{O}_\pm - O_\pm^*\|_1 \leq C\sqrt{|S|\log(m)/(n\bar{s})}$. In addition, we have seen that $F^* = F$ and $T^* = \frac{1-\hat{\tau}}{1-\tau}\hat{\rho}T$ in (C.38). It follows that with probability $1 - o(1)$,

$$\|\hat{F} - F\|_1 \leq C\sqrt{\frac{|S|\log(m)}{n\bar{s}}}, \quad \|\hat{T} - T\|_1 \leq \left|1 - \frac{1-\hat{\tau}}{1-\tau}\hat{\rho}\right| \cdot \|T\|_1 + C\sqrt{\frac{|S|\log(m)}{n\bar{s}}}. \quad (\text{C.46})$$

Note that $1 - \frac{1-\hat{\tau}}{1-\tau}\hat{\rho} = \frac{1-\hat{\tau}}{1-\tau}(1 - \hat{\rho}) + \frac{\hat{\tau}-\tau}{1-\tau}$ and $\|T\|_1 = (1-\tau)\|O_+ - O_-\|_1 \leq 2(1-\tau)$. We plug them into the bound for $\|\hat{T} - T\|_1$ to get

$$\|\hat{T} - T\|_1 \leq 2|1 - \hat{\rho}| + 2|\hat{\tau} - \tau| + C\sqrt{\frac{|S|\log(m)}{n\bar{s}}}. \quad (\text{C.47})$$

By definition, $\tau = n^{-1} \sum_{i=1}^n p_i$ and $\hat{\tau} = n^{-1} \sum_{i=1}^n \hat{p}_i$. By Cauchy-Schwarz inequality,

$$|\hat{\tau} - \tau| \leq \frac{1}{n} \sum_{i=1}^n |\hat{p}_i - p_i| \leq \sqrt{\frac{1}{n} \sum_{i=1}^n (p_i - \hat{p}_i)^2}. \quad (\text{C.48})$$

Using the definition of $\hat{\rho}$ and Cauchy-Schwarz inequality again,

$$\begin{aligned} |1 - \hat{\rho}| &= \frac{1}{\sum_{i=1}^n (\hat{p}_i - \hat{\tau})^2} \cdot \left| \sum_{i=1}^n (\hat{p}_i - \hat{\tau}) [(\hat{p}_i - p_i) - (\hat{\tau} - \tau)] \right| \\ &= \frac{1}{\sum_{i=1}^n (\hat{p}_i - \hat{\tau})^2} \cdot \left| \sum_{i=1}^n (\hat{p}_i - \hat{\tau})(\hat{p}_i - p_i) \right| \\ &\leq \frac{1}{\sum_{i=1}^n (\hat{p}_i - \hat{\tau})^2} \cdot \sqrt{\sum_{i=1}^n (\hat{p}_i - \hat{\tau})^2} \sqrt{\sum_i (\hat{p}_i - p_i)^2}, \end{aligned}$$

where the second line is because $\sum_i (\hat{p}_i - \hat{\tau}) = 0$. By the assumption, $\sum_i (\hat{p}_i - \hat{\tau})^2 \geq a_1 n$ for a constant $a_1 > 0$. It follows that

$$|1 - \hat{\rho}| \leq C \sqrt{\frac{1}{n} \sum_{i=1}^n (p_i - \hat{p}_i)^2}. \quad (\text{C.49})$$

We insert (C.48)-(C.49) into (C.47), and then plug (C.47) into (C.46). It gives the claim. $\square$

C.7 Proof of Theorem 3

Proof. Note that by Theorem 1, $\mathbb{P}(\widehat{S} = S) = 1 - o(1)$. Hence, without loss of generality, we assume $\widehat{S} = S$ throughout this proof.

Recall that p and p^* are the true sentiment and rescaled sentiment of the new document, respectively. The penalized MLE in (11) also uses the notation p . We now replace it by p_0 , to differentiate from the true sentiment p . In other words, we write

$$\widehat{p} = \arg \max_{p_0 \in [0,1]} \left\{ s^{-1} \sum_{j \in S} d_j \log \left(p_0 \widehat{O}_{+,j} + (1 - p_0) \widehat{O}_{-,j} \right) + \lambda [\widehat{\tau} \log(p_0) + (1 - \widehat{\tau}) \log(1 - p_0)] \right\}.$$

We re-parametrize $(p_0, \widehat{p}, p)$ by $\eta_0 = p_0 - \widehat{\tau}$, $\widehat{\eta} = \widehat{p} - \widehat{\tau}$, and $\eta = p - \tau$. Since $\widehat{F} = \widehat{\tau} \widehat{O}_+ + (1 - \widehat{\tau}) \widehat{O}_-$ and $\widehat{T} = (1 - \widehat{\tau})(\widehat{O}_+ - \widehat{O}_-)$, we have

$$p_0 \widehat{O}_+ + (1 - p_0) \widehat{O}_- = p_0 (\widehat{F} + \widehat{T}) + (1 - p_0) \left(\widehat{F} - \frac{\widehat{\tau}}{1 - \widehat{\tau}} \widehat{T} \right) \widehat{T} = \widehat{F} + \frac{\eta_0}{1 - \widehat{\tau}} \widehat{T}.$$

Therefore, with the re-parametrization, $\widehat{\eta} = \arg \max_{\eta_0 \in [-\widehat{\tau}, 1 - \widehat{\tau}]} \widehat{\ell}(\eta_0)$, where

$$\widehat{\ell}(\eta_0) \equiv s^{-1} \sum_{j \in S} d_j \log \left(\widehat{F}_j + (1 - \widehat{\tau})^{-1} \eta_0 \widehat{T}_j \right) + \lambda [\widehat{\tau} \log(\widehat{\tau} + \eta_0) + (1 - \widehat{\tau}) \log(1 - \widehat{\tau} - \eta_0)]. \quad (\text{C.50})$$

Let $\eta^* = p^* - \widehat{\tau}$. Then, $|\widehat{p} - p^*| = |\widehat{\eta} - \eta^*|$. It is sufficient to bound $|\widehat{\eta} - \eta^*|$.

In preparation, we study $\widehat{\ell}'(\eta_0)$ and $\widehat{\ell}''(\eta_0)$. By direct calculations,

$$\begin{aligned} \widehat{\ell}'(\eta_0) &= \sum_{j \in S} \frac{d_j \widehat{T}_j}{s[(1 - \widehat{\tau}) \widehat{F}_j + \eta_0 \widehat{T}_j]} - \frac{\lambda \eta_0}{(\widehat{\tau} + \eta_0)(1 - \widehat{\tau} - \eta_0)}, \\ \widehat{\ell}''(\eta_0) &= - \sum_{j \in S} \frac{d_j \widehat{T}_j^2}{s[(1 - \widehat{\tau}) \widehat{F}_j + \eta_0 \widehat{T}_j]^2} - \frac{\lambda [\widehat{\tau}(1 - \widehat{\tau}) + \eta_0^2]}{(\widehat{\tau} + \eta_0)^2 (1 - \widehat{\tau} - \eta_0)^2}. \end{aligned} \quad (\text{C.51})$$

First, consider $\widehat{\ell}'(\eta_0)$. By Models (1)-(2) and the definition of (η, F, T) , $d \sim \text{Multinomial}(s, F + \frac{1}{1 - \tau} \eta T)$. Hence, $\mathbb{E} d_j = s(F_j + \frac{1}{1 - \tau} \eta T_j)$. Additionally, by Theorem 2, $\widehat{F}_j \approx F_j$, and $\widehat{T}_j \approx T_j^*$, where $T_j^* := \frac{1 - \widehat{\tau}}{1 - \tau} \widehat{\rho} T_j$. In light of this, we introduce

$$\widehat{f}(\eta_0) = \sum_{j \in S} \frac{d_j \widehat{T}_j}{s[(1 - \widehat{\tau}) \widehat{F}_j + \eta_0 \widehat{T}_j]}, \quad f(\eta_0) = \sum_{j \in S} \frac{[F_j + (1 - \tau)^{-1} \eta T_j] T_j^*}{(1 - \widehat{\tau}) F_j + \eta_0 T_j^*},$$

where $\widehat{f}(\eta_0)$ is the first term in $\widehat{\ell}'(\eta_0)$, and $f(\eta_0)$ is its counterpart by replacing $(d_j, \widehat{F}_j, \widehat{T}_j)$ by $(\mathbb{E} d_j, F_j, T_j^*)$. Let E be the event that

$$|\widehat{F}_j - F_j| \leq C \sqrt{F_j \log(m)/(n \bar{s})}, \quad |\widehat{T}_j - T_j^*| \leq C \sqrt{F_j \log(m)/(n \bar{s})}, \quad (\text{C.52})$$

simultaneously for all $j \in S$. In the proof of Theorem 2, we have seen that $\mathbb{P}(E) = 1 - o(1)$. We now condition on training data and assume that the realized $(\hat{F}, \hat{T})$ satisfy (C.52). Note that

$$\begin{aligned} |\hat{f}(\eta_0) - f(\eta_0)| &= \left| \sum_{j \in S} \frac{s^{-1} d_j \hat{T}_j}{(1 - \hat{\tau}) \hat{F}_j + \eta_0 \hat{T}_j} - \sum_{j \in S} \frac{s^{-1} (\mathbb{E} d_j) T_j^*}{(1 - \hat{\tau}) F_j + \eta_0 T_j^*} \right| \\ &\leq \left| \sum_{j \in S} \frac{s^{-1} (d_j - \mathbb{E} d_j) \hat{T}_j}{(1 - \hat{\tau}) \hat{F}_j + \eta_0 \hat{T}_j} \right| + \left| \sum_{j \in S} \frac{s^{-1} (\mathbb{E} d_j) (\hat{T}_j - T_j^*)}{(1 - \hat{\tau}) \hat{F}_j + \eta_0 \hat{T}_j} \right| \\ &\quad + \left| \sum_{j \in S} s^{-1} (\mathbb{E} d_j) T_j^* \left(\frac{1}{(1 - \hat{\tau}) \hat{F}_j + \eta_0 \hat{T}_j} - \frac{1}{(1 - \hat{\tau}) F_j + \eta_0 T_j^*} \right) \right|. \end{aligned}$$

First, by our assumptions, $\hat{\rho}|T_j| \leq (1 - c_1)(1 - \tau)F_j$. Hence, $(1 - \hat{\tau})F_j + \eta_0 T_j^* = (1 - \hat{\tau})[(1 - \tau)F_j + \eta_0 \hat{\rho}T_j] \geq (1 - \hat{\tau})[(1 - \tau)F_j - \hat{\rho}|T_j|] \geq (1 - \hat{\tau}) \cdot c_1(1 - \tau)F_j \geq c_1 a_0^2 F_j$ (where a_0 is the same as in Assumption 2(a)). Second, by (C.52) and the assumption $n\bar{s}F_j/\log(m) \rightarrow \infty$, we have: $\max\{|\hat{F}_j - F_j|, |\hat{T}_j - T_j^*|\} = o(F_j)$. It implies $(1 - \hat{\tau})\hat{F}_j + \eta_0 \hat{T}_j \gtrsim (1 - \hat{\tau})F_j + \eta_0 T_j^* \gtrsim c_1 a_0^2 F_j$. Finally, $s^{-1}\mathbb{E}d_j = F_j + (1 - \tau)^{-1}\eta T_j = O(F_j)$. Plugging them into the above inequality gives

$$\begin{aligned} |\hat{f}(\eta_0) - f(\eta_0)| &\leq \left| \sum_{j \in S} \frac{s^{-1} (d_j - \mathbb{E} d_j) \hat{T}_j}{(1 - \hat{\tau}) \hat{F}_j + \eta_0 \hat{T}_j} \right| + C \sum_{j \in S} \sqrt{\frac{F_j \log(m)}{n\bar{s}}}, \\ &\leq \left| \sum_{j \in S} \frac{s^{-1} (d_j - \mathbb{E} d_j) \hat{T}_j}{(1 - \hat{\tau}) \hat{F}_j + \eta_0 \hat{T}_j} \right| + C \sqrt{\frac{|S| \log(m)}{n\bar{s}}}, \end{aligned} \quad (\text{C.53})$$

where the last inequality is due to $\sum_{j \in S} \sqrt{F_j} \leq \sqrt{|S| \sum_j F_j} \leq \sqrt{|S|}$. We then bound the first term in (C.53). We condition on the training data. Since d is independent of the training data, it is still valid that $d \sim \text{Multinomial}(s, F + \frac{1}{1-\tau}\eta T)$. Let $\{b_\ell\}_{\ell=1}^s$ be *iid* random vectors, with $b_\ell \sim \text{Multinomial}(1, F + \frac{1}{1-\tau}\eta T)$. Conditioning on training data, d has the same distribution as $\sum_{\ell=1}^s b_\ell$. It follows that

$$\sum_{j \in S} \frac{s^{-1} (d_j - \mathbb{E} d_j) \hat{T}_j}{(1 - \hat{\tau}) \hat{F}_j + \eta_0 \hat{T}_j} \stackrel{(d)}{=} \sum_{\ell=1}^s \xi_\ell, \quad \text{with} \quad \xi_\ell \equiv \sum_{j \in S} \frac{s^{-1} \hat{T}_j}{(1 - \hat{\tau}) \hat{F}_j + \eta_0 \hat{T}_j} (b_{\ell,j} - \mathbb{E} b_{\ell,j}).$$

Conditional on the training data, $(\hat{F}, \hat{T}, \hat{\rho})$ are non-random. Hence, $\{\xi_\ell\}_{\ell=1}^s$ are *iid* variables with mean zero. We compute the variance of ξ_ℓ . Let $v \in \mathbb{R}^{|S|}$ be such that $v_j = \frac{s^{-1} \hat{T}_j}{(1 - \hat{\tau}) \hat{F}_j + \eta_0 \hat{T}_j}$. Then, $\xi_\ell = v'(b_\ell - \mathbb{E} b_\ell)$. By elementary properties of multinomial distributions, the covariance matrix of ξ_ℓ is $\text{diag}(F + \frac{1}{1-\tau}\eta T) - (F + \frac{1}{1-\tau}\eta T)(F + \frac{1}{1-\tau}\eta T)' \preceq \text{diag}(F + \frac{1}{1-\tau}\eta T)$. It follows

that $\text{Var}(\xi_\ell) \leq \sum_{j \in S} v_j^2 (F_j + \frac{1}{1-\tau} \eta T_j)$. Plugging in the expression of v_j gives

$$\text{Var}(\xi_\ell) \leq \sum_{j \in S} \frac{\widehat{T}_j^2 \cdot [F_j + (1-\tau)^{-1} \eta T_j]}{s^2[(1-\widehat{\tau})\widehat{F}_j + \eta_0 \widehat{T}_j]^2} \leq C \sum_{j \in S} \frac{|T_j^*|^2}{s^2 F_j} \leq C \sum_{j \in S} \frac{|O_{+,j} - O_{-,j}|^2}{s^2 (O_{+,j} + O_{-,j})} \leq C s^{-2} \widehat{\rho}^2 \Theta,$$

where we have used $|F_j + (1-\tau)^{-1} \eta T_j| = O(F_j)$, $(1-\widehat{\tau})\widehat{F}_j + \eta_0 \widehat{T}_j \gtrsim c_1 a_0^2 F_j$ (see the paragraph above (C.53)), $|\widehat{T}_j| \lesssim |T_j^*| + o(F_j)$, $F_j \geq C^{-1}(O_{+,j} + O_{-,j})$, and $|T_j^*| \leq C\widehat{\rho}|O_{+,j} - O_{-,j}|$ (the last two inequalities are from Assumption 2(a)). Additionally, since $|v_j| \leq C s^{-1} F_j^{-1} |T_j^*| \leq C s^{-1} \widehat{\rho}$, and $b_{\ell,j}$ is a random vector whose entries sum to 1, we have $|\xi_\ell| \leq C s^{-1} \widehat{\rho}$. We apply (C.21) and find that, for any fixed η_0 , conditioning on the training data, with probability $1 - \epsilon$,

$$\left| \sum_{j \in S} \frac{s^{-1}(d_j - \mathbb{E}d_j) \widehat{T}_j}{\widehat{F}_j + \eta_0 \widehat{T}_j} \right| \leq C \left(\sqrt{\frac{\widehat{\rho}^2 \Theta \log(\epsilon^{-1})}{s}} + \frac{\widehat{\rho} \log(\epsilon^{-1})}{s} \right) \leq C \widehat{\rho} \sqrt{\frac{\Theta \log(\epsilon^{-1})}{s}}, \quad (\text{C.54})$$

where the last inequality is due to $s\Theta \rightarrow \infty$. A combination of (C.53) and (C.54) gives the following result: Conditioning on the training data where the event E occurs (this event, as defined in (C.52), is about the training data; hence, it does not affect the probability here), for any given $\eta_0 \in [-1, 1]$, with probability $1 - \epsilon$,

$$|\widehat{f}(\eta_0) - f(\eta_0)| \leq C \left(\widehat{\rho} \sqrt{\frac{\Theta \log(\epsilon^{-1})}{s}} + \sqrt{\frac{|S| \log(m)}{n\bar{s}}} \right), \quad (\text{C.55})$$

where the constant C does not depend on η_0 . We now study $f(\eta_0)$. By definitions, $T^* = \frac{1-\widehat{\tau}}{1-\tau} \widehat{\rho} T$, and $\eta^* = \widehat{\rho}^{-1} \eta$. It follows that

$$\begin{aligned} f(\eta_0) &= \frac{1}{1-\tau} \sum_{j \in S} \frac{[(1-\tau)F_j + \eta T_j] \widehat{\rho} T_j}{(1-\tau)F_j + \eta_0 \widehat{\rho} T_j} \\ &= \frac{1}{1-\tau} \sum_{j \in S} \widehat{\rho} T_j + \frac{1}{1-\tau} \sum_{j \in S} \frac{(\eta T_j - \eta_0 \widehat{\rho} T_j) \widehat{\rho} T_j}{(1-\tau)F_j + \eta_0 \widehat{\rho} T_j} \\ &= 0 + (\widehat{\rho}^{-1} \eta - \eta_0) \sum_{j \in S} \frac{\widehat{\rho}^2 T_j^2}{(1-\tau)F_j + \eta_0 \widehat{\rho} T_j} \\ &= (\eta^* - \eta_0) \sum_{j \in S} \frac{\widehat{\rho}^2 T_j^2}{(1-\tau)F_j + \eta_0 \widehat{\rho} T_j}, \end{aligned} \quad (\text{C.56})$$

where the third line is due to $\sum_{j \in S} T_j = 0$. In light of (C.56) and the form of $\widehat{\ell}'(\eta_0)$ in (C.51), we introduce

$$\psi(\eta_0) := (\eta^* - \eta_0) \sum_{j \in S} \frac{\widehat{\rho}^2 T_j^2}{(1-\tau)F_j + \eta_0 \widehat{\rho} T_j} - \frac{\lambda \eta_0}{(\widehat{\tau} + \eta_0)(1 - \widehat{\tau} - \eta_0)}.$$

By plugging (C.55) and (C.56) into $\widehat{\ell}'(\eta_0)$ in (C.51), we obtain: conditioning on training data where the event E occurs, for a fixed $\eta_0 \in (-1, 1)$, with probability $1 - \epsilon$,

$$|\widehat{\ell}'(\eta_0) - \psi(\eta_0)| \leq C \left(\widehat{\rho} \sqrt{\frac{\Theta \log(\epsilon^{-1})}{s}} + \sqrt{\frac{|S| \log(m)}{n\bar{s}}} \right). \quad (\text{C.57})$$

Second, consider $\widehat{\ell}''(\eta_0)$. Since $d_j \geq 0$ and $(1 - \widehat{\tau})\widehat{F}_j + \eta_0\widehat{T}_j = O(F_j)$ for $\eta_0 \in [-\widehat{\tau}, 1 - \widehat{\tau}]$ (the constant does not depend on η_0), the first term of $\widehat{\ell}''(\eta_0)$ in (C.51) satisfies that

$$\sup_{\eta_0 \in [-\widehat{\tau}, 1 - \widehat{\tau}]} \left\{ - \sum_{j \in S} \frac{s^{-1} d_j \widehat{T}_j^2}{[(1 - \widehat{\tau})\widehat{F}_j + \eta_0\widehat{T}_j]^2} \right\} \leq -b_0 \sum_{j \in S} \frac{d_j \widehat{T}_j^2}{s \widehat{F}_j^2}, \quad \text{for a constant } b_0 > 0.$$

Similar to (C.54), we can show that, conditioning on the training data, with probability $1 - \epsilon$,

$$\left| \sum_{j \in S} \frac{(d_j - \mathbb{E}d_j) \widehat{T}_j^2}{s \widehat{F}_j^2} \right| \leq C \widehat{\rho}^2 \sqrt{\frac{\Theta \log(\epsilon^{-1})}{s}}.$$

Combing the above and noting that $s^{-1} \mathbb{E}d_j = F_j + (1 - \tau)^{-1} \eta T_j \geq c_1 F_j$, we have

$$\sup_{\eta_0 \in [-\widehat{\tau}, 1 - \widehat{\tau}]} \left\{ - \sum_{j \in S} \frac{s^{-1} d_j \widehat{T}_j^2}{[(1 - \widehat{\tau})\widehat{F}_j + \eta_0\widehat{T}_j]^2} \right\} \leq -b_0 c_1 \sum_{j \in S} \frac{F_j \widehat{T}_j^2}{4 \widehat{F}_j^2} + C \widehat{\rho}^2 \sqrt{\frac{\Theta \log(\epsilon^{-1})}{s}}. \quad (\text{C.58})$$

On the event E (see (C.52)), $|\widehat{T}_j^2 - (T_j^*)^2| = |\widehat{T}_j + T_j^*| |\widehat{T}_j - T_j^*| \lesssim 2|T_j^*| \sqrt{F_j \log(m)/(n\bar{s})}$, and similarly, $|\widehat{F}_j^2 - F_j^2| \leq C F_j \sqrt{F_j \log(m)/(n\bar{s})}$. In addition, $|T_j^*| = \frac{1 - \widehat{\tau}}{1 - \tau} \widehat{\rho} |T_j| \geq a_0 \widehat{\rho} |O_{+,j} - O_{-,j}|$, where a_0 is as in Assumption 2(a), $|T_j^*| = O(F_j)$, and $F_j \leq O_{+,j} + O_{-,j}$. It follows that,

$$\begin{aligned} - \sum_{j \in S} \frac{F_j \widehat{T}_j^2}{\widehat{F}_j^2} &\leq - \sum_{j \in S} \frac{F_j |T_j^*|^2}{F_j^2} + \sum_{j \in S} F_j |T_j^*|^2 \left| \frac{1}{F_j^2} - \frac{1}{\widehat{F}_j^2} \right| + \sum_{j \in S} \frac{F_j |\widehat{T}_j^2 - (T_j^*)^2|}{\widehat{F}_j^2} \\ &\leq - \sum_{j \in S} \frac{a_0^2 \widehat{\rho}^2 |O_{+,j} - O_{-,j}|^2}{O_{+,j} + O_{-,j}} + C \sum_{j \in S} \sqrt{\frac{F_j \log(m)}{n\bar{s}}} \\ &\leq -a_0^2 \widehat{\rho}^2 \Theta + C \sqrt{\frac{|S| \log(m)}{n\bar{s}}}, \end{aligned}$$

where the last line uses $\sum_{j \in S} \sqrt{F_j} \leq \sqrt{|S| \sum_{j \in S} F_j} \leq \sqrt{|S|}$. We plug it into (C.58) to get

$$\sup_{\eta_0 \in [-\widehat{\tau}, 1 - \widehat{\tau}]} \left\{ - \sum_{j \in S} \frac{s^{-1} d_j \widehat{T}_j^2}{[(1 - \widehat{\tau})\widehat{F}_j + \eta_0\widehat{T}_j]^2} \right\} \leq -b_1 \widehat{\rho}^2 \Theta + C \left(\widehat{\rho}^2 \sqrt{\frac{\Theta \log(\epsilon^{-1})}{s}} + \sqrt{\frac{|S| \log(m)}{n\bar{s}}} \right), \quad (\text{C.59})$$

for a constant $b_1 = a_0^2 b_0 c_1$. Introduce

$$\varphi(\eta_0) := -b_1 \widehat{\rho}^2 \Theta - \frac{\lambda[\widehat{\tau}(1 - \widehat{\tau}) + \eta_0^2]}{(\widehat{\tau} + \eta_0)^2 (1 - \widehat{\tau} - \eta_0)^2}$$

We plug (C.59) into the expression of $\widehat{\ell}''(\eta_0)$ in (C.51) to obtain: conditioning on the training data where the event E occurs, with probability $1 - \epsilon$, simultaneously for all $\eta_0 \in (-\widehat{\tau}, 1 - \widehat{\tau})$,

$$\widehat{\ell}''(\eta_0) \leq \varphi(\eta_0) + C \left(\widehat{\rho}^2 \sqrt{\frac{\Theta \log(\epsilon^{-1})}{s}} + \sqrt{\frac{|S| \log(m)}{n\bar{s}}} \right). \quad (\text{C.60})$$

Different from (C.57), the bound holds simultaneously for all $\eta_0 \in (-\widehat{\tau}, 1 - \widehat{\tau})$. This is because in the derivation of (C.60) we do not use large-deviation inequalities directly but only base on the event E (see (C.52)). Since this event does not depend on η_0 , (C.60) holds for all η_0 .

With (C.57) and (C.60), we are ready to show the claim. Since $\lambda > 0$, from the second term in the expression of $\widehat{\ell}(\eta_0)$, $\widehat{\ell}(\eta_0) \rightarrow -\infty$ as $\eta_0 \rightarrow -\widehat{\tau}$ or $\eta_0 \rightarrow 1 - \widehat{\tau}$. Hence, the maximum can only be attained in the interior of $(-\widehat{\tau}, 1 - \widehat{\tau})$. By Lagrange's mean value theorem, there exists $\widetilde{\eta}$, which is between η^* and $\widehat{\eta}$, such that

$$0 = \widehat{\ell}'(\widehat{\eta}) = \widehat{\ell}'(\eta^*) + \widehat{\ell}''(\widetilde{\eta}) \cdot (\widehat{\eta} - \eta^*).$$

It follows that

$$|\widehat{\eta} - \eta^*| \leq \frac{|\widehat{\ell}'(\eta^*)|}{|\widehat{\ell}''(\widetilde{\eta})|} \leq \frac{|\widehat{\ell}'(\eta^*)|}{\left(\inf_{\eta_0 \in (-\widehat{\tau}, 1 - \widehat{\tau})} |\widehat{\ell}''(\eta_0)| \right)}. \quad (\text{C.61})$$

Recall that $err_n = \Theta^{-1}(\sqrt{|S| \log(m)/(n\bar{s})} + \sqrt{\Theta/s})$ and note that $\widehat{\rho} = O(1)$. Then, in both (C.57) and (C.60), the term in the brackets can be written as $C_\epsilon \Theta err_n$, where C_ϵ is a generic constant depending on ϵ , whose meaning varies from occurrence to occurrence. In addition, since $\eta^* = \widehat{\rho}^{-1} \eta = \widehat{\rho}^{-1}(p - \tau)$, our assumption gives $-\widehat{\tau} + c_2 \leq \eta^* \leq 1 - \widehat{\tau} - c_2$. It follows from (C.57) and (C.60) that

$$\begin{aligned} |\widehat{\ell}(\eta^*)| &\leq \frac{\lambda |\eta^*|}{(\widehat{\tau} + \eta^*)(1 - \widehat{\tau} - \eta^*)} + C_\epsilon \Theta err_n \leq \frac{\lambda |\eta^*|}{c_2^2} + C_\epsilon \Theta err_n, \\ \inf_{\eta_0 \in (-\widehat{\tau}, 1 - \widehat{\tau})} |\widehat{\ell}''(\eta_0)| &\geq \inf_{\eta_0 \in (-1, 1)} \left\{ b_1 \widehat{\rho}^2 \Theta + \frac{\lambda [\widehat{\tau}(1 - \widehat{\tau}) + (\eta^*)^2]}{(\widehat{\tau} + \eta^*)^2 (1 - \widehat{\tau} - \eta^*)^2} \right\} - C_\epsilon \Theta err_n \\ &\geq b_1 \rho_0^2 \Theta + \frac{\lambda a_0^2}{c_2^4} - C_\epsilon \Theta err_n. \end{aligned}$$

From (C.57) and (C.60), the above inequalities hold with probability $1 - \epsilon$ when conditioning on the training data where the event E occurs. Since the right hand side does not depend on training data, the above holds with the same probability marginally (i.e., without conditioning), provided that the event E occurs. We have seen that $\mathbb{P}(E) = 1 - o(1)$. Therefore, the

above inequalities hold with probability $1 - \epsilon$. We plug them into (C.61) and find that, with probability $1 - \epsilon$,

$$\begin{aligned}
|\hat{\eta} - \eta^*| &\leq \frac{c_2^{-2}\lambda|\eta^*| + C_\epsilon\Theta \cdot \text{err}_n}{c_2^{-4}a_0^2\lambda + b_1\rho_0^2\Theta - C_\epsilon\Theta\text{err}_n} \\
&\leq C_\epsilon \begin{cases} \Theta^{-1}\lambda|\eta^*| + \text{err}_n, & \text{if } \lambda \leq \Theta, \\ |\eta^*| + \lambda^{-1}\Theta\text{err}_n, & \text{if } \lambda > \Theta, \end{cases} \\
&\leq C_\epsilon \min\left\{1, \frac{\lambda}{\Theta}\right\} \times |\eta^*| + C_\epsilon \min\left\{1, \frac{\Theta}{\lambda}\right\} \times \text{err}_n.
\end{aligned} \tag{C.62}$$

The claim follows by noting that $|\eta^*| = \hat{\rho}^{-1}|p - \tau| \leq \rho_{\min}^{-1}|p - \tau|$. $\square$

C.8 Proof of Theorem 4

Proof. Since $\{p_i\}_{i=1}^L$ are drawn from a continuous density, with probability 1, their values are distinct from each other. The Spearman's correlation coefficient has an equivalent form:

$$SR(\hat{p}, p) = 1 - \frac{6}{L(L^2 - 1)} \sum_{i=1}^L (\hat{r}_i - r_i)^2, \tag{C.63}$$

where r_i is the rank of p_i among $\{p_j\}_{j=1}^L$, which also equals to the rank of p_i^* among $\{p_j^*\}_{j=1}^L$, and $\hat{r}_i$ is the rank of $\hat{p}_i$ among $\{\hat{p}_j\}_{j=1}^L$. By definition,

$$r_i = \frac{1}{2} \sum_{j=1}^L \text{sgn}(p_i^* - p_j^*) + \frac{L+1}{2}, \quad \hat{r}_i = \frac{1}{2} \sum_{j=1}^L \text{sgn}(\hat{p}_i - \hat{p}_j) + \frac{L+1}{2},$$

where the sign function takes values in $\{0, \pm 1\}$. We condition on $\{p_i\}_{i=1}^L$ and let $\epsilon = L^{-2}$ in Theorem 3 (the statement of this theorem assumes that ϵ is a fixed constant, but the proof can be easily extended to allow for $\epsilon \rightarrow 0$) and apply the probability union bound. It yields that, with probability $1 - L^{-1}$,

$$\max_{1 \leq i \leq L} |\hat{p}_i - p_i^*| \leq \delta, \quad \text{where } \delta = \frac{C}{\sqrt{\Theta}} \left(\frac{\sqrt{|S| \log(m)}}{\sqrt{n s \Theta}} + \frac{\sqrt{\log(L)}}{\sqrt{s}} \right). \tag{C.64}$$

Let D be the event that (C.64) holds. For each $1 \leq i \leq L$, define the index set

$$B_i(3\delta) = \{1 \leq j \leq L : j \neq i, |p_j^* - p_i^*| \leq 3\delta\}.$$

On the event D , for $j \notin B_i(3\delta)$, $|p_i^* - p_j^*| > 3\delta$, while $|\hat{p}_i - p_i^*| \leq \delta$ and $|\hat{p}_j - p_j^*| \leq \delta$; hence, $(\hat{p}_i - \hat{p}_j)$ must have the same sign as $(p_i^* - p_j^*)$. It follows that

$$|\hat{r}_i - r_i| \leq \frac{1}{2} \sum_{j \in B_i(3\delta)} (|\text{sgn}(p_i^* - p_j^*)| + |\text{sgn}(\hat{p}_i - \hat{p}_j)|) \leq |B_i(3\delta)|.$$

We plug it into (C.63) and note that $|\hat{r}_i - r_i|^2 \leq L|\hat{r}_i - r_i|$. It yields

$$1 - SR(\hat{p}, p) \leq \frac{6}{L^2 - 1} \sum_{i=1}^L |\hat{r}_i - r_i| \leq \frac{6L}{L^2 - 1} \max_{1 \leq i \leq L} |B_i(3\delta)|. \quad (\text{C.65})$$

More precisely, conditioning on $\{p_i\}_{i=1}^L$, (C.65) holds with probability $1 - L^{-1}$.

We now bound $|B_i(3\delta)|$, taking into account the randomness of $\{p_i\}_{i=1}^L$. Each p_i^* is a non-random linear function of p_i . Therefore, the distribution of $\{p_i\}_{i=1}^L$ yields that $\{p_i^*\}_{i=1}^L$ are *iid* drawn from a continuous density on $[\tau + \rho_{\min}(-\hat{\tau} + c_2), \tau + \rho_{\min}(1 - \hat{\tau} - c_2)]$ (note: $\hat{\rho}$ depends on the training data, hence, it is independent of p_i^* 's). Since this is a compact set, the probability density function must be Lipschitz. Fix i and write

$$|B_i(3\delta)| = \sum_{j \neq i} 1 \left\{ p_j^* \in [p_i^* - 3\delta, p_i^* + 3\delta] \right\}.$$

Conditioning on p_i^* , the other p_j^* 's are *iid* drawn from a Lipschitz probability density. As a result, each other p_j^* has a probability of $O(\delta)$ to fall within a distance of 3δ to p_i^* , i.e., $|B_i(3\delta)|$ is the sum of $(L - 1)$ *iid* Bernoulli variables with a success probability of $O(\delta)$. Then, $\mathbb{E}|B_i(3\delta)| = O(L\delta)$. Moreover, by the Bernstein's inequality (C.21), with probability $1 - L^{-2}$,

$$|B_i(3\delta)| \leq CL\delta + C\sqrt{L\delta \log(L)} + C\log(L).$$

Combining it with the probability union bound, with probability $1 - L^{-1}$, the above inequality holds simultaneously for all $1 \leq i \leq L$. We then plug it into (C.65) and get

$$1 - SR(\hat{p}, p) \leq C\delta + C\sqrt{\frac{\delta \log(L)}{L}} + \frac{C\log(L)}{L} \leq C\max\left\{\delta, \frac{\log(L)}{L}\right\}. \quad (\text{C.66})$$

Under our assumption, the right hand side of (C.66) is $o(1)$. The claim follows immediately. $\square$

D Summary of the Algorithm

For convenience of readers, we now state each step of SESTM as a separate algorithm. Algorithm D.1 outlines the specific procedure to build $\hat{S}$, Algorithm D.2 describes how to obtain the two sentiment topics $\hat{O}_{\pm}$, and Algorithm D.3 is about how to score the sentiment of a new testing document (when y_i 's have no ties, $\hat{\tau} = 1/2$, and the penalty term has a simpler form).

Algorithm D.1.

S1. For each word $1 \leq j \leq m$, let

$$f_j = \frac{\# \text{ articles including word } j \text{ AND having } \text{sgn}(y) = 1}{\# \text{ articles including word } j}.$$

S2. For a proper threshold $\alpha_+ > 0$, $\alpha_- > 0$, and $\kappa > 0$ to be determined, construct

$$\hat{S} = \{j : f_j \geq 1/2 + \alpha_+\} \cup \{j : f_j \leq 1/2 - \alpha_-\} \cap \{j : k_j \geq \kappa\},$$

where k_j is the total count of articles in which word j appears.

Algorithm D.2.

S1. Sort the returns $\{y_i\}_{i=1}^n$ in ascending order. For each $1 \leq i \leq n$, let

$$\hat{p}_i = \frac{\text{rank of } y_i \text{ in all returns}}{n}. \quad (\text{D.67})$$

S2. For $1 \leq i \leq n$, let $\hat{s}_i$ be the total counts of words from $\hat{S}$ in article i , and let $\hat{d}_i = \hat{s}_i^{-1} d_{i, [\hat{S}]}$.

Write $\hat{D} = [\hat{d}_1, \hat{d}_2, \dots, \hat{d}_n]$. Construct

$$\hat{O} = \hat{D} \hat{W}' (\hat{W} \hat{W}')^{-1}, \quad \text{where} \quad \hat{W} = \begin{bmatrix} \hat{p}_1 & \hat{p}_2 & \cdots & \hat{p}_n \\ 1 - \hat{p}_1 & 1 - \hat{p}_2 & \cdots & 1 - \hat{p}_n \end{bmatrix}. \quad (\text{D.68})$$

Set negative entries of $\hat{O}$ to zero and re-normalize each column to have a unit ℓ^1 -norm.

We use the same notation $\hat{O}$ for the resulting matrix. We also use $\hat{O}_{\pm}$ to denote the two columns of $\hat{O} = [\hat{O}_+, \hat{O}_-]$.

Algorithm D.3.

S1. Let $\hat{s}$ be the total count of words from $\hat{S}$ in the new article. Obtain $\hat{p}$ by

$$\hat{p} = \arg \max_{p \in [0,1]} \left\{ \hat{s}^{-1} \sum_{j=1}^{\hat{s}} d_j \log \left(p \hat{O}_{+,j} + (1-p) \hat{O}_{-,j} \right) + \lambda \log(p(1-p)) \right\},$$

where d_j , $\hat{O}_{+,j}$, and $\hat{O}_{-,j}$ are the j th entries of the corresponding vectors, and $\lambda > 0$ is a tuning parameter.

Remark: The screening statistic f_j in Algorithm D.1 differs slightly from the one used in our main paper: here, the “count of word j ” is replaced by the “number of articles containing word j .” For words of moderate or low frequency, the two versions yield similar values. For high-frequency words, however, the version in Algorithm D.1 helps reduce noise and provides greater robustness. The two statistics share similar theoretical properties, but the regularity conditions are more elegant and transparent for the version in the main paper, as both the “count of word j ” and the “number of articles containing word j ” are modeled as binomial random variables, yet the binomial parameters for the latter have a clearer form. For these reasons, we present the theory using the version in the main paper but adopt this slightly modified version for empirical analysis.

E Additional Empirical Results

In this section, we present further analysis of the empirical study discussed in the main text.

E.1 Summary Statistics and Data Pre-processing Details

For some of our analyses we study the association between news text and intradaily returns. For this purpose, we merge articles with transaction prices from the NYSE Trade and Quote (TAQ) database. Open-to-open and intraday returns are only used in our out-of-sample analysis from February 2004 to April 2020. We start the out-of-sample testing period from February 2004 because, starting in January 17, 2004, the Newswire data is streamlined and comes exclusively from one data source. Prior to that, Newswires data are derived from multiple news sources, which among other things can lead to redundant coverage of the same event. Although it does not affect in-sample training and validation, whose sole purpose is to associate words with positive or negative news, this could have an adverse impact on our out-of-sample analysis that is best suited for “fresh” news. Table E.1 lists step-by-step details for our sample filters.

The top panel of Figure E.1 plots the average number of articles in each half-hour interval throughout the day. News articles arrive more frequently prior to the market open and close. The bottom panel plots the average number of articles per day over a year. It shows holiday effects, as well as quarterly earnings season effects corresponding to a rise in article counts

Table E.1: Summary Statistics

Filter	Remaining Sample Size	Observations Removed
Total Number of Dow Jones Newswire Articles	36,283,597	
Combine chained articles	25,743,474	10,540,123
Remove articles with no stocks tagged	16,023,999	9,719,475
Remove articles with more than one stocks tagged	11,988,392	4,035,607
Number of articles whose tagged stocks have three consecutive daily returns from CRSP between Jan 1989 and Dec 2015	7,521,196	
Number of articles whose tagged stocks have open-to-open returns from CRSP since Feb 2004	6,656,843	
Number of articles whose tagged stocks have high-frequency returns from TAQ since Feb 2004	6,605,647	

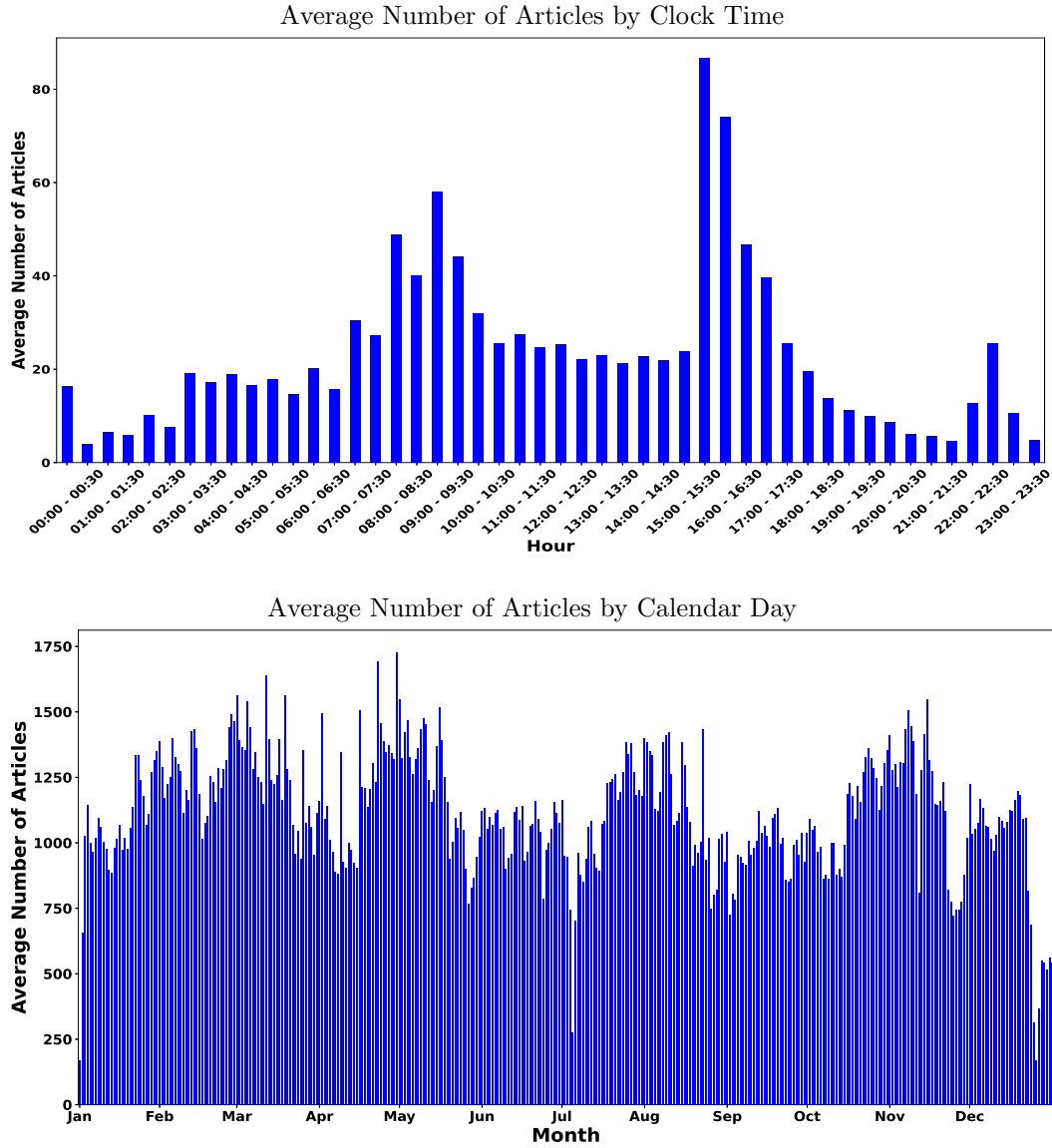
Note: In this table, we report the impact of each filter we apply on the number of articles in our sample. The sample period ranges from January 1, 1989 to April 30, 2020. The CRSP three-day returns are only used in training and validation steps, so we apply the CRSP filter only for articles dated from January 1, 1989 to December 31, 2015. The open-to-open returns and intraday returns are used in out-of-sample periods from February 1, 2004 to April 30, 2020.

around February, May, August, and November. Figure E.2 plots the total number of news articles per year in our sample. There is a steady increase in the number of articles until around 2007. Some news volume patterns reflect structural changes in news data sources and some reflect variation in the number of listed stocks. According to the *Dow Jones Newswires* user guide, there were three historical merges of news sources which occurred on October 31, 1996, November 5, 2001, and January 16, 2004, respectively.

To pre-process these articles, the first step is to remove proper nouns. Next, we follow common steps from the natural language processing literature to clean and structure news articles.¹ The first step is normalization, including 1) changing all words in the article to

¹We use the natural language toolkit (NLTK) in Python to preprocess the data.

Figure E.1: Average Article Counts

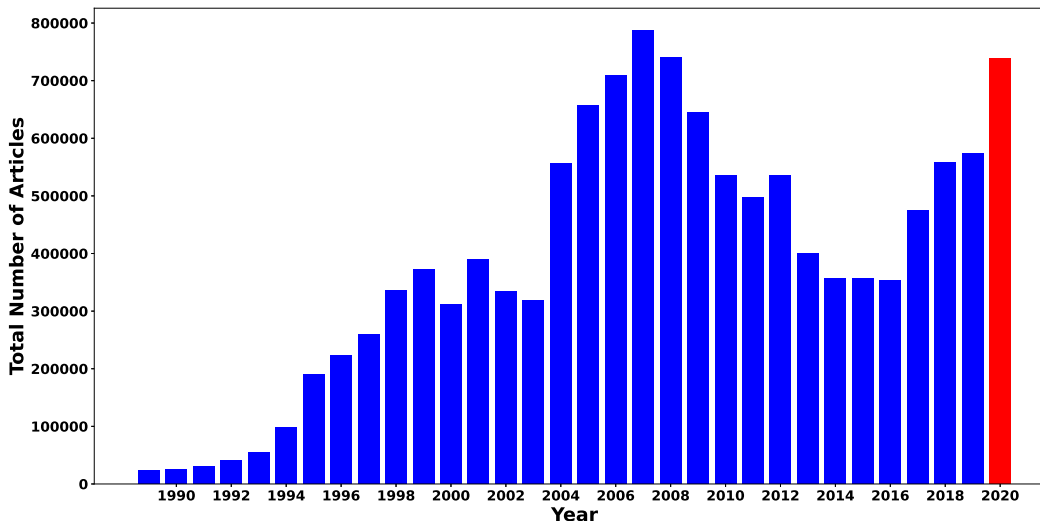


Note: The top figure plots the average numbers of articles per half an hour (24 hour EST time) from January 1, 1989 to April 30, 2020. The bottom figure plots the average numbers of articles per calendar day. Averages are taken over the full sample from January 1, 1989 to April 30, 2020.

lower case letters; 2) expanding contractions such as “haven’t” to “have not”; and 3) deleting numbers, punctuations, special symbols, and non-English words.² The second step uses stemming and lemmatizing to group together different forms of a word to analyze them as a single

²The list of English words is available from item 61 on http://www.nltk.org/nltk_data/.

Figure E.2: Annual Time Series of the Total Number of Articles



Note: This figure plots the annual time series of the total number of articles from January 1987 to April 2020. We only provide an estimate for 2020 (highlighted in red), by annualizing the total number of articles of the four months we observe, since we do not have a whole year’s data for this year.

root word, e.g., “disappointment” to “disappoint,” “likes” to “like,” and so forth.³ The third step is tokenization, which splits each article into a list of words. The fourth step removes common stop words such as “and”, “the”, “is”, and “are.”⁴ Finally, we translate each article into a vector of word counts, which constitutes its so-called “bag of words” representation. The resulting number of words in our dictionary is 39,673.

We also obtain a list of 2,337 negative words (Fin-Neg) and 353 positive words (Fin-Pos) from the Loughran-McDonald (LM) Sentiment Word Lists for comparison purposes.⁵

³The lemmatization procedure uses WordNet (<https://wordnet.princeton.edu/>) as a reference database. The stemming procedure uses the package “porter2stemmer” on <https://pypi.org/project/porter2stemmer/>. Frequently, the stem of an English word is not itself an English word; for example, the stem of “accretive” and “accretion” is “accret.” In such cases, we replace the root with the most frequent variant of that stem in our sample (e.g., “accretion”) among all words sharing the same stem, which aids interpretability of estimation output.

⁴We use the list of stopwords available from item 70 on http://www.nltk.org/nltk_data/.

⁵The LM word lists [5] also include 285 words in Fin-Unc, 731 words in Fin-Lit, 19 strong modal words and 27 weak words. We only present results based on Fin-Neg and Fin-Pos. Other dictionaries are less relevant to sentiment.

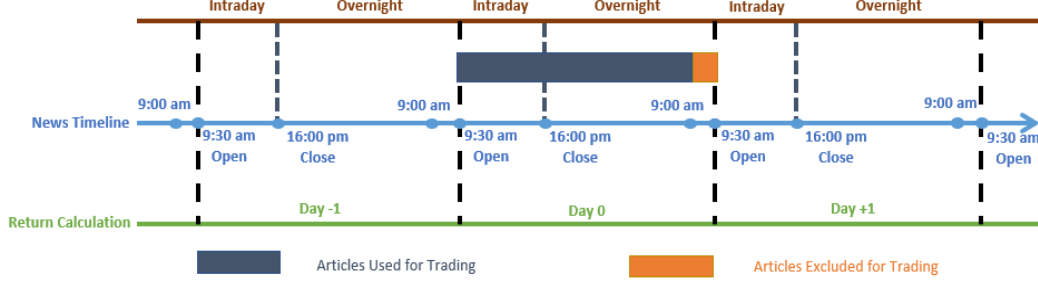


Figure E.3: News timeline and trading activities. We exclude news from 9:00 am to 9:30 am EST from trading (our testing exercise), although these news are still used for training and validation purposes. For news that occur on day 0, we build positions at the market opening on day 1, and rebalance at the next market opening, holding the positions of the portfolio within the day. We call this portfolio day+1 portfolio. Similarly, we can define day 0 and day-1, day ± 2 , $\dots$, day ± 10 portfolios.

LM show that the Harvard-IV dictionary misclassifies words when gauging tone in financial applications, and propose their own dictionary for use in business and financial contexts.

E.2 Timing Considerations for Portfolio Formation

Figure E.3 summarizes the news and trading timing of our approach. We are careful to form portfolios only at the market open each day for two reasons. First, overnight news can be challenging to act on prior to the morning open as this is the earliest time most traders can access the market. Second, with the exception of funds that specialize in high-frequency trading, funds are unlikely to change their positions continuously in response to intraday news because of their investment styles and investment process constraints. Following a similar choice of [8], we exclude articles published between 9:00am and 9:30am EST. By imposing that trade occurs at the market open and with at least a half-hour delay, we hope to better match realistic considerations like allowing funds time to calculate their positions in response to news and allowing them to trade when liquidity tends to be highest.

E.3 Robustness Check on Return Predictions

We provide a battery of results in Table E.2 that investigate the sensitivity of our results to varying some of the implementation choices we make in the portfolio formation step. To begin with, we expand the number of stocks for trading each day from 100 to 200. This slightly reduces average daily returns: 28 bps (EW) and 8 bps (VW), but increases Sharpe ratios to 5.68

(EW) and 1.62 (VW). Next, we exclude all news that occurs around earning announcement days since these news are more likely associated with fundamental information. Concretely, we select again 100 stocks on each day t , but avoid those whose earning announcements are scheduled on day $t - 1$, day t , and day $t + 1$.⁶ The resulting average returns are 36 bps (EW) and 9 bps (VW), suggesting that a significant portion of the information captured by our sentiment scores is not directly derived from earnings reports.

Additionally, we experiment with using residual returns rather than raw returns as our supervising y variable in model (4). Residuals are obtained from time-series regressions of raw returns over a two-year rolling window with respect to either the CAPM model, or the Fama-French three-factor model, or the CAPM model augmented with 17 industry portfolios.⁷ The results are comparable to our benchmark case, indicating that the out-of-sample portfolio performance remains robust to the choice of labels used in the training sample.

Last but not least, we consider a version of our model built on phrases and words together, and again find very similar quantitative results. We discuss our experiments with phrases further below. Like Sharpe ratios and average returns, the turnover and regression results throughout all variations are almost identical to the benchmark case, hence they are omitted from Table E.2. Overall, our portfolio formation strategy delivers robust out-of-sample performance for all of the implementation choices we examine.

Our baseline bag-of-words approach with single words ignores the context in which a word appears. To investigate the benefit of semantic information in word dependencies, we augment our bag-of-words by including two-word phrases (bigrams) in addition to single words (unigrams). Specifically, we start with all possible bigrams appearing in the original articles, remove infrequent phrases whose component words are not common (i.e., below 90% quantile of word counts) and, among remaining phrases, only keep those whose mutual information

⁶Earnings announcement dates are from Compustat.

⁷These 17 industry portfolios (equal-weighted) are obtained from Kenneth French’s website.

Table E.2: Performance of Alternative Daily News Sentiment Portfolios

Formation	Benchmark		Trading 200 Stocks		w/o. Earning Days		w/ Phrases	
	Sharpe Ratio	Average Return	Sharpe Ratio	Average Return	Sharpe Ratio	Average Return	Sharpe Ratio	Average Return
EW L-S	5.21	36	5.68	28	5.21	36	4.99	36
EW L	2.15	19	1.91	17	2.10	19	2.16	20
EW S	1.52	16	1.12	11	1.57	17	1.48	16
VW L-S	1.56	10	1.62	8	1.54	9	1.70	11
VW L	1.04	9	0.95	8	0.96	8	1.01	9
VW S	0.09	1	-0.04	0	0.08	1	0.18	2

Formation	CAPM Resid.		FF3 Resid.		CAPM+ Ind. Resid.	
	Sharpe Ratio	Average Return	Sharpe Ratio	Average Return	Sharpe Ratio	Average Return
EW L-S	5.16	35	5.33	35	5.84	40
EW L	2.13	19	2.09	18	2.24	20
EW S	1.54	17	1.60	17	1.85	20
VW L-S	1.80	12	1.73	11	1.74	12
VW L	1.20	10	1.11	9	1.13	10
VW S	0.12	2	0.15	2	0.17	2

Note: The table reports the performance of equal-weighted (EW) and value-weighted (VW) long-short (L-S) portfolios and their long (L) and short (S) legs. The performance measures include annualized Sharpe ratio and expected returns. We have presented results of the benchmark portfolio in Table 2. Additionally, we report results for alternative portfolios constructed similarly: trading up to 200 stocks per day, excluding firms near their earnings announcement days, using phrases (bigrams) in addition to words (unigrams), using residuals from time series regressions of raw returns (with respect to the CAPM model, the Fama-French three-factor model, or the CAPM model augmented with 17 industry portfolios as factors) as y in model (4).

scores rank in the top 1%.⁸ Including phrases dramatically increases the dimensionality of the problem: the dictionary includes 39,673 words plus 56,812 phrases.

Nevertheless, our methodology scales very well with respect to the expanded dictionary. Following the same aforementioned steps,⁹ we find that several phrases are selected in the set

⁸Mutual information score represents the extent to which words co-occur relative to the frequencies they appear separately. Accordingly, we calculate this score as the ratio of the frequency of a phrase (bigram) over the frequencies of its two component words. Mutual information score is potentially problematic if a component word is rather infrequent. As described in the text, we only consider mutual information scores for the universe of phrases whose component words are very common (above 90% quantile of word counts).

⁹We use the same grid values for α_+ , α_- , and κ for both words and phrases, with the total number of

Table E.3: Price Response On Days -1 , 0 , and $+1$

Formation	Sharpe	Turnover	Average	FF3		FF5		FF5+MOM	
	Ratio		Return	α	R^2	α	R^2	α	R^2
Day -1									
L-S	3.56	95.53%	25	26	0.27	26	0.46	26	0.72
L	1.56	95.92%	14	14	1.00	14	1.17	14	1.32
S	1.01	95.14%	12	11	0.35	11	0.67	11	0.67
Day 0									
L-S	7.92	95.57%	90	90	0.38	90	0.98	90	1.00
L	5.73	95.93%	53	52	7.55	53	7.62	53	7.88
S	2.67	95.21%	37	38	4.65	37	5.25	37	5.30
Day +1									
L-S	5.21	95.55%	36	36	1.26	36	1.48	36	1.79
L	2.15	95.93%	19	18	38.88	18	39.49	18	40.33
S	1.52	95.17%	16	18	34.50	18	35.43	18	36.73
Day -1 to +1									
L-S	12.05	95.62%	154	155	0.83	154	1.27	154	1.31
L	5.25	95.95%	85	82	21.06	83	21.67	83	22.74
S	3.56	95.28%	70	72	18.67	71	19.78	71	20.77

Note: The table repeats the analysis of Table 2 for the equal-weighted long-short (L-S) portfolios plotted in Figure E.4, as well as their long (L) and short (S) legs. Sharpe ratios are annualized, while returns and alphas are in basis points per day.

of sentiment-charged terms (see Table E.6 in Appendix E.7). On the positive side we have *repurchase program*, *stock repurchase*, *tender offer*, *definitive agreement*, *investment trust*, *cash dividend*, *estate investment*, *free cash* among the most common phrases that appear at least 10 times out of 17 models; whereas on the negative side, phrases such as *negatively impact* and *registration statement* appear 14 times. Interestingly, as shown in Table E.2, we find that including phrases does not meaningfully impact the performance of our out-of-sample one-day portfolios.

phrases replacing the total number of words in the calculation for κ . To reduce computational burden, we apply the same grid value for both words and phrases, resulting in a total of 60 sets of tuning parameters.

E.4 Speed of Information Assimilation

The analysis in Figure 5 and Table 2 studies how sentiment on day t relates to returns on day $t + 1$. In the next two subsections, we investigate the timing of price responses to news sentiment with finer resolution and with an event calendar.

E.4.1 Lead-lag Relationship Among News and Prices

In our training sample, we estimate SESTM from the three-day return beginning the day before an article is published and ending the day after. In Figure E.4, we separately investigate the subsequent out-of-sample association between news sentiment on day t and returns on day $t - 1$ (from open $t - 1$ to open t), day t , and day $t + 1$. We report this association in the economic terms of trading strategy performance. The association between sentiment and the $t + 1$ return is identical to that in Figure 5, and is rightly interpreted as performance of an implementable (out-of-sample) trading strategy. For the association with returns on days $t - 1$ and t , the interpretation is different. These are not implementable strategies because the timing of the news article would not generally allow a trader to take a position and exploit the return at time t (and certainly not at $t - 1$). They are instead interpreted as out-of-sample correlations between article sentiment and realized returns, converted into economic return units. They are out-of-sample because the fitted article sentiment score, $\hat{p}_i$, is based on a model estimated from an entirely distinct data set (that pre-dates the arrival of article i and returns $y_{i,t-1}$, $y_{i,t}$, and $y_{i,t+1}$). Table E.3 reports summary statistics for these portfolios, including their annualized Sharpe ratios, average returns, alphas, and turnover. For this analysis, we specialize to equally weighted portfolios.

The Day -1 strategy (dashed black line) shows the association between news article sentiment and the stock return one day prior to the news. This strategy thus quantifies the extent to which our sentiment score picks up on stale news. On average, prices move ahead of news in our sample, as indicated by the infeasible annualized Sharpe ratio of 3.56. Thus we see that

much of the daily news flow is prompted by or repeats information already known to market participants.

The Day 0 strategy (dashed red line) shows the association between news and returns on the same day. This strategy assesses the extent to which our sentiment score captures fresh news that has not previously been incorporated into prices. The Day 0 strategy provides the clearest out-of-sample validation that our sentiment score accurately summarizes fresh, value-relevant information in news text. In particular, price responses are most concentrated on the same day that the news arrives, as reflected by the same-day infeasible annualized Sharpe ratio of 7.92.

The Day +1 strategy (solid black line) shows the association between news on day t and returns on the subsequent day. It thus quantifies the extent to which information in our sentiment score is impounded into prices with a delay. This corresponds exactly to the implementable trading strategy shown in Figure 5. The excess performance of this strategy, summarized in terms of an annualized Sharpe ratio of 5.21 (and shown to be all alpha in Table 2), supports the maintained alternative hypothesis.

We next analyze trading strategies that trade in response to news sentiment with various time delays. We focus on one-day open-to-open returns initiated anywhere from one to 10 days following the announcement. Figure E.5 reports average returns in basis points per day with shaded 95% confidence intervals (constructed via Newey-West standard errors). It shows the long-short portfolio as well as the long and short legs separately. For the long-short strategy, sentiment information is essentially fully incorporated into prices by the start of Day +3. For the individual sides of the trade, the long leg appears to achieve full price incorporation within two days, while the short leg takes one extra day.

E.4.2 High Frequency Trading

We also consider very rapid price responses via intra-day high frequency trading that takes a position every 30 minutes after the article’s time stamp, and holds positions for only 30 minutes. For intra-day news, trading starts the moment news arrive. For overnight news, trading occurs at the next market open following the news announcements, as this is, generally speaking, the earliest time these news can be traded. Figure E.6 demonstrates a steady decay in average returns for both long and short positions reflecting the decreasing reward to incorporating news in prices as time lapses. These returns are in terms of basis points per 30 minutes, so an 12 bps return for the long-short portfolio within the first 30 minutes is exorbitant in daily or annual terms.

E.4.3 Fresh News and Stale News

The evidence in Section E.4.1 indicates that a substantial fraction of news is “old news” and already impounded in prices by the time an article is published. The assimilation analysis of Figure E.5 thus pools together both fresh and stale news. In order to investigate the difference in price response to fresh versus stale news, we conduct separate analyses for articles grouped by the novelty of their content.

We construct a measure of article novelty as follows. For each article for firm i on day t , we calculate its cosine similarity with all articles about firm i on the five trading days prior to t (denoted by the set $\chi_{i,t}$). Novelty of recent news is judged based on its most similar preceding article, thus we define article novelty as

$$\text{Novelty}_{i,t} = 1 - \max_{j \in \chi_{i,t}} \left(\frac{d_{i,t} \cdot d_j}{\|d_{i,t}\| \|d_j\|} \right).$$

Figure E.7 splits out our news assimilation analysis by article novelty. We partition news into two groups. The “fresh” news group contains articles novelty score of 0.75 or more, while

“stale” news has novelty below 0.75.¹⁰ It shows that the one-day price response (from market open after news arrival to the open the following day) of the long-short portfolio formed on fresh news (solid blue line) is about 50% higher than that response to stale news (solid red line). Furthermore, it takes four days for fresh news to be fully incorporated in prices (i.e., the day five average return is statistically indistinguishable from zero), or twice as long as the two days it takes for prices to complete their response to stale news. This result extends the evidence from [7] that market responses to stale news are muted but nonetheless significant.

E.4.4 Stock Heterogeneity Analysis: Size and Volatility

Figure E.7 investigates differential price responses to different types of news. In this section, we investigate differences in price assimilation with respect to heterogeneity among stocks.

The first dimension of stock heterogeneity that we analyze is market capitalization. Larger stocks represent a larger share of the representative investor’s wealth and command a larger fraction of investors’ attention or information acquisition effort [e.g., 10, 9]. In Figure E.8, we analyze the differences in price adjustment based on firm size by sorting stocks into big and small groups (based on NYSE median market capitalization each period). Prices of large stocks respond by 6 basis points on the first day after news arrival, and their price response is complete after one day (the day two effect is insignificantly different from zero). The price response of small stocks is 60 basis points on the first day after news arrival, more than ten times larger, and it takes three days for their news to be fully incorporated into prices.

The second dimension of heterogeneity that we investigate is stock volatility. Volatility embodies a limit-to-arbitrage, as higher volatility dissuades traders from taking a position based on their information, all else equal. At the same time, higher stock volatility represents more uncertainty about asset outcomes. With more uncertainty, there are potentially

¹⁰The average article novelty in our sample is approximately 0.75. The conclusions from Figure E.7 are insensitive to the choice of cutoff.

larger profits to be earned by investors with superior information, which incentivizes informed investors to allocate more attention to volatile stocks all else equal. But higher uncertainty may also reflect that news about the stock is more difficult to interpret, manifesting in slower incorporation into prices. The direction of this effect on price assimilation is ambiguous.

Figure E.9 shows the comparative price response of high versus low volatility firms.¹¹ The price response to SESTM sentiment on the first day following news arrival is 25 basis points for low volatility firms, but 46 basis points for high volatility firms. It is worth noting that in both Figure E.8 and Figure E.9 the largest differences in price response appear on the short side, consistent with short sales frictions being particularly pertinent among small and risky firms.

E.5 Transaction Costs

Our trading strategy performance analysis thus far ignores transaction costs. This is because the portfolios above are used primarily to give economic context and a sense of economic magnitude to the strength of the predictive content of each sentiment measure. The profitability of the trading strategy net of costs is neither here nor there for assessing sentiment predictability. Furthermore, the comparative analysis of SESTM, LM, RavenPack, and BERT is apples-to-apples in the sense that all four strategies face the same trading cost environment and have similar turnover.

That said, evaluating the usefulness of news article sentiment for practical portfolio choice is a separate question and is interesting in its own right. But the practical viability of our sentiment strategies is difficult to ascertain from preceding tables due to their large turnover.

¹¹Specifically, we calculate idiosyncratic volatility from residuals of a market model using the preceding 250 daily return observations. We then estimate the conditional idiosyncratic volatility via exponential smoothing according to the formula $\sigma_t = \sum_{i=0}^{\infty} (1 - \delta)\delta^i u_{t-1-i}^2$ where u is the market model residual and δ is chosen so that the exponentially-weighted moving average has a center of mass $(\delta/(1 - \delta))$ of 60 days .

Table E.4: Performance of SESTM Long-Short Portfolios Net of Transaction Costs

γ	Turnover	Gross		Net	
		Return	Sharpe Ratio	Return	Sharpe Ratio
0.1	0.16	4.29	1.96	0.14	0.06
0.2	0.18	8.97	3.41	4.33	1.65
0.3	0.27	13.31	4.23	6.41	2.04
0.4	0.37	17.32	4.74	7.96	2.18
0.5	0.46	21.00	5.05	9.17	2.21
0.6	0.56	24.41	5.23	10.09	2.16
0.7	0.66	27.57	5.32	10.76	2.08
0.8	0.76	30.51	5.34	11.21	1.96
0.9	0.86	33.26	5.30	11.47	1.83

Note: The table reports the performance of equally-weighted long-short EWCT portfolios based on SESTM scores. The EWCT parameter is γ . Average returns are reported in basis points per day and Sharpe ratios are annualized. Portfolio average daily turnover is calculated as $\frac{1}{2T} \sum_{t=1}^T \left(\sum_i \left| w_{i,t+1} - \frac{w_{i,t}(1+y_{i,t+1})}{1+\sum_j w_{j,t}y_{j,t+1}} \right| \right)$.

In this section, to better understand the relevance of SESTM’s predictability gains for practical asset management, we investigate the performance of sentiment-based trading strategies while taking into account trading costs.

To approximate the net performance of a strategy, we assume that each asset incurs a daily transaction cost, that is 10bps for large stocks, and 20bps for smaller ones (bellow NYSE 20% breakpoints). The choice approximates the average trading cost experienced by large asset managers, as reported in [3], while also accounting for differential costs of large and small stocks.

We propose a novel trading strategy that directly reduces portfolio turnover and hence trading costs. Specifically, we design a strategy that i) turns over (at most) a fixed proportion of the existing portfolio every period and ii) assigns weights to stocks that decay exponentially with the time since the stock was in the news. These augmentations effectively extend the stock holding period from one day to multiple days. We refer to this approach as an exponentially-weighted calendar time (EWCT) portfolio.

On the first day of trading, we form an equal-weighted portfolio that is long the top N stocks in terms of news sentiment on a given day and short N stocks with the most negative

news sentiment. A single parameter (γ) determines the extent of the turnover constraint. Each subsequent day t , we liquidate a fixed proportion γ of all existing positions, and reallocate that γ proportion to an equal-weighted long-short portfolio based on day t news. For a stock i experiencing large positive sentiment news on day t , its weight changes according to $w_{i,t} = \frac{\gamma}{N} + (1 - \gamma)w_{i,t-1}$. For a stock i in the long-side of the portfolio at day $t - 1$ but with no news on date t , its portfolio weight decays to $w_{i,t} = (1 - \gamma)w_{i,t-1}$. The analogous weight transitions apply to the short leg of the strategy.

To see this more clearly, consider an example with three stocks, A , B , and C , in a broader cross section of stocks. Suppose at time t that A has a weight of zero ($w_{A,t} = 0$) while B and C had their first and only positive news five and ten days prior, respectively (that is, $w_{B,t} = (1 - \gamma)^4 \gamma / N$ and $w_{C,t} = (1 - \gamma)^9 \gamma / N$). Now suppose that, at time $t + 1$, positive news articles about stocks A and C propel them into the long side of the sentiment strategy, and neither A , B , nor C experiences news coverage thereafter. The weight progression of A , B , and C is the following:

	t	$t + 1$	$t + 2$	$t + 3$	...
w_A	0	$\frac{\gamma}{N}$	$\frac{\gamma}{N}(1 - \gamma)$	$\frac{\gamma}{N}(1 - \gamma)^2$	...
w_B	$\frac{\gamma}{N}(1 - \gamma)^4$	$\frac{\gamma}{N}(1 - \gamma)^5$	$\frac{\gamma}{N}(1 - \gamma)^6$	$\frac{\gamma}{N}(1 - \gamma)^7$	...
w_C	$\frac{\gamma}{N}(1 - \gamma)^9$	$\frac{\gamma}{N}(1 + (1 - \gamma)^{10})$	$\frac{\gamma}{N}(1 - \gamma)(1 + (1 - \gamma)^{10})$	$\frac{\gamma}{N}(1 - \gamma)^2(1 + (1 - \gamma)^{10})$	...

The portfolio weights for A and C spike upon news arrival and gradually revert to zero. The turnover parameter simultaneously governs both the size of the weight spike at news arrival (the amount of portfolio reallocation) as well as the exponential decay rate for existing weights. For high values of γ , new information is immediately assigned a large weight in the portfolio and old information is quickly discarded, generating large portfolio turnover. In contrast, low values of γ reduce turnover both by limiting the amount of wealth reallocated to the most recent news and by holding onto past positions for longer, which in turn increases the effective holding period of the strategy. Finally, note that the EWCT strategy guarantees daily turnover is never larger than γ . When a stock is already in a portfolio and a new article

arrives with the same sign as recent past news (as in the example of stock C) the actual turnover will be less than γ .

Table E.4 reports the performance of EWCT portfolios as we vary turnover limits from mild ($\gamma = 0.9$) to heavily restricted ($\gamma = 0.1$). Moving down the rows we see that a more severe turnover restriction drags down the gross Sharpe ratio of the trading strategy, indicating a loss in predictive information due to signal smoothing. This drag is offset by a reduction in trading costs. The net Sharpe ratio peaks at 2.21 when $\gamma = 0.5$. That is, with a moderate amount of turnover control (and concomitant signal smoothing), the gain from reducing transaction costs outweighs the loss in predictive power. In sum, Table E.4 demonstrates an attractive risk-return tradeoff to investing based on news sentiment even after accounting for transactions costs.

E.6 RavenPack

The data we use are composite sentiment scores from RavenPack News Analytics 4 (RPNA4) DJ Edition Equities. The underlying news data for this version of RavenPack should be identical to the collection of Dow Jones articles that we use to build SESTM. However, the observation count that we see in RavenPack is somewhat larger than the number of observations we can construct from the underlying Dow Jones news. The discrepancy arises from the black-box transformations that RavenPack applies during its analytics process. Ultimately, what we observe in RavenPack is their collection of article-level scores that is indexed by stock ticker and time, and it is not possible to accurately map RavenPack observations back to the original news. As a result, we cannot pin down the precise source of the difference in observation counts between our two data sets. The most likely explanation is that RavenPack uses a proprietary algorithm to assign ticker tags to articles, while we rely on the tags assigned directly by Dow Jones.

Figure E.10 shows the differences in observation counts in our data set (the complete set

of Dow Jones Newswires from 1984 through April 2020) versus RavenPack (only used in our out-of-sample analysis from 2004). We restrict all counts to those having a uniquely matched stock identifier in CRSP with which we can find open-to-open returns. We see that the article counts for Newswires and RavenPack are quite close except for the middle part of our sample. When we map Newswires to CRSP, we use articles' stock identifier tags, which are provided by Dow Jones.

E.7 Additional Exhibits

Table E.5: List of Top 50 Positive/Negative Sentiment Words

Positive					Negative				
Word	Score	Samples	LM	HIV	Word	Score	Samples	LM	HIV
repurchase	0.568	17	0	0	downgrade	0.391	17	1	0
declare	0.540	17	0	0	hurt	0.419	17	1	1
upgrade	0.536	17	0	1	fell	0.430	17	0	0
beat	0.557	16	0	0	warn	0.434	17	1	0
boost	0.532	16	1	1	slowdown	0.436	17	1	0
jump	0.551	15	0	0	lower	0.438	17	0	1
exceed	0.531	15	0	0	resign	0.440	17	1	0
raise	0.527	15	0	0	soft	0.445	17	0	0
strong	0.520	15	1	0	delay	0.446	17	1	1
definitive	0.525	14	0	1	weak	0.447	17	1	0
board	0.524	14	0	1	slow	0.456	17	1	0
split	0.522	14	0	0	cut	0.457	17	1	1
strength	0.519	14	1	0	drop	0.461	17	0	1
approval	0.526	13	0	1	decrease	0.470	17	0	1
dividend	0.525	13	0	0	disappointing	0.389	16	1	0
rise	0.524	13	0	0	loss	0.465	15	1	1
excellent	0.522	13	1	1	negative	0.454	14	1	1
authorize	0.534	12	0	0	fall	0.459	14	0	1
coverage	0.529	12	0	0	parentheses	0.462	14	0	0
momentum	0.526	12	0	0	incur	0.463	14	0	0
square	0.522	12	0	0	cite	0.465	14	0	0
shareholder	0.520	12	0	0	decline	0.469	14	1	1
double	0.518	12	0	0	bad	0.470	14	1	1
solid	0.517	12	0	0	parenthesis	0.434	13	0	0
climb	0.535	11	0	0	longer	0.464	13	0	0
attract	0.522	11	0	1	fail	0.466	13	1	1
complement	0.520	11	0	1	allege	0.468	13	1	1
presence	0.520	11	0	0	pursue	0.470	13	0	0
diversify	0.519	11	0	0	lawsuit	0.471	13	0	0
record	0.516	11	0	0	restate	0.473	13	1	0
block	0.516	11	0	0	registration	0.474	13	0	0
payable	0.532	10	0	0	miss	0.426	12	1	1
branch	0.526	10	0	0	warrant	0.461	12	0	0
trust	0.524	10	0	1	compliance	0.467	12	0	0
electric	0.520	10	0	0	reverse	0.469	12	0	0
foot	0.520	10	0	0	lack	0.470	12	1	1
ownership	0.518	10	0	0	however	0.476	12	0	0
defense	0.518	10	0	1	inventory	0.461	11	0	0
stake	0.515	10	0	0	pressure	0.463	11	0	0
diverse	0.515	10	0	0	shelf	0.464	11	0	0
excite	0.515	10	0	1	lose	0.467	11	1	1
private	0.513	10	0	0	difficult	0.467	11	1	1
top	0.513	10	0	0	problem	0.468	11	1	1
estate	0.523	9	0	0	impairment	0.469	11	1	0
outstanding	0.522	9	0	1	low	0.469	11	0	1
driven	0.522	9	0	0	lender	0.470	11	0	0
favorable	0.519	9	1	1	reduce	0.471	11	0	0
military	0.517	9	0	0	defer	0.472	11	1	0
own	0.514	9	0	0	transition	0.474	11	0	0
premier	0.514	9	1	1	compensation	0.475	11	0	0

Note: The table shows the list of top 50 words with positive and negative sentiment based on screening from the 17 training and validation samples. These 50 words are selected by first sorting on the number of samples (out of 17) in which the word was selected, and then sorting on their average sentiment score. A label 1 in Columns “LM” or “HIV” indicates this word also appears in the LM or Harvard-IV dictionaries.

Table E.6: List of Top 50 Positive/Negative Sentiment Phrases (including Words)

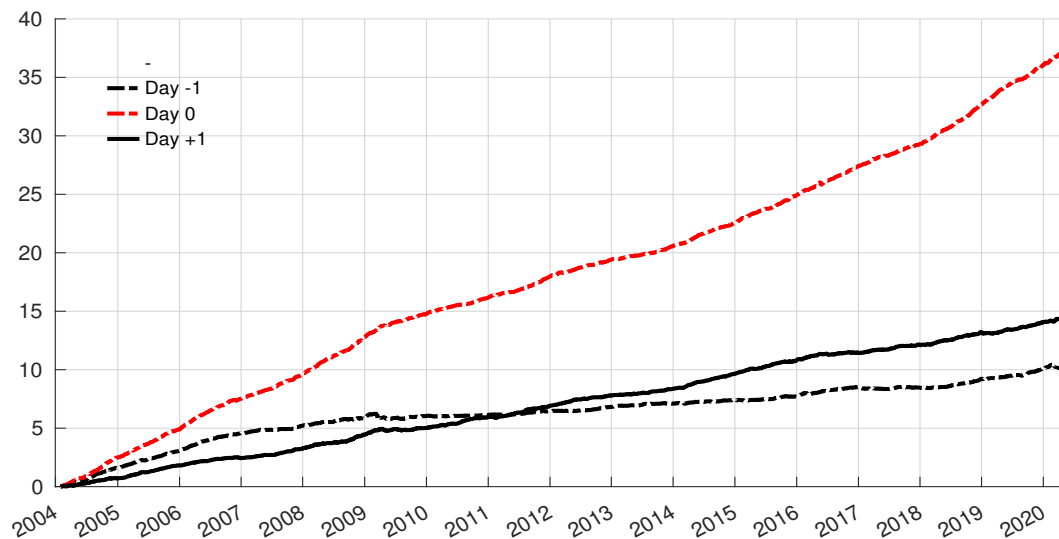
Positive					Negative				
Word	Score	Samples	LM	HIV	Word	Score	Samples	LM	HIV
repurchase	0.568	17	0	0	disappointing	0.390	17	1	0
repurchase program	0.562	16	0	0	downgrade	0.392	17	1	0
declare	0.542	16	0	0	hurt	0.420	17	1	1
upgrade	0.536	16	0	1	fell	0.431	17	0	0
beat	0.558	15	0	0	lower	0.438	17	0	1
jump	0.550	15	0	0	resign	0.441	17	1	0
climb	0.539	14	0	0	soft	0.446	17	0	0
stock repurchase	0.558	13	0	0	weak	0.448	17	1	0
tender offer	0.532	13	0	0	warn	0.435	16	1	0
definitive agreement	0.532	13	0	0	slowdown	0.440	16	1	0
boost	0.532	13	1	1	drop	0.462	16	0	1
exceed	0.531	13	0	0	delay	0.444	15	1	1
surge	0.545	11	0	1	slow	0.456	15	1	0
investment trust	0.531	11	0	0	miss	0.425	14	1	1
cash dividend	0.530	11	0	0	cut	0.457	14	1	1
momentum	0.527	11	0	0	negatively impact	0.460	14	0	0
customary	0.539	10	0	0	parentheses	0.461	14	0	0
tender	0.536	10	0	0	registration statement	0.462	14	0	0
estate investment	0.532	10	0	0	negative	0.455	13	1	1
accretion	0.532	10	0	0	interim	0.456	13	0	0
dividend	0.527	10	0	0	non cash	0.464	13	0	0
free cash	0.526	10	0	0	loss	0.465	13	1	1
lift	0.532	9	0	0	parenthesis	0.434	12	0	0
advisor	0.525	9	0	0	warrant	0.462	12	0	0
open market	0.561	8	0	0	liability payable	0.467	12	0	0
buy back	0.539	8	0	0	decrease	0.468	12	0	1
financial advisor	0.536	8	0	0	deficit	0.442	11	1	1
raise	0.532	8	0	0	weaken	0.457	11	1	1
branch	0.529	8	0	0	fall	0.458	11	0	1
definitive	0.526	8	0	1	doubtful	0.463	11	1	1
board	0.526	8	0	1	cite	0.465	11	0	0
square	0.525	8	0	0	fail	0.467	11	1	1
double digit	0.524	8	0	0	gross profit	0.471	11	0	0
split	0.523	8	0	0	private placement	0.472	11	0	0
excellent	0.522	8	1	1	shelf registration	0.447	10	0	0
strong	0.521	8	1	0	offer solicitation	0.451	10	0	0
electric	0.521	8	0	0	deterioration	0.452	10	1	0
shareholder	0.521	8	0	0	incur	0.461	10	0	0
treasury	0.559	7	0	0	temporary	0.461	10	0	0
surpass	0.549	7	1	1	decline	0.469	10	1	1
upward	0.546	7	0	1	primarily due	0.469	10	0	0
authorize	0.538	7	0	0	restate	0.475	10	1	0
payable	0.538	7	0	0	net proceeds	0.475	10	0	0
coverage	0.534	7	0	0	shortfall	0.338	9	1	0
discretion	0.528	7	0	1	auditor	0.433	9	0	0
petroleum	0.527	7	0	0	unlawful	0.447	9	1	1
agricultural	0.525	7	0	0	base compensation	0.451	9	0	0
rise	0.525	7	0	0	unlawful prior	0.452	9	0	0
strength	0.522	7	1	0	shelf	0.462	9	0	0
diversify	0.520	7	0	0	doubtful account	0.463	9	0	0

Note: The table shows the list of top 50 phrases (including words) with positive and negative sentiment based on screening from the 17 training and validation samples. These 50 phrases are selected by first sorting on the number of samples (out of 17) in which the phrase was selected, and then sorting on their average sentiment score. A label 1 in Columns “LM” or “HIV” indicates this phrase also appears in the LM or Harvard-IV dictionaries.

References

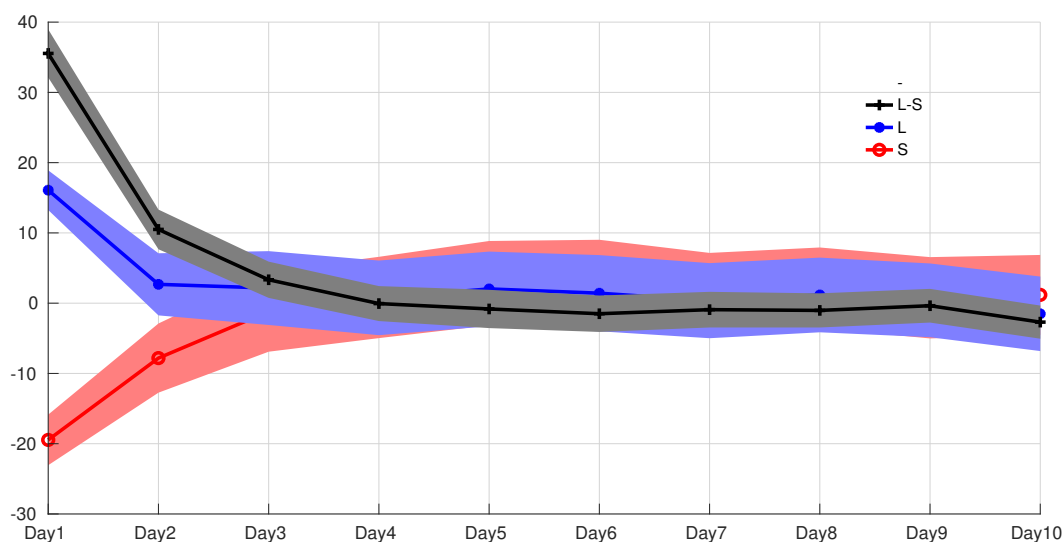
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Figure E.4: Price Response On Days -1 , 0 , and $+1$



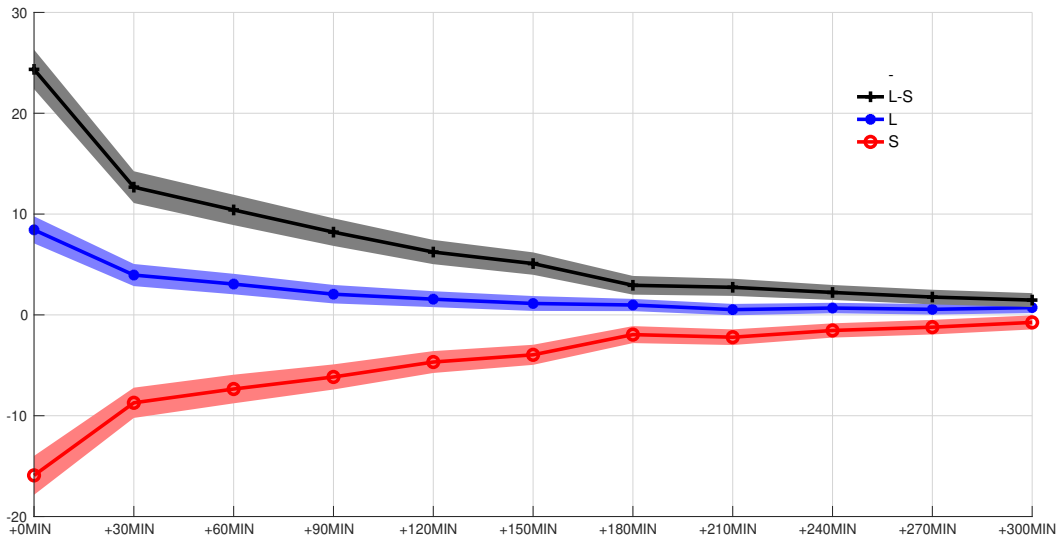
Note: This figure compares the out-of-sample cumulative log returns of long-short portfolios sorted on sentiment scores. The Day -1 strategy (dashed black line) shows the association between news and returns one day prior to the news; the Day 0 strategy (dashed red line) shows the association between news and returns on the same day; and the Day $+1$ strategy (solid black line) shows the association between news and returns one day later. The Day -1 and Day 0 strategy performance is out-of-sample in that the model is trained on a sample that entirely precedes portfolio formation, but these are not implementable strategies because the timing of the news article would not necessarily allow a trader to take such positions in real time. They are instead interpreted as out-of-sample correlations between article sentiment and realized returns in economic return units. The Day $+1$ strategy corresponds to the implementable trading strategy shown in Figure 5. All strategies are equal-weighted.

Figure E.5: Speed of News Assimilation



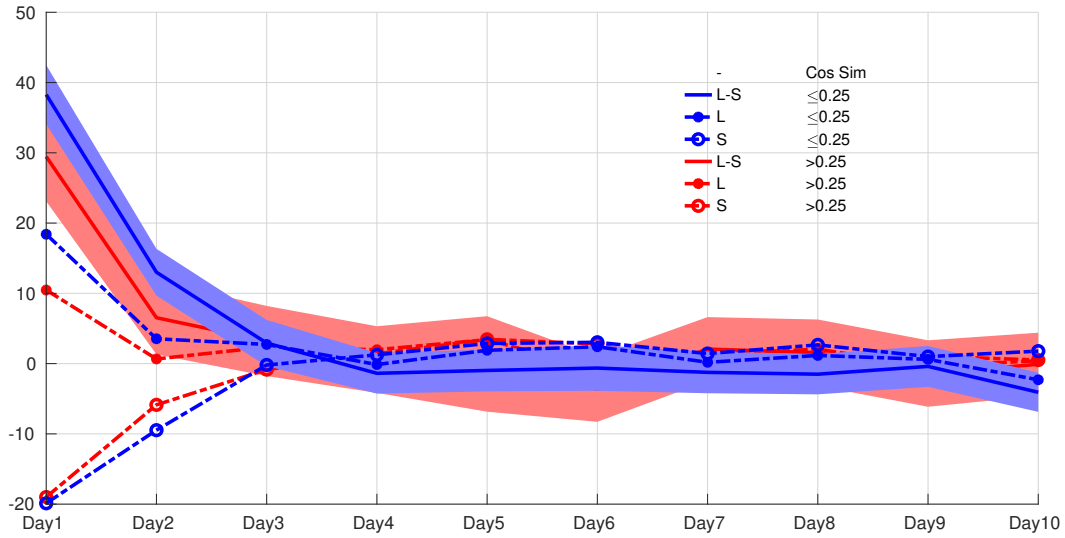
Note: This figure compares average one-day holding period returns to the news sentiment trading strategy as a function of when the trade is initiated. We consider daily open-to-open returns initiated from one to 10 days following the announcement. We report equal-weighted portfolio average returns (in basis points per day) in excess of an equal-weighted version of the S&P 500 index, with 95% confidence intervals given by the shaded regions. The confidence intervals around the average excess returns are constructed using Newey-West standard errors. We consider the long-short (L-S) portfolio as well as the long (L) and short (S) legs separately.

Figure E.6: Speed of News Assimilation (Intraday Trading)



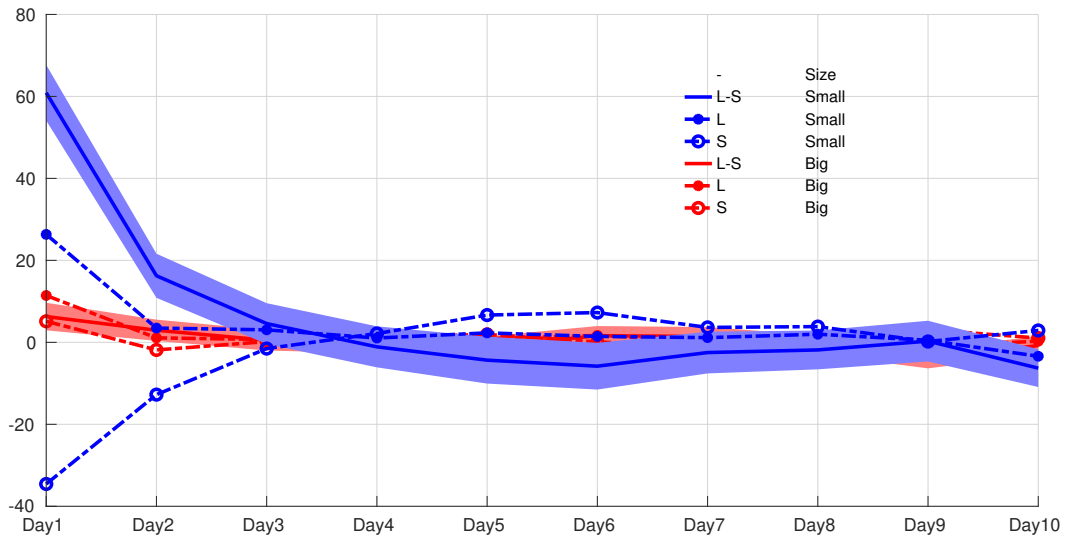
Note: This figure compares average 15-minute holding period returns to the news sentiment trading strategy as a function of when the trade is initiated. We consider 30-minute returns initiated from the earliest time point feasible to trade (+0 MIN) to 5 hours (+ 300 MIN) following the announcement. We report equal-weighted portfolio average returns (in basis points per 30 minutes), with 95% confidence intervals given by the shaded regions. The confidence intervals around the average returns are constructed using Newey-West standard errors. We consider the long-short (L-S) portfolio as well as the long (L) and short (S) legs separately.

Figure E.7: Speed of News Assimilation (Fresh Versus Stale News)



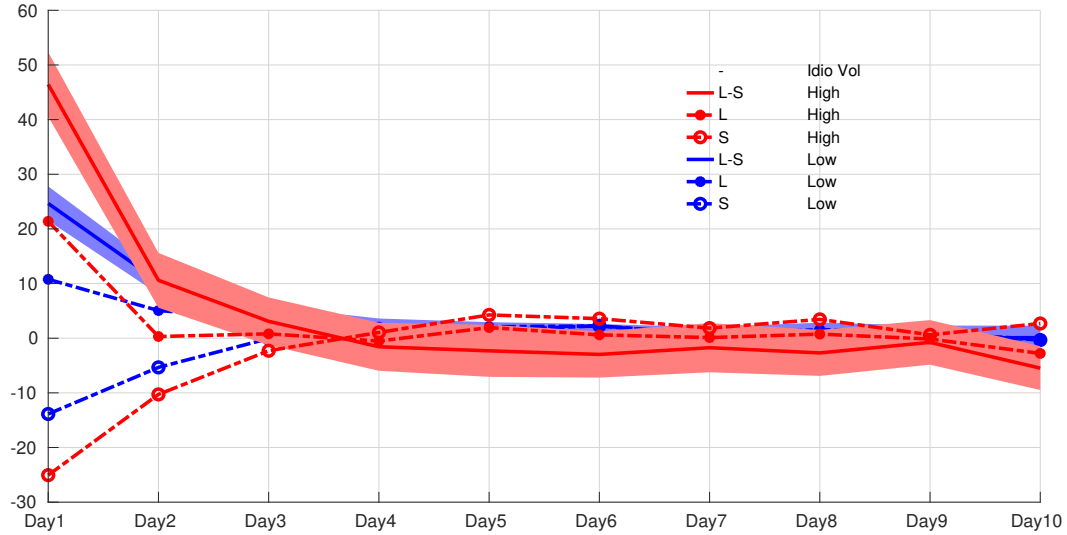
Note: See Figure E.5. This figure divides stock-level news events based on maximum cosine similarity with the stock's prior news.

Figure E.8: Speed of News Assimilation (Big Versus Small Stocks)



Note: See Figure E.5. This figure divides stock-level news events based on stocks' market capitalization. The big/small breakpoint is defined as the NYSE median market capitalization each period.

Figure E.9: Speed of News Assimilation (High Versus Low Volatility Stocks)



Note: See Figure E.5. This figure divides stock-level news events based on stocks' idiosyncratic volatility. The high/low volatility breakpoint is defined as the cross-sectional median volatility each period.

Figure E.10: Dow Jones Newswire and RavenPack Observation Counts

